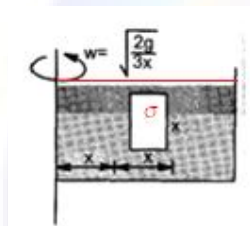
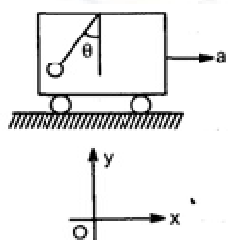


axis. The value of $\frac{a_x}{g}$, where a_x is the horizontal acceleration of the cube away from the axis, is



σ = density of solid cube ρ = density of the liquid

(D) The string of a pendulum makes an angle θ with vertical when the cart moves with an acceleration a . If the cart is filled with liquid of density $\rho = \sigma$ (density of the bob), the value of the angle of inclination with vertical becomes θ' , then $\frac{\tan \theta}{\tan \theta'}$ is



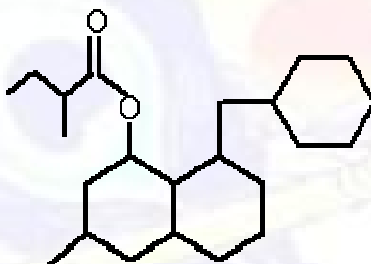
(S) $\frac{2\sigma + \rho}{2(\sigma - \rho)}$

SECTION – I
(SINGLE INTEGER ANSWER TYPE)

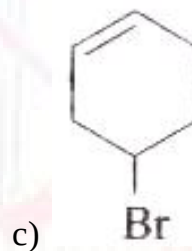
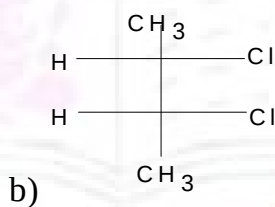
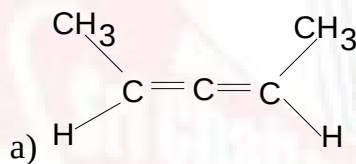
This section contains 8 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

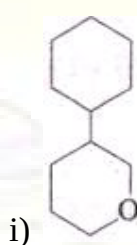
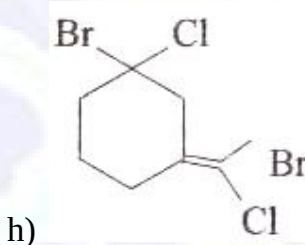
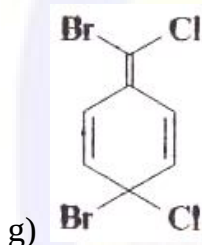
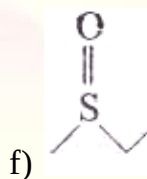
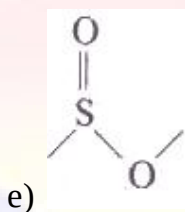
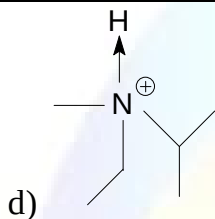
Marking scheme +4 for correct answer, 0 if not attempted and 0 in all other cases.

21. How many chiral centres are present in the following compound?



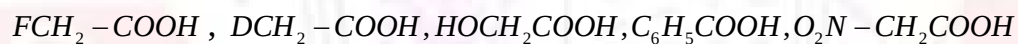
22. Examine the following structural formulae and select those that are chiral.





23. A 2:3 racemic mixture (by weight) has an observed specific rotation of 16° . The specific rotation of the optically pure component is $x \times 10$ degrees where x is

24. How many of the following acids are more acidic than acetic acid?



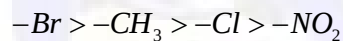
25. Total number of stereo isomers $(CH_3CH(D))_3CH$ for the given compound is/are

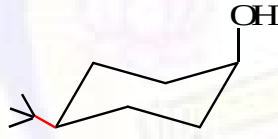
26.  on monochlorination give X number of isomeric products which on fractional distillation give Y number of isomeric products. Find the value of $|X - 2Y|$?

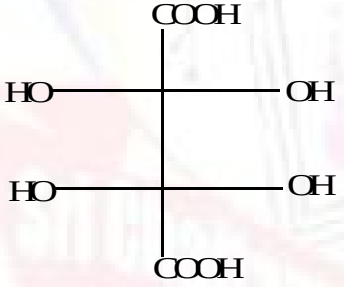
27. How many statements are correct?

1. Trans cyclooctene is chiral due to presence of chiral (stereo) axis.

2. Bulkiness order at ortho positions in Biphenyl system follow

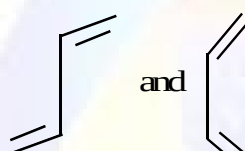
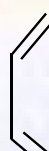


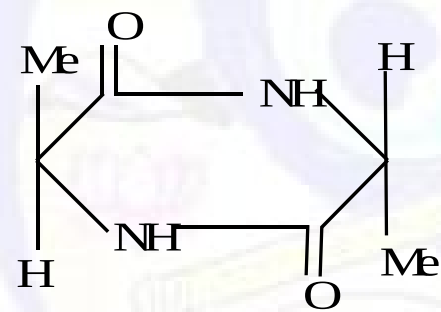
3.  is an optically active compound

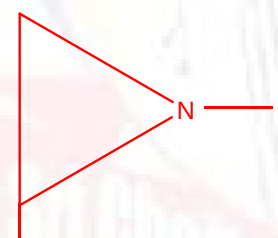
4.  is a meso compound due to plane of symmetry

28. Find the number of incorrect statements

I. Meso tartaric acid has different melting point than (+) Tartaric acid

II.  and  are conformational diastereomers

III.  is optically active molecule

IV.  this molecule has chiral centre and shows optical activity

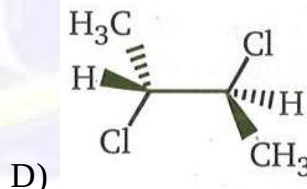
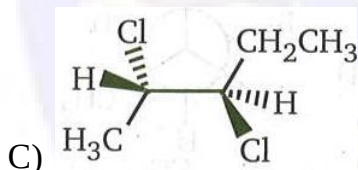
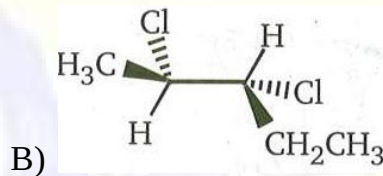
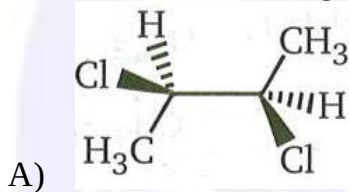
SECTION - II

(ONE OR MORE CORRECT ANSWER TYPE)

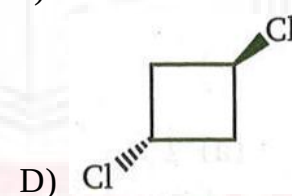
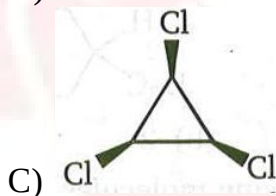
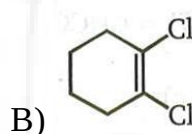
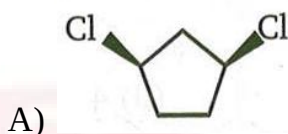
This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.

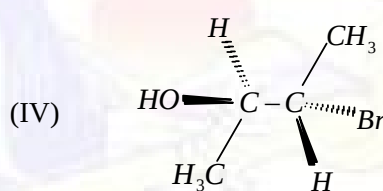
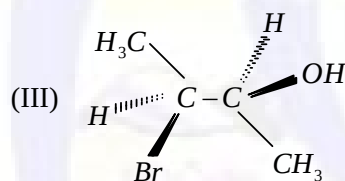
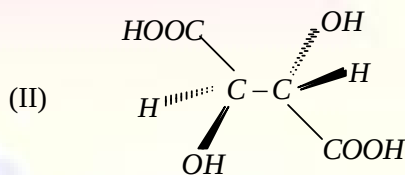
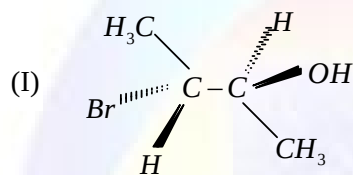
29. Which of the following are not meso compounds?



30. Which of the following compounds has a zero dipole moment?



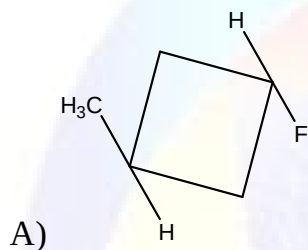
31. In the following structures



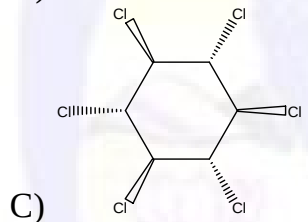
find the correct statements.

- A) Structure II is a meso compound
- B) Structures I & III are diastereomers
- C) Structures III & IV are diastereomers
- D) Structures I & IV are enantiomers

32. Plane of symmetry present in

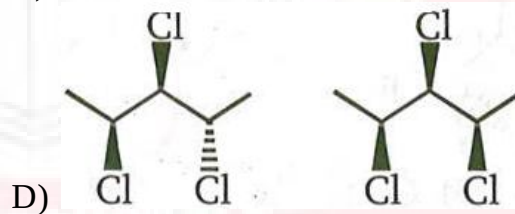
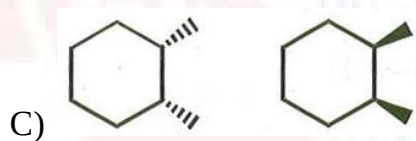
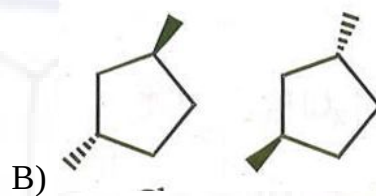
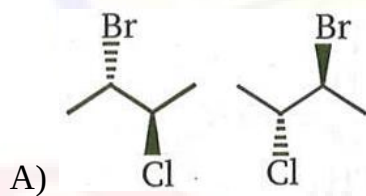


B) 2R, 3S 2,3 dichlorobutane

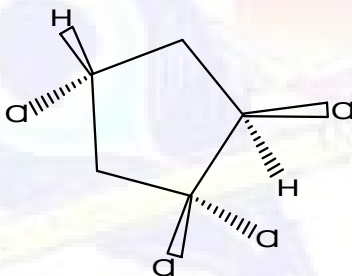


D) $(ph)_3CH$

33. Which of the following pairs of compounds are not pair of enantiomers?

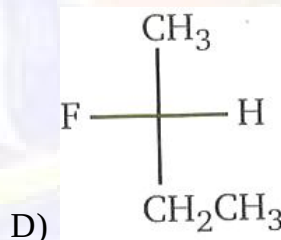
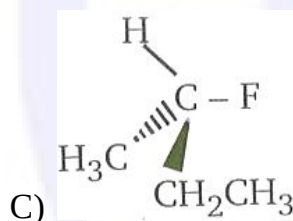
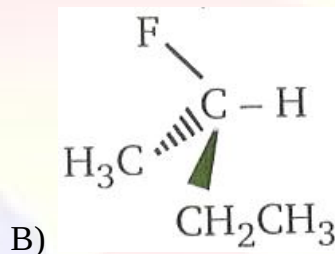
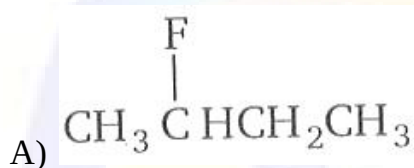


34. S-Malic acid at a concentration of $5.0g\ mL^{-1}$ in a 10cm polarimeter cell gave an observed optical rotation of 75° at $25^\circ C$. What is the specific rotation of S-Malic acid?
- A) $> 25^\circ$ B) $> 50^\circ$ C) $< 25^\circ$ D) $< 10^\circ$
35. CIP-stereochemical designations used for the following substance



- A) 2R B) 3S C) 4R D) 4S
36. Allenes of the type $abC=C=Cab$ (a,b are non stereogenic different ligands) are optically active due to
- A) chiral centre B) chiral axis
C) chiral plane D) chiral helix

37 The structure of (S) -2- fluorobutane is best represented by



38 Which of the following are methods of optical resolution ?

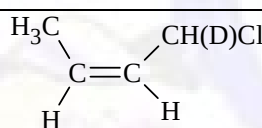
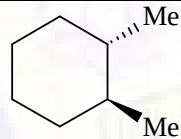
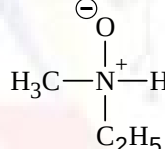
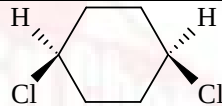
- A) crystallisation
- B) Fractional distillation
- C) Diastereomeric complex formation
- D) enantioselective chromatography

SECTION - III

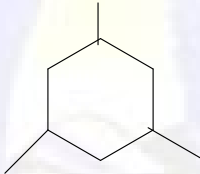
(MATRIX MATCHING ANSWER TYPE)

This section contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and four statements (P, Q, R and S) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question darken the bubbles corresponding to Q and R in the OMR sheet. For each correct matching **will be awarded +2 marks ONLY and 0 if not attempted and -1 in all other cases.**

39. Match the Following:-

	Column – I		Column – II
(A)		(P)	Optically active
(B)		(Q)	Optically inactive
(C)		(R)	Plane of symmetry
(D)		(S)	Number of chiral centre = 1
		(T)	Number of of stereocentres = 2

40. Match the Column (I) and (II)

Column – I	Column – I I(Stereo Isomers)
A) $H_3C(OH)HC-CH=CH-CH(OH)CH_3$	P) 2
B) 	Q) 4
C) $CH_3-CH=CH-CH=CH-CH_2-CH_3$	R) 6
D) $CH_3-CH=CH-CH=CH-CH-Ph$	S) 8
	T) 12

SECTION – I
(SINGLE INTEGER ANSWER TYPE)

This section contains 8 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +4 for correct answer, 0 if not attempted and 0 in all other cases.

41. Let $f(x) = 2x^3 - 3(2+p)x^2 + 12px + \ln(16-p^2)$. If $f(x)$ has exactly one local maxima and one local minima, then the number of integral values of p is
42. A population $P(t)$ of 1000 bacteria introduced into nutrient medium grows according to law $P(t) = 1000 + \frac{1000t}{100+t^2}$. The maximum size of bacterial population is equal to N then sum of the digits in N is
43. The function $f(x) = \sqrt{ax^3 + bx^2 + cx + d}$ is/has its non zero local minimum and local maxima values at $x = -2$ and $x = 2$ respectively. Given 'a' is root of the equation $x^2 - x - 6 = 0$. The value of $\left(\frac{a+b+c}{11}\right)$ is equal to
44. Let $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + x + 2$ and $g'(x) = (x^2 - 9)(x^2 - 4x + 3)(x^2 - 3x + 2)(x^2 - 2x - 3)$. If n_1, n_2 and n_3 denote number of points of local minima, number of points of local maxima and number of points of inflection of the function $f(g(x))$ then find the value of $(n_1 + n_2 + n_3)$.

45. For constant number 'a', consider the function $f(x) = ax + \cos 2x + \sin x + \cos x$ on $\mathbb{R}$ (the set of real numbers) such that $f(u) < f(v)$ for $u < v$. If the range of 'a' for any real numbers u, v is $\left[\frac{m}{n}, \infty\right)$, (m, n or co-prime) then find value of $m - 2n = \dots$
46. Let $A(1, -1), B(4, -2)$ and $C(9, 3)$ be the vertices of the triangle ABC. A parallelogram AFDE is drawn with vertices D, E and F on the line segments BC, CA and AB respectively. Find the maximum area of parallelogram AFDE.
47. If $f(x) = \left(\sqrt{4-x^2} - 3\right)^2 + \left(\sqrt{4-x^2} + 1\right)^3$, then the maximum value of $f(x)$ is N then $\frac{N}{7} =$
48. If absolute maximum value of $f(x) = \frac{1}{|x-4|+1} + \frac{1}{|x+8|+1}$ is $\frac{p}{q}$, (p, q are coprime) then $p - q$ is

SECTION – II**(ONE OR MORE CORRECT ANSWER TYPE)**

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.

49. Let $f(x) = x^2 e^{-x^2}$ then

A) $f(x)$ has local maxima at $x = -1$ and $x = 1$

B) $f(x)$ has local minima at $x = 0$

C) $f(x)$ is strictly decreasing on $x \in \mathbb{R}$

D) Range of $f(x)$ is $\left[0, \frac{1}{e}\right]$

50. Let $f(x) = \begin{cases} 2x - 4; & (x \leq 2) \\ -x^2 + \frac{k^3(k-1)^2}{k^2 - k - 2} + 4; & (x > 2) \end{cases}$, $f(x)$ attains local maximum at $x = 2$,

if k lies in

A) $(0, 1)$

B) $(3, \infty)$

C) $(-\infty, -1)$

D) $(1, 2)$

51. Consider $f(x) = \begin{cases} x^3 - 3x + 3 & ; \quad x \in [-2, 0) \\ x + 1 & ; \quad x \in [0, 5] \\ 1 - e^{-x} & ; \quad x > 5 \end{cases}$ then

- A) $f(x)$ has local maxima at two points
- B) $f(x)$ has no local minima
- C) Global minima of $f(x)$ does not exist
- D) Global maxima value of $f(x)$ is 6

52. The function $f(x) = x^{1/3}(x-1)$

- A) has 2 inflection points
- B) is strictly increasing for $x > \frac{1}{4}$ and strictly decreasing for $x < \frac{1}{4}$
- C) is concave down in $\left(\frac{-1}{2}, 0\right)$
- D) $f''(x) = 0$, at $x = \frac{1}{2}$

53. The extremum value of the function $f(x) = \frac{1}{\sin x + 4} - \frac{1}{\cos x - 4}$, $x \in R$, is
- A) $\frac{4}{8-\sqrt{2}}$ B) $\frac{2\sqrt{2}}{8-\sqrt{2}}$ C) $\frac{2\sqrt{2}}{4\sqrt{2}+1}$ D) $\frac{4\sqrt{2}}{8+\sqrt{2}}$
54. Which of the following is true?
- A) The least natural number 'a' for which $x + ax^{-2} > 2, \forall x \in (0, \infty)$ is: 2
- B) The least value of the expression $x^2 + 4y^2 + 3z^2 - 2x - 2y - 6z + 14$ is: 2
- C) If $\sin\theta + \cos\theta = 1$ then the minimum value of $(1 + \operatorname{cosec}\theta)(1 + \sec\theta)$ is: 9
- D) If $a, b \in R$ distinct numbers satisfying, $|a-1| + |b-1| = |a| + |b| = |a+1| + |b+1|$ then the minimum value of $|a-b|$ is 2.
55. The function $f(x) = 2|x| + |x+2| - \|x+2\| - 2|x\|$ has a local minimum or a local maximum at $x = \dots\dots$
- A) -2 B) $-\frac{2}{3}$ C) 2 D) $\frac{2}{3}$

56. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all 4 – corners. If the total area of removed squares is 100. The resulting box has maximum volume. The lengths of the sides of the rectangular sheet are
- A) 24 B) 32 C) 45 D) 60
57. Let $f(x) = (x-1)^4(x-2)^n; n \in N$ then $f(x)$ has
- A) a maximum at $x=1$, if n is odd B) a maximum at $x=1$, if n is even
- C) a minimum at $x=1$, if n is even D) a minima at $x=2$, if n is even
58. Let S be a square of unit area. Consider any quadrilateral which has one vertex on each side of S , If a, b, c and d denote the lengths of the sides of the quadrilateral.
- A) minimum value of $a^2 + b^2 + c^2 + d^2 = 2$
- B) maximum value of $a^2 + b^2 + c^2 + d^2 = 4$
- C) minimum value of $a^2 + b^2 + c^2 + d^2 = 1$
- D) maximum value of $a^2 + b^2 + c^2 + d^2 = 3$

SECTION - III**(MATRIX MATCHING ANSWER TYPE)**

This section contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and four statements (P, Q, R and S) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question darken the bubbles corresponding to Q and R in the OMR sheet. For each correct matching **will be awarded +2 marks ONLY and 0 if not attempted and -1 in all other cases.**

59.

	Column – I		Column – II
A)	$f(x) = (x-1)^3(x-2)^5$ has/is	P)	Local maxima
B)	$f(x) = \begin{cases} \sin\left(\frac{\pi x}{4}\right) & ; x \leq 2 \\ 9-4x & ; x > 2 \end{cases}$ is/has in $(0,3)$	Q)	Local minima
C)	$f(x) = \{2x\}$ ($\{.\}$ denotes fractional part of x) is/has in $(0,1)$	R)	Continuous
D)	$f(x) = \begin{cases} x-2 -2 & ; x < 2 \\ [x] & ; x \geq 2 \end{cases}$ (where $[.] =$ G.I.F), then $f(x)$ is/has in $(-1,4)$	S)	Non - differentiable
		T)	differentiable

60.

	Column – I		Column – II
A)	$f(x) = 2x^3 - 3(a-3)x^2 + 6ax + a + 2$, where $a \in R$ the set of all values of 'a' for which $f(x)$ has no point of extrema is	P)	$(-\infty, -3) \cup \left(3, \frac{29}{7}\right)$
B)	If the function $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$ has a positive point of maximum, then $a \in$	Q)	$(-\infty, -3) \cup (3, \infty)$
C)	The set of values of 'a' for which the function $f(x) = \frac{ax^3}{3} + (a+2)x^2 + (a-1)x + 2$ possess a negative point of inflection	R)	$a \in [1, 9]$
D)	If $f(x) = x^3 - (9-a)x^2 + 3(9-a^2)x + 7$ has points of extrema which are of opposite sign, then parameter 'a' is belongs to	S)	$(-\infty, -2) \cup (0, \infty)$
		T)	$(-\infty, -4)$



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A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

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Sec: Jr. Super60-I

2015-P1

Date: 28-10-17

Time: 07:30 AM to 10:30 AM

WAT-23

Max.Marks: 264

KEY SHEET

PHYSICS

1	1	2	1	3	2	4	2	5	4
6	0	7	5	8	8	9	CD	10	D
11	BC	12	ABC	13	BC	14	ABCD	15	AD
16	ACD	17	AB	18	AC	19	A-Q B-R C-R D-R	20	A-S B-R C-P D-Q

CHEMISTRY

21	6	22	8	23	8	24	4	25	4
26	0	27	1	28	1	29	ABC	30	D
31	ABCD	32	ABC	33	ACD	34	C	35	AC
36	B	37	C	38	ACD	39	A-PST B-QRT C-PS D-QRT	40	A-R B-P C-R D-S

MATHS

41	6	42	6	43	2	44	5	45	1
46	5	47	4	48	1	49	ABD	50	ACD
51	ACD	52	ABC	53	AC	54	ACD	55	AB
56	AC	57	ACD	58	AB	59	A-QRT B-PRS C-QS D-PQS	60	A-R B-PT C-ST D-PQT

SOLUTIONS**PHYSICS**

$$1. \quad P_x = (RC\theta + RS\theta)\rho_2 g$$

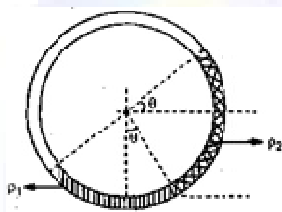
$$P_y = (RC\theta - RS\theta)\rho_1 g$$

$$\text{But } P_x = P_y$$

$$\frac{C\theta + S\theta}{C\theta - S\theta} = \frac{\rho_2}{\rho_1}$$

$$\tan \theta = \frac{\rho_2 - \rho_1}{\rho_2 + \rho_1}$$

$$\frac{\theta_1}{\theta_2} = 1$$



$$2. \quad A = \pi R^2$$

$$xA\sigma = m + L\rho A$$

$$\frac{x}{2} = \frac{(AL\rho)\frac{L}{2}}{m + A\rho L}$$

$$\text{Solving } m = (\pi R^2) L \rho \left(\sqrt{\frac{\sigma}{\rho}} - 1 \right)$$

$$3. \quad F_{\text{Net}} = A \left[\rho L - (A_2 \rho + A \rho) \frac{L}{2} \right] g$$

$$(A\rho L)a = A\rho L \left[1 - \left(\frac{3}{2} \right) \right] g$$

$$a = \frac{-g}{2} \text{ i.e. } \frac{g}{2} \text{ up}$$

$$4. \quad g^h = \frac{1}{2}(Rw)^2$$

$$\text{Solving } h = 2 \text{ cm}$$

$$5. \quad \Delta\rho_{\text{max}} = \rho(\sqrt{2})Rg$$

$$= 9600 \text{ pascal}$$

6. FBD of Block

$$F_{Net} = 0(\text{Vessel frame})$$

$$T = \left[\frac{3V}{5}(10^3) + \frac{2V}{5}(1.5 \times 10^3) - V(0.8 \times 10^3) \right] (g + a)$$

$$= 6N$$

7.
$$F_{Ap} - F_{Flat} = \frac{2}{3} \pi R^3 \rho g$$

$$F_{Flat} - F_{down} = \frac{2}{3} \pi R^3 \rho g$$

$$\text{Solving } \frac{F_{up}}{F_{down}} = 5$$

8. Conceptual

9. Conceptual

10. Conceptual

11. Conceptual

12.

13.
$$P_q = 2H\rho g + H(2\rho)g = 4H\rho g$$

Where liquids are mixed

$$(\pi R^2) 2H\rho + \pi (2R^2) H 2\rho = [\pi R^2 (2H) + (2\pi R^2) H] \rho^1$$

$$= \rho^1 = \frac{3\rho}{2}$$

$$P_{\text{f}} = (3H) \left(\frac{3\rho}{2} \right) g = \frac{9H\rho g}{2}$$

Bottom

$$\Delta\rho = \frac{H\rho g}{2}$$

14.
$$P_{\text{Bot}} = 2h\rho g$$

$$F_{\text{Bot}} = 2h\rho g A_2$$

$$F_x = \rho_x (A) = h\rho g (A_2 - A_1)$$

15.

16. Conceptual

$$17. \quad T_{\text{IRON}} = m(g+a)\left(1 - \frac{\sigma}{\rho}\right)$$

$$T_{\text{wood}} = m(g+a)\left(\frac{\sigma}{\rho} - 1\right)$$

18. Conceptual

19. Conceptual

$$20. \quad (A) \leftrightarrow mg = \max = \frac{mg}{2} \left[1 + \frac{\rho}{\sigma}\right]$$

$$\frac{mg}{2} \left(1 - \frac{\rho}{\sigma}\right)_{\max} = mg \left(1 - \frac{\rho}{\sigma}\right)$$

$$\frac{a_x}{a_y} = \frac{\sigma + \rho}{2(\sigma - \rho)}$$

$$(B) \max = mg$$

$$\max = mg \left(1 - \frac{\rho}{\sigma}\right) \frac{a_x}{a_y} = \frac{\sigma}{\sigma - \rho}$$

$$(c) ma = m \frac{3x}{2} w^2 - m \frac{\rho}{\sigma} g = mg \left(1 - \frac{\rho}{\sigma}\right)$$

$$a_x = g \left(1 - \frac{\rho}{\sigma}\right)$$

$$(D) \tan \theta = \frac{ma}{mg}$$

$$\tan \theta^1 = \frac{ma \left(1 - \frac{\sigma}{\rho}\right)}{mg \left(1 - \frac{\sigma}{\rho}\right)} = \frac{\tan \theta}{\tan \theta^1} = 1$$

CHEMISTRY

21. Conceptual

22. Conceptual

$$23. \quad ee = \frac{1}{5} \times 100 = 20\%$$

$$\therefore 20 = \frac{16}{\alpha} \times 100 \Rightarrow \alpha = 80^\circ$$

$$= 8 \times 10 \Rightarrow x = 8$$

24. Except $DCH_2 - COOH$

25. 4 stereo isomers are possible

RRR

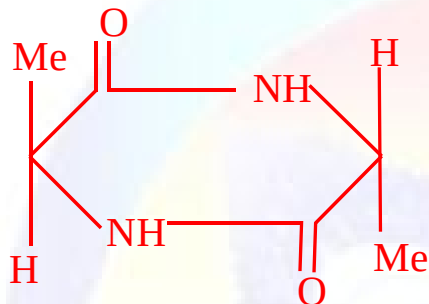
SSS

RRS

SSR

26. $X=20$, $Y=10$

27. only option 2 is correct



28. is optically active molecule is wrong as it has centre of symmetry

29. Conceptual

30. Conceptual

31. Structure I is $(2S,3R)$ -3-Bromo-2-Butanol

Structure II is $(2R,3S)$ -tartaric acid

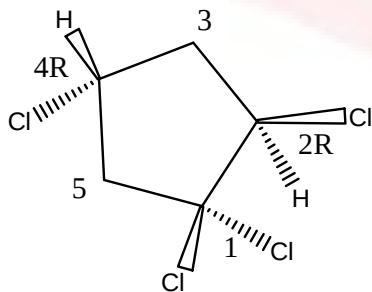
Structure III is $(2S,3S)$ -3-Bromo-2-Butanol

Structure IV is $(2R,3S)$ -3-Bromo-2-Butanol

32. D has no P.O.S (Propellar shape)

33. Option B is pair of enantiomers

34.
$$\text{Specific rotation} = \frac{\text{Observed rotation}}{\text{Concentration}(g\ mL^{-1}) \times \text{Length}(dm)}$$



35.

36. Allenes have axial chirality.

37. Conceptual

38. Optical resolution methods.

39. Conceptual.

40. Conceptual.

MATHS41. We have, $f(x) = 2x^3 - 3(2+p)x^2 + 12px + \ln(16-p^2)$

$$\text{Now, } f'(x) = 6(x^2 - (2+p)x + 2p)$$

$$\therefore f'(x) = 0 \Rightarrow x = 2, p$$

As, $f(x)$ has exactly one local maxima and one local minimaSo, $p \neq 2$

$$\text{Also, } 16 - p^2 > 0$$

$$\Rightarrow p^2 - 16 < 0$$

$$\Rightarrow (p+4)(p-4) < 0$$

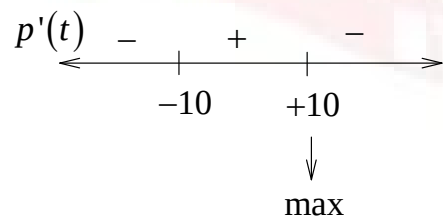
$$\Rightarrow -4 < p < 4$$

 $\therefore$ The possible integral values of p are $-3, -2, -1, 0, 1, 3$ 42. We have, $P(t) = 1000 + \frac{1000t}{100+t^2}$

$$\Rightarrow P'(t) = \frac{1000(100-t^2)}{(100+t^2)^2}$$

$$\therefore \text{Put } P'(t) = 0$$

$$\Rightarrow t = \pm 10$$



$$\text{So, } P_{\max}(t=10) = 1050 = N$$

sum of the digits of $N = 6$

43. Since minimum occurs before maximum, so $a < 0$

Also, a is root of $x^2 - x - 6 = 0$

$$\Rightarrow a = -2$$

$$\text{Let } g(x) = ax^2 + bx^2 + cx + d$$

$$\Rightarrow g(x) = -2x^3 + bx^2 + cx + d$$

$$\text{So, } g'(x) = -6x^2 + 2bx + c = -6(x+2)(x-2)$$

$$\Rightarrow b = 0, c = 24$$

$$\Rightarrow a + b + c = -2 + 0 + 24 = 22$$

(Clearly, the equation $ax^2 + bx + c = 0$ is $-2x^2 + 0 \cdot x + 24 = 0 \Rightarrow x = \pm 2\sqrt{3}$, so roots of above equation are opposite in sign)

44. $f'(x) = x^2 + x + 1 > 0 \forall x \in \mathbb{R}$

$$g'(x) = (x-3)^3(x+3)(x-1)^2(x-2)(x+1)$$

$$\begin{array}{c} + \quad - \quad + \quad + \quad - \quad + \\ \hline -3 \quad -1 \quad 1 \quad 2 \quad 3 \end{array}$$

$$n(x) = f(g(x))$$

$$n'(x) = \underbrace{f'(g(x))}_{+ve} g'(x)$$

$$\therefore n_1 = 2, n_2 = 2, n_3 = 1$$

$$\Rightarrow n_1 + n_2 + n_3 = 5$$

45. We have $f(x) = ax + \cos 2x + \sin x + \cos x \Rightarrow f'(x) = a - 2\sin 2x + \cos x - \sin x$

As $f'(x) \geq 0$ for any real number $x \Rightarrow a \geq 2\sin 2x + \sin x - \cos x$

$$\text{Let } t = \sin x - \cos x, 1 - \sin 2x = t^2 \Rightarrow \sin 2x = 1 - t^2$$

$$\therefore a \geq -2t^2 + t + 2, \text{ where } -\sqrt{2} \leq t \leq \sqrt{2}$$

$$a \geq -2t^2 + t + 2 \Big|_{\max}, \text{ but } 2 - 2t^2 + t \Big|_{\max} = \frac{17}{8}, \text{ when } t = \frac{1}{4}$$

$$a \geq \frac{17}{8} \Rightarrow a \in \left[\frac{17}{8}, \infty \right)$$

Hence, $(m - 2n) = 1$

46. Let $AF = x = DE$ and $AE = y = DF$

As $\triangle CAB$ is $\triangle CED$

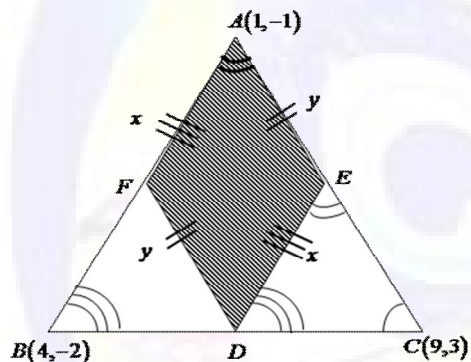
$$\text{So, } \frac{CE}{CA} = \frac{DE}{AB} \Rightarrow \frac{b-y}{b} = \frac{x}{c} \Rightarrow y = b \left(1 - \frac{x}{c} \right)$$

(Here $BC = a, AC = b$ and $AB = c$)

Now, area of parallelogram AFDE

$$= S = (AF)(EM) = xy \sin A \Rightarrow S = x b \left(1 - \frac{x}{c} \right) \sin A \quad [\text{Note: } \sin A \text{ is fixed}]$$

Now, differentiating both sides of equation (i) with respect to x , we get



$$\frac{dS}{dx} = \frac{b}{c} (c - 2x) \sin A = 0 \Rightarrow x = \frac{c}{2}$$

$$\text{Also, } \left. \frac{d^2 S}{dx^2} \right|_{x=\frac{c}{2}} = \frac{-2b}{c} < 0$$

So, S is maximum when $x = \frac{c}{2}$

$$\text{Now, } S_{\max} = \frac{1}{4} bc \sin A$$

$$= \frac{1}{2} \left(\frac{1}{2} bc \sin A \right) = \frac{1}{2} [\text{area}(\triangle ABC)] = \frac{1}{4} \begin{vmatrix} 1 & -1 & 1 \\ 4 & -2 & 1 \\ 9 & 3 & 1 \end{vmatrix} = 5 \text{ (sq)}$$

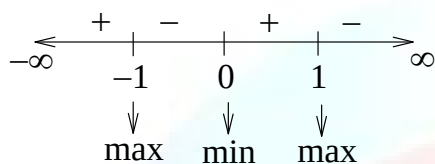
$$47. \quad f(x) = \left(\sqrt{4-x^2} - 3 \right)^2 + \left(\sqrt{4-x^2} + 1 \right)^3$$

$$= \max \left\{ 10, 28, \frac{256}{27} \right\} = 28$$

$$p - q = 1$$

49. $f(x) = x^2 e^{-x^2}$

$$f'(x) = 2xe^{-x^2}(1 - x^2)$$



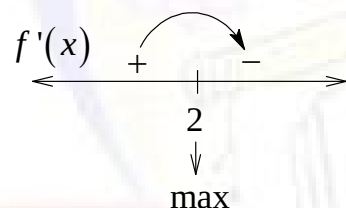
$f(x)$ has local maximum at $x = -1$ & 1 and local minimum at $x = 0$

$$f(0) = 0, f(1) = \frac{1}{e}, \quad \text{As } x \rightarrow \infty \Rightarrow f(x) \rightarrow 0$$

So, Range of $f(x)$ is $\left[0, \frac{1}{e}\right]$

50. When $f(x)$ is continuous at $x = 2$

$$f(2^-) = f(2^+) \Rightarrow 0 = \frac{k^3(k-1)^2}{k^2 - k - 2} \Rightarrow k = 0, 1$$



When $f(x)$ is discontinuous at $x = 2$

$$f'(x) = \begin{cases} 2; & (x \leq 2) \\ -2x; & (x > 2) \end{cases}$$

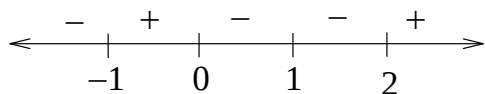
$f'(x)$ does not exist at $x = 2$, and $f'(x)$ changes from $+$ to $-$

$\Rightarrow f(x)$ attains maximum at $x = 2$

$$\Rightarrow \underbrace{\lim_{x \rightarrow 2^-} f(x) = f(2)}_0 > \lim_{x \rightarrow 2^+} f(x)$$

$$\Rightarrow 0 > \frac{k^3(k-1)^2}{(k-2)(k+1)}$$

$$\Rightarrow k^3(k-1)^2(k-2)(k+1) < 0 \text{ and } k \neq 2, -1$$



$$k \in (-\infty, -1) \cup [0, 2)$$

51. $f(x)$ is continuous $[-2, 0)$

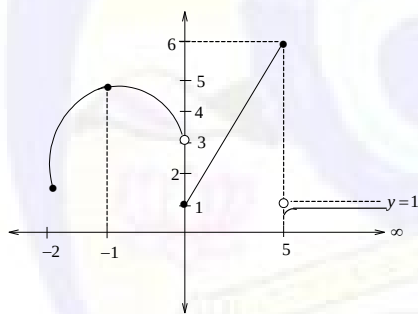
$\Rightarrow f(x)$ has local maximum, at $x = -1$, and discontinuous at $x = 5$

But $f(x)$ has local maximum at $x = 5$

$$(f(5) > f(5^+))$$

$\rightarrow$ global minima of $f(x)$ does not exist

$\rightarrow$ global maxima = 6



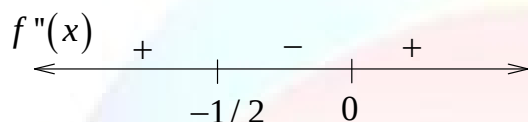
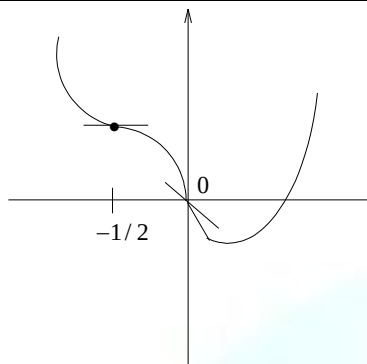
52. $y = x^{1/3}(x-1) \Rightarrow \frac{dy}{dx} = \frac{1}{3x^{2/3}}(4x-1)$, hence f is increasing for $x > \frac{1}{4}$ and f is

decreasing for $x < \frac{1}{4}$ then $f(x)$ has local minimum at $x = \frac{1}{4}$

$$f'(x) = \frac{4x-1}{3x^{2/3}} \text{ [non existence at } x = 0; \text{ Vertical Tangent]}$$

$$f''(x) = \frac{2}{9x^{5/3}} \left(\frac{2x+1}{x} \right)$$

$$f''(x) = 0 \Rightarrow \text{at } x = -\frac{1}{2} \text{ (inflection point)}$$



$$53. \quad f(x) = \frac{1}{\sin x + 4} - \frac{1}{\cos x - 4} \Rightarrow f'(x) = \frac{-\cos x}{(\sin x + 4)^2} - \frac{\sin x}{(\cos x - 4)^2}$$

$$f'(x) = 0 \Rightarrow (\sin x + \cos x) [1 - \sin x \cos x + 16 + 8(\sin x - \cos x)] = 0$$

$$\Rightarrow \sin x + \cos x = 0$$

$$\Rightarrow \tan x = -1$$

$$\Rightarrow x = \frac{3\pi}{4} \text{ or } \frac{7\pi}{4}$$

$$\text{Global minimum : } x = 2n\pi + \left(\frac{3\pi}{4}\right)$$

$$\text{Global maximum : } x = 2n\pi + \left(\frac{7\pi}{4}\right)$$

$$M = \frac{4}{8 - \sqrt{2}}, m = \frac{4}{8 + \sqrt{2}} = \frac{2\sqrt{2}}{4\sqrt{2} + 1}$$

$$54. \quad a) \quad x + \frac{a}{x^2} > 2, \forall x \in (0, \infty)$$

$$\Rightarrow \frac{x}{2} + \frac{x}{2} + \frac{a}{x^2} > 2$$

$$\therefore AM \geq GM$$

$$\Rightarrow \frac{\frac{x}{2} + \frac{x}{2} + \frac{a}{x^2}}{3} \geq \left(\frac{x}{2} \cdot \frac{x}{2} \cdot \frac{a}{x^2} \right)^{1/3}$$

$$\Rightarrow x + ax^{-2} \geq 3 \frac{a^{1/3}}{4^{1/3}} > 2$$

$$\Rightarrow \frac{3^3 a}{4} > 8 \Rightarrow a > \frac{32}{27}$$

$$\begin{aligned} \text{b) } f(x, y, z) &= x^2 - 2x + 1 + 4y^2 - 12y + 9 + 3z^2 - 6z + 3 + 1 \\ &= (x-1)^2 + (2y-3)^2 + 3(z-1)^2 + 1 \end{aligned}$$

$$\text{Hence least value of } f(x, y, z) = 1, \text{ when } x=1, y=\frac{3}{2}, z=1$$

$$\text{c) } A.M \geq G.M$$

$$\Rightarrow \frac{\sin x + \cos x}{2} \geq \sqrt{\sin \theta \cos \theta} \Rightarrow \sin \theta \cos \theta \leq \frac{1}{4} \Rightarrow xy \leq \frac{1}{4}$$

$$\text{Let } \sin \theta = x, \cos \theta = y \text{ and } (1 + \operatorname{cosec} \theta)(1 + \sec \theta) \geq p$$

$$\Rightarrow \left(1 + \frac{1}{x}\right) \left(1 + \frac{1}{y}\right) \geq p$$

$$\Rightarrow (x+1)(1+y) \geq pxy$$

$$\Rightarrow 1+x+y+xy \geq pxy$$

$$\Rightarrow 2 \geq (p-1)xy$$

$$\Rightarrow xy \leq \frac{2}{p-1} \quad (\text{But } xy \leq \frac{1}{4})$$

$$\Rightarrow \frac{2}{p-1} \leq \frac{1}{4} \Rightarrow p-1 \leq 8 \Rightarrow p \leq 9$$

$$\text{d) Let } a < b \text{ and } f(x) = |x-a| + |x-b|, \forall x \in R$$

So, $f(x)$ is decreasing in $(-\infty, a]$, constant in $[a, b]$ and increasing in $[b, \infty)$ We

$$\text{have, } f(0) = f(1) = f(-1)$$

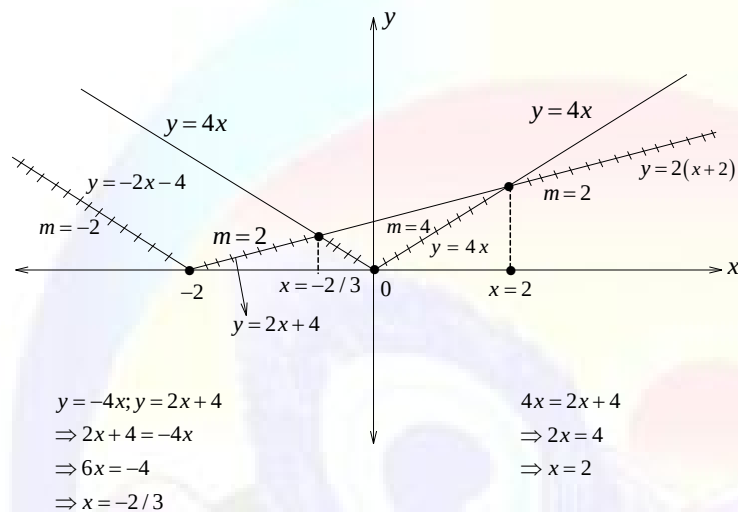
$$\Rightarrow \{-1, 0, 1\} \in [a, b] \therefore |a-b|_{\min} = 2$$

$$55. \quad \frac{f+g-|f-g|}{2} = \begin{cases} \frac{f+g-(f-g)}{2} = g(x) = \min(f, g) & (\text{if } f > g) \\ \frac{f+g+f-g}{2} = f(x) = \min(f, g) & (\text{if } f < g) \end{cases}$$

$$\left[\frac{f+g-|f-g|}{2} = \min(f(x), g(x)) \right]$$

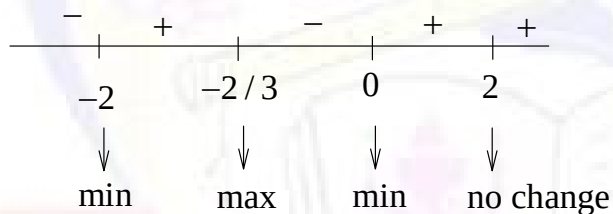
$$\frac{2|x|+|x+2|-||x+2|-2|x||}{2} = \min(|x+2|, 2|x|)$$

$$\Rightarrow 2|x|+|x+2|-||x+2|-2|x|| = \min(2|x+2|, 4|x|)$$

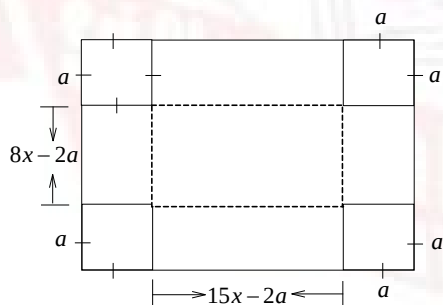


no of points of local maximum/minimum = 3

(by observation of slopes 'm')



56.



$$V = lbh = (15x - 2a)(8x - 2a)(a)$$

$$V = 120ax^2 - 46a^2x + 4a^3$$

$$\frac{dv}{da} = 120x^2 - 2.46ax + 12a^2$$

$$= 2 \left[6a^2 - 46ax + 60x^2 \right]$$

$$\frac{d^2v}{da^2} = 12a - 46x$$

$$\frac{dv}{da} = 0 \Rightarrow 6a^2 - 46ax + 60x^2 = 0$$

$$\Rightarrow 6x(x-3) - 5(x-3) = 0$$

$$\Rightarrow x = 3, \frac{5}{6}$$

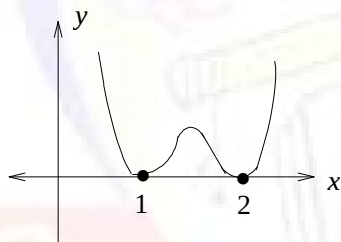
$$\text{At } x = 3, \frac{d^2v}{da^2} = 60 - 46 \times 3 < 0 \Rightarrow V \text{ is maximum, when } x = 3$$

$$\text{At } x = \frac{5}{6}, \frac{d^2v}{da^2} = 60 - 46 \times \frac{5}{6} > 0 \Rightarrow V \text{ is minimum, when } x = \frac{5}{6}$$

$\therefore$ Thus sides are $8x = 24$ and $15x = 45$

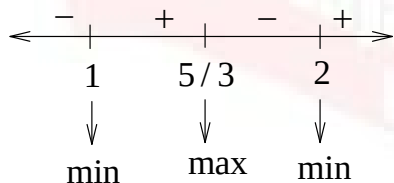
57. Graph of $f(x)$,

When n is even :

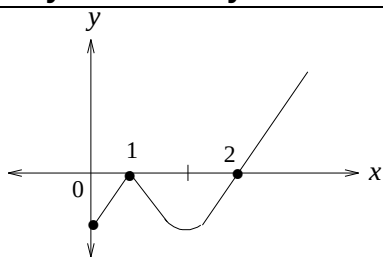


$$f(x) = (x-1)^4(x-2)^2$$

$$f'(x) = 2(x-1)^3(x-2)[3x-5]$$

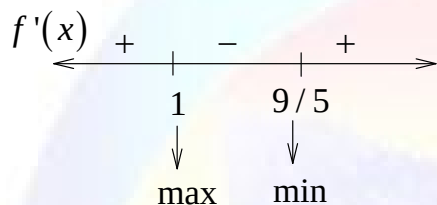


When n is odd:

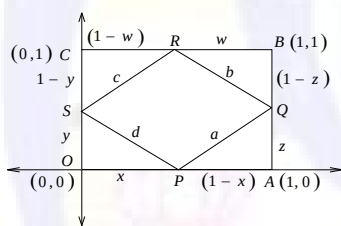


$$f(x) = (x-1)^4(x-2)$$

$$f'(x) = (x-1)^3(5x-9)$$



58.



$$a^2 + b^2 + c^2 + d^2 = \{x^2 + (1-x)^2\} + \{y^2 + (1-y)^2\} + \{x^2 + (1-x)^2\} + \{w^2 + (1-w)^2\}$$

Where $0 \leq x, y, z, w \leq 1$

Let us consider a for

$$f(x) = x^2 + (1-x)^2, 0 \leq x \leq 1$$

$$f'(x) = 2x - 2(1-x)$$

$$f'(x) = 0, \text{ for max/min, when } x = \frac{1}{2}$$

 $\therefore f(x)$ is minimum at $x = \frac{1}{2}$, and maximum at $x = 1$

$$2 \leq a^2 + b^2 + c^2 + d^2 \leq 4$$

59. a) $f(x) = (x-1)^3(x-2)^5$

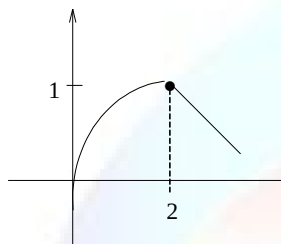
$$f'(x) = (x-1)^2(x-2)^4(8x-11)$$

at $x = \frac{11}{8}$ minimum

$f|x|$ is continuous and differentiable

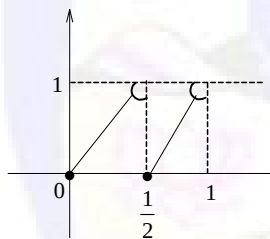
B) $f(x)$ has local maximum at $x=2$ and continuous on $(0,2)$

$f(x)$ is non different at $x=2$ $(B \rightarrow p, r, s)$



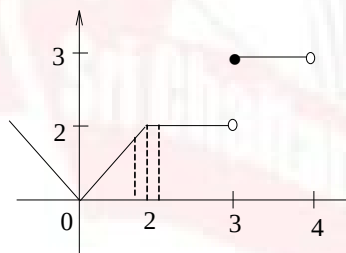
C) $f(x)$ is non differentiable at $x = \frac{1}{2}$ and $f(x)$ has local minimum at $x = \frac{1}{2}$

$(C \rightarrow q, s)$



$$f\left(\frac{1}{2}\right) < f\left(\frac{1}{2}^{-}\right) \text{ and } f\left(\frac{1}{2}\right) < f\left(\frac{1}{2}^{+}\right)$$

D) $D \rightarrow (q, s)$



$f(x)$ is non differentiable at $x=0, 2, 3$ and $f(x)$ has minimum at $x=0$

60. A) $f(x)$ has no point of extrema

$\Leftrightarrow f(x)$ is increasing on $\mathbb{R}$

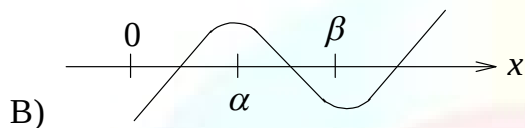
$\Leftrightarrow f'(x) \geq 0, \forall x \in \mathbb{R}$

$$\Leftrightarrow x^2 - (a-3)x + a \geq 0, \forall x \in R$$

$$\Leftrightarrow \Delta \leq 0, \Rightarrow (a-3)^2 - 4a \leq 0$$

$$\Rightarrow (a-9)(a-1) \leq 0$$

$$\Rightarrow 1 \leq a \leq 9$$



$$f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$$

$$f'(x) = 3x^2 + 6(a-7)x + 3(a^2-9)$$

$$\Rightarrow \Delta > 0, f'(0) > 0 \text{ and sum of the roots} > 0$$

$$\text{Condition (1): } 36(a-7)^2 - 4 \times 3 \times 3(a^2-9) > 0$$

$$\Rightarrow -14a + 58 > 0 \Rightarrow a < \frac{29}{7}$$

$$\text{Condition (2): } f'(0) > 0$$

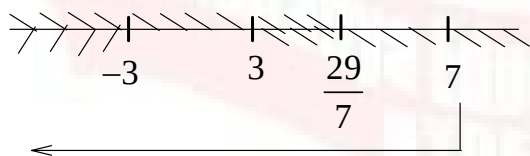
$$\Rightarrow a^2 - 9 > 0$$

$$\Rightarrow a \in (-\infty, -3) \cup (3, \infty)$$

$$\text{Condition (3): } S_1 > 0 \Rightarrow \frac{-6(a-7)}{3} > 0$$

$$\Rightarrow a - 7 < 0$$

$$\Rightarrow a < 7$$



$$a \in (-\infty, -3) \cup \left(3, \frac{29}{7}\right)$$

$$\text{C) } f(x) = \frac{ax^3}{3} + (a+2)x^2 + (a-1)x + 2$$

$$f'(x) = ax^2 + 2(a+2)x + (a-1)$$

$$f''(x) = 2ax + 2(a+2) = 0 \Rightarrow x = \frac{-(a+2)}{a}$$

($\because$ since negative point of inflection $\frac{d^2y}{dx^2} = 0$, we get x and put $x < 0$)

$$\Rightarrow \frac{-(a+2)}{a} < 0$$

$$\Rightarrow a \in (-\infty, -2) \cup (0, \infty)$$

(or)

For negative point of inflection $\Rightarrow \frac{-b}{2a} < 0$

$$D) f(x) = x^3 - (9-a)x^2 + 3(9-a^2)x + 7$$

$$f'(x) = 3x^2 - 2(9-a)x + 3(9-a^2)$$

Since x_1 and x_2 are opposite signs

$$\Rightarrow x_1 x_2 < 0$$

$$\Rightarrow \frac{3(9-a^2)}{3} < 0$$

$$\Rightarrow a^2 - 9 > 0$$

$$\Rightarrow a \in (-\infty, -3) \cup (3, \infty)$$

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

WAT-24

Date: 11-11-17

Max.Marks:240

Name of the Student: _____

H.T. NO:

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2011-PAPER -II

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

11-11-17_Jr.Super60-I_Jee-Adv_WAT-24_WEEKEND SYLLABUS

MATHS: SYLLABUS COVERED FROM_ 06-11-17 TO 10-11-2017

PHYSICS: SYLLABUS COVERD FROM _06-11-17 TO 10-11-2017

CHEMISTRY: Resonance, Aromaticity Electromeric effect , Electrophiles and nucleophiles,
Hyperconjugation

JEE-ADVANCE-2011-P2-Model**Time: 03:00 Hr's****IMPORTANT INSTRUCTIONS****Max Marks: 240****CHEMISTRY:**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 8)	Questions with Single Correct Choice	3	-1	8	24
Sec – II (Q.N : 9 – 12)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III (Q.N : 13 – 18)	Questions with Integer Answer Type	4	0	6	24
Sec – IV (Q.N : 19 – 20)	Matrix Matching Type	8	0	2	16
Total				20	80

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 21 – 28)	Questions with Single Correct Choice	3	-1	8	24
Sec – II (Q.N : 29 – 32)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III (Q.N : 33 – 38)	Questions with Integer Answer Type	4	0	6	24
Sec – IV (Q.N : 39 – 40)	Matrix Matching Type	8	0	2	16
Total				20	80

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 41 – 48)	Questions with Single Correct Choice	3	-1	8	24
Sec – II (Q.N : 49 – 52)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III (Q.N : 53 – 58)	Questions with Integer Answer Type	4	0	6	24
Sec – IV (Q.N : 59 – 60)	Matrix Matching Type	8	0	2	16
Total				20	80

Sec: Jr. Super60-I**space for rough work****Page 2**

SECTION-I

(SINGLE CORRECT CHOICE TYPE)

Section-I (Single Correct Answer Type, Total Marks: 24) contains 8 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**. For each question you will be awarded 3 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks if no bubble is darkened. In all other cases, minus one (-1) mark will be awarded.

1. Which of the following has maximum number of hyper conjugative structures?

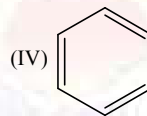
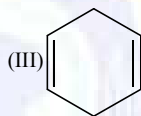
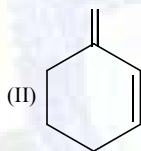
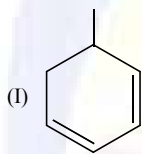
A) Isopropyl carbocation

B) Tertiary butyl carbocation

C) N-propyl carbocation

D) Benzyl carbocation

2. The order of decreasing stability is –



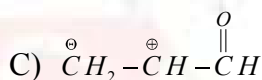
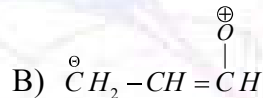
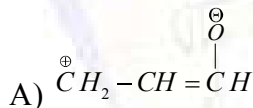
A) IV > I > II > III

B) I > IV > III > II

C) I > II > IV > III

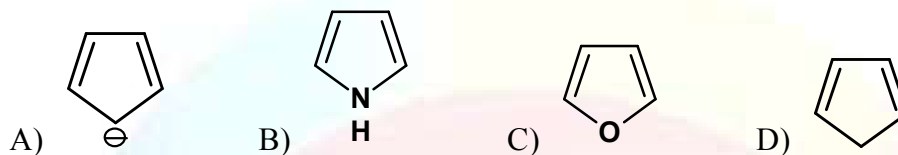
D) IV > II > I > III

3. Which of the following is the least stable resonance structure of $\text{CH}_2=\text{CH}-\text{CHO}$



D) All are equally stable

4. Which of the following possesses highest resonance energy

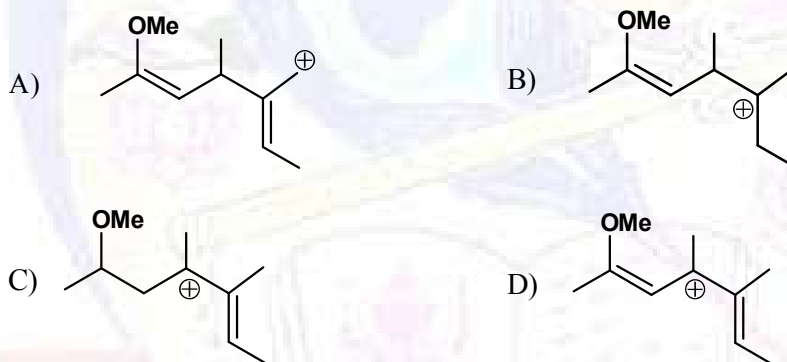


5. The correct stability order of the following resonance structures is

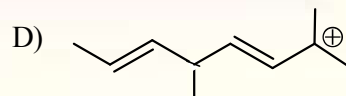
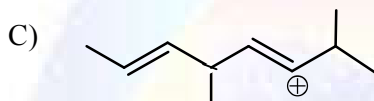
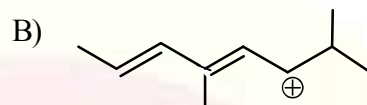
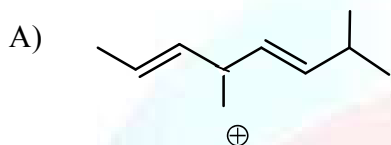


A) I>II>IV>III B) I>III>II>IV C) II>I>III>IV D) III>I>IV>II

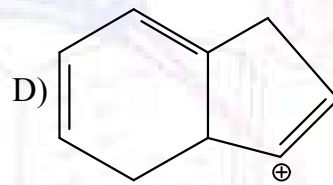
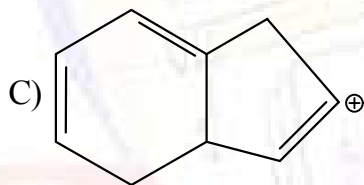
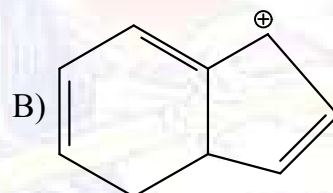
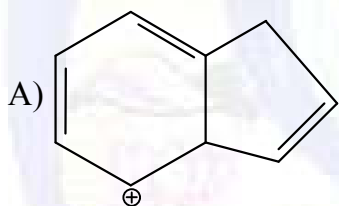
6. Which of the following is the most stabilized carbocation?



7. Which carbocation is most stable amongst the following?

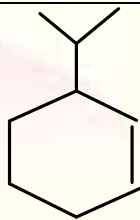


8. Which carbocation is most stabilized?

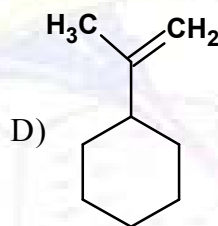
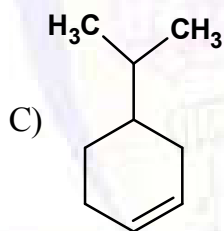
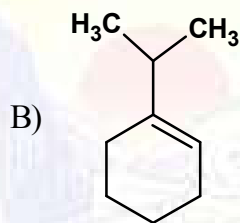
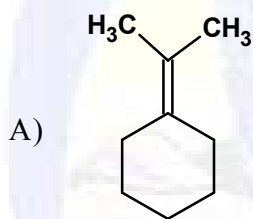


**SECTION-II
(MORE THAN ONE TYPE)**

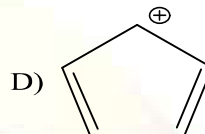
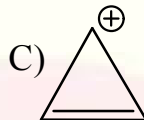
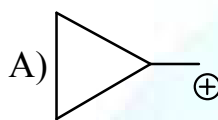
Section - II (Multiple Correct Answers Type, Total Marks: 16) contains 4 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE may be correct. For each question you will be awarded 4 marks if you darken ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. There are no negative marks in this section.



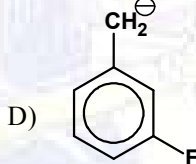
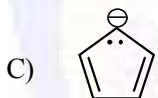
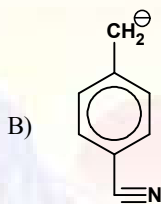
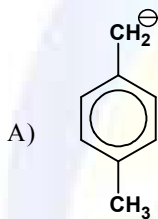
9. Which of the following alkenes are more stable than



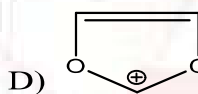
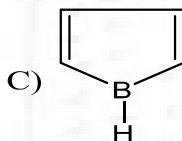
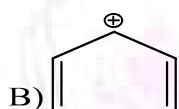
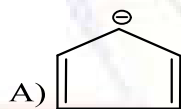
10. Among the following which is/are more stable than $C_6H_5\overset{\oplus}{C}H_2$



11. Among the following which are more stable carbanions than $PhCH_2^-$?



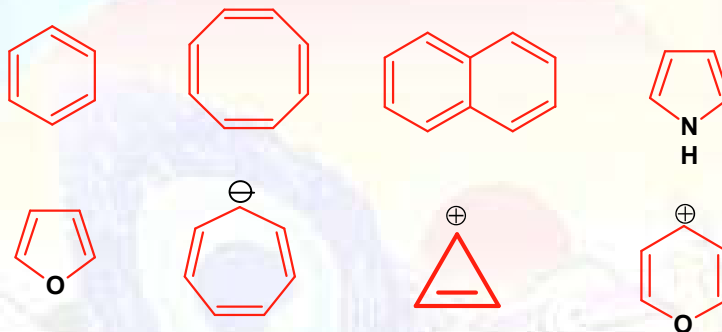
12. Among the given molecules, anti-aromatic



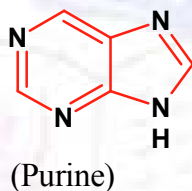
SECTION-III
(INTEGER TYPE)

Section-III (Integer Answer Type, Total Marks: 24) contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS. For each question you will be awarded 4 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks otherwise. There are no negative marks in this section.

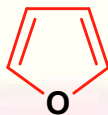
13. Find out number of aromatic compounds or ion from following.



14. The purine heterocycle occurs mainly in the structure of DNA. Identify number of 'N' atoms having localized lone pair of electrons.

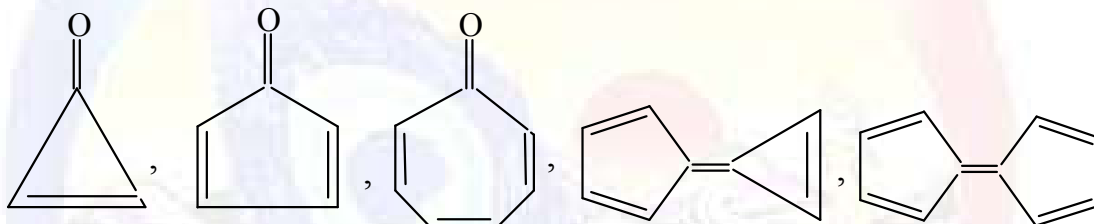


15. How many resonating structures are possible for the compound?

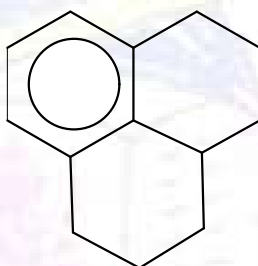


(Furan)

16. Find out number of compounds which have stable ionic structure, from following.



17. Find out number of benzylic hydrogen in.....



18. Number of maximum benzenoid structures which can be present in a particular resonating structures of phenanthracene are

SECTION-IV
(Matrix Matching Type)

Section-IV (Matrix-Match Type, Total Marks: 16) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and five statements (p, q, r, s and t) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 2 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. Thus, each question in this section carries a maximum of 8 marks. There are no negative marks in this section.

19. Match the following Column-I (groups) and column-II (effect shown by the groups).

Column-I(groups)	column-II (effect shown by the groups)
A) $-NO_2$	P) - m effect
B) $-O^-$	Q) + m effect
C) $-O-CH_3$	R) + I effect
D) $-C \equiv N$	S) - I effect

20. Match the following Column-I and Column-II.

Column-I**SPECIES**

(A) Carbanion(CH_3^-)

(B) SO_3

(C) $\text{C}_6\text{H}_5-\text{CH}_2^+$

(D) Ethene

Column-II**PROPERTIES**

(P) Electrophile

(Q) Nucleophile

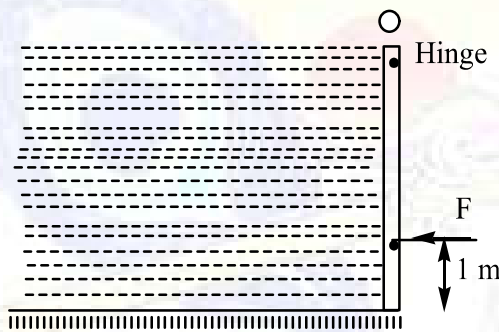
(R) sp^2 hybridised central atom

(S) sp^3 hybridised central atom

SECTION-1
(SINGLE CORRECT CHOICE TYPE)

Section-I (Single Correct Answer Type, Total Marks: 24) contains 8 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**. For each question you will be awarded 3 marks if you darken **ONLY** the bubble corresponding to the correct answer and zero marks if no bubble is darkened. In all other cases, minus one (-1) mark will be awarded.

21. A square gate of size $4m \times 4m$ is hinged at topmost point. A liquid of density ρ fills the space left of it. The horizontal force (F) which acting 1m from lowest point can hold the gate stationary is: (There is no friction between gate and floor)



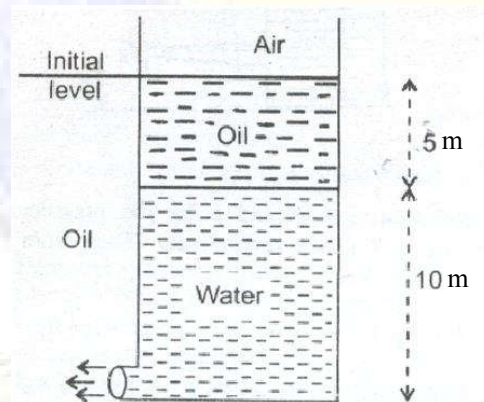
A) $\frac{256}{3} \rho g$

B) $\frac{256}{9} \rho g$

C) $\frac{128}{9} \rho g$

D) $\frac{128}{3} \rho g$

22. A tank with a small circular hole at the bottom contains oil on top of water. It is immersed in a large tank of same oil. Initially oil levels are same in both the tanks and water flows through the hole. If the ratio of cross-sectional areas of tank to that of the hole is 10^3 , determine the time in which the flow stops. Density of oil = 800 kg/m^3 and that of water = 1000 kg/m^3 . Assume that level of oil outside the tank above the orifice does not change)



A) $\frac{32}{\sqrt{2}} \text{ s}$

B) $32\sqrt{2} \text{ s}$

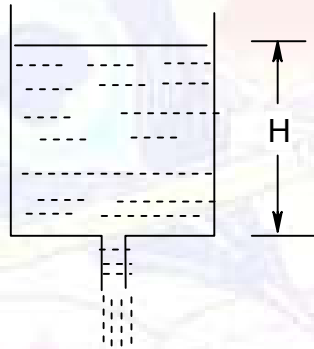
C) $\sqrt{\frac{2}{5}} \text{ s}$

D) 32 s

23. A cylindrical tank of base area A has a small hole of area ' a ' at the bottom. At time $t = 0$, a tap starts to supply water into the tank at a constant $\alpha m^3/s$. Then find the maximum level of water in the tank.

A) $\frac{\alpha^2}{ga^2}$ B) $\frac{2\alpha^2}{ga^2}$ C) $\frac{3}{4} \frac{\alpha^2}{ga^2}$ D) $\frac{\alpha^2}{2ga^2}$

24. A tank of cross sectional area A_0 is filled with a liquid. A small orifice of area A ($A \ll A_0$) is present at the bottom of tank. At a moment the height of liquid in the tank is ' H '. What is acceleration of top layer of liquid at this moment?

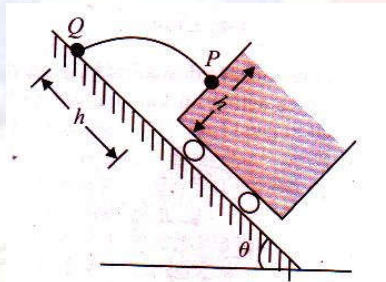


A) g B) $\frac{A}{A_0} g$ C) $\left(\frac{A}{A_0}\right)^2 g$ D) zero

25. A hole is made at the bottom of a tank filled with water (density $= 10^3 \text{ kg/m}^3$). If the total pressure at the bottom of the tank is 3 atm ($1 \text{ atm} = 10^5 \text{ N/m}^2$), then the velocity of efflux is

A) $\sqrt{400} \text{ m/s}$ B) $\sqrt{200} \text{ m/s}$ C) $\sqrt{600} \text{ m/s}$ D) $\sqrt{500} \text{ m/s}$

26. A cubical tank carrying a non-viscous liquid (water, say) moves down a smooth incline plane freely. When we open the hole made at the mid-point P of the face of the tank, it strikes the point Q of the inclined plane. Speed of the tank at the time of opening the valve is



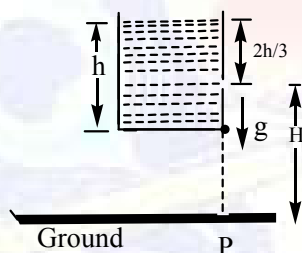
- A) $\sqrt{gh \cos \theta} \left[2 + \frac{\tan \theta}{2} \right]$ B) $\sqrt{gh \cos \theta} \left(\frac{\tan \theta}{2} \right)$
 C) $\sqrt{gh \cos \theta} (1 + \tan \theta)$ D) None of the above

27. Water (density ρ) flows through a uniform pipe bent an angle α with a velocity v .

What is the force exerted on the bend of the pipe if cross-sectional area of the pipe is S ?

- A) $2\rho v^2 S \sin \frac{\alpha}{2}$ B) $2\rho v^2 S \cos \alpha$ C) $2\rho v^2 S \sin \alpha$ D) $2\rho v^2 S \sin \alpha \cos \alpha$

28. An open vessel full of water is falling freely under gravity. There is a small hole in one face of the vessel, as shown in the figure. The water which comes out from the hole at the instant when hole is at height H above the ground, strikes the ground at a distance of x from P . Which of the following is correct for the situation described?

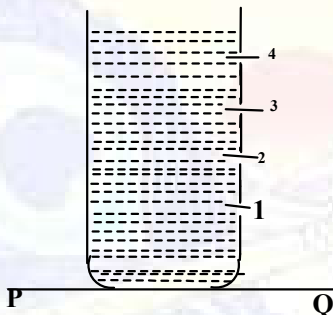


- A) The value of x is $2\sqrt{\frac{2hH}{3}}$
 B) The value of x is $\sqrt{\frac{4hH}{3}}$
 C) The value of x can't be computed from information provided.
 D) The question is irrelevant as no water comes out from the hole

SECTION-II
(MORE THAN ONE TYPE)

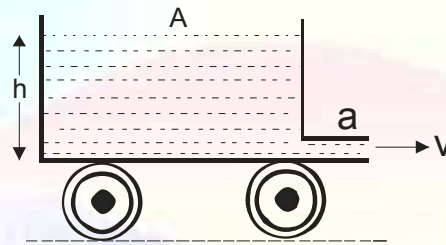
Section - II (Multiple Correct Answers Type, Total Marks: 16) contains 4 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE may be correct. For each question you will be awarded 4 marks if you darken ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. There are no negative marks in this section.

29. A cylindrical vessel of 90 cm height is kept filled upto the brim. It has four holes 1,2,3,4 which are respectively at heights 20 cm, 30cm, 40 cm and 50 cm from the horizontal floor PQ. The water falling at the maximum horizontal distance from the vessel comes from

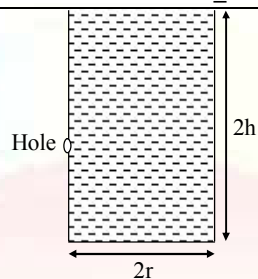


- A) hole number 4 B) hole number 3
C) hole number 2 D) hole number 1
30. A container having a hole at the bottom is free to move on a frictionless horizontal surface. As the liquid comes out, container moves in backward direction with an

acceleration α and finally acquired velocity v (when all liquid drains out) Neglect the mass of container. The correct options are



- A) α is independent of height of liquid in vessel.
 - B) α depends only A and a and g.
 - C) v depends upon α and h
 - D) v independent of α
31. A light cylindrical vessel of radius r is kept on a rough horizontal surface so that it cannot slide but can topple. It is filled with water upto a height ' $2h$ ' and a small hole of area ' a ' is punched in it so that the water coming out of it falls at the maximum distance from its wall along horizontal surface. Water comes out horizontally from the hole. The value of h for which the cylinder topples is/are



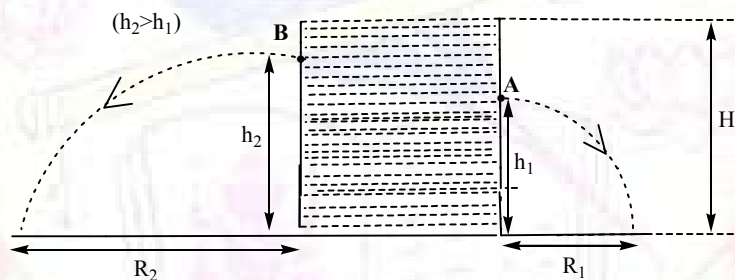
A) $\frac{\pi r^3}{2a}$

B) $\frac{2\pi r^3}{a}$

C) $\frac{3\pi r^3}{2a}$

D) $\frac{4\pi r^3}{3a}$

32. A tank is placed on a horizontal surface and is full of water up to height 'H'. Two very small holes are created simultaneously on the vertical side of tank at height h_1 & h_2 from bottom of tank as shown in figure and R_1 & R_2 are the ranges corresponding heights h_1 & h_2 respectively. Choose the correct statement(s).

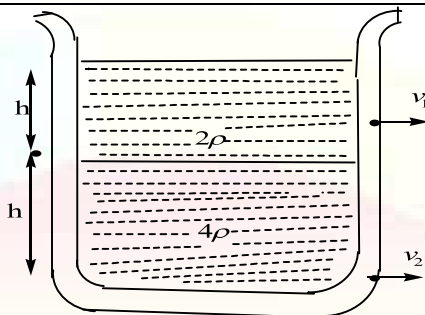


- A) If $R_1 = R_2$ then $H = h_1 + h_2$
- B) If t_1 & t_2 are the time taken by water ejecting from holes 'A' and 'B' to reach on horizontal surface, then $t_1 = t_2$ if $R_1 = R_2$
- C) The water ejected from both holes strikes on horizontal surface with velocity $\sqrt{2gH}$
- D) If the ratio of the velocity head of water just after ejecting from holes A to B is 3:2, then $H = (3h_2 - 2h_1)$

SECTION-III
(INTEGER TYPE)

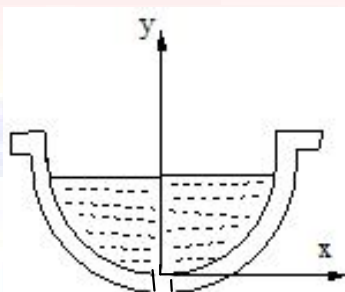
Section-III (Integer Answer Type, Total Marks: 24) contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS. For each question you will be awarded 4 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks otherwise. There are no negative marks in this section.

33. Equal volumes of two immiscible liquids of densities 2ρ and 4ρ are filled in a vessel as shown in figure. Two small holes are punched at depths $h/2$ and $2h$ from the surface of lighter liquid. If v_1 and v_2 are the velocities of efflux at these two holes, then $\frac{v_2}{v_1}$ is k. Find k^2 _____



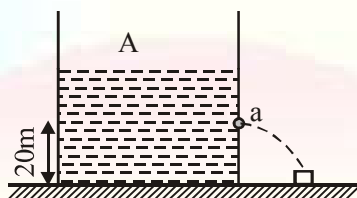
34. Water level is maintained in a cylindrical vessel placed on horizontal floor up to a fixed height 'H'. A tiny hole of area 'A' is punched in the side wall at a height 'y' from the bottom of the vessel. The emerging stream strikes the ground at a horizontal distance $y(\neq 0)$ from cylinder. To maintain the level in the vessel at 'H', the rate (volume /sec) of addition of water is $A \sqrt{\frac{2gH}{n}}$. Find n
35. Water is used as the manometric liquid in a pitot tube mounted in an aircraft to measure air speed. If the maximum difference in height between the liquid columns is 0.1 m, the maximum air speed that can be measured is $5z$ m/s (nearly). Find z. Density of air $d_{air} = 1.3 \text{ kg/m}^3$; $(g = 10 \text{ m/s}^2)$

36. A small hole is made at the bottom of a symmetrical jar as shown in fig. A liquid is filled into the jar up to a certain height. The rate of fall of liquid surface is independent of the level of liquid in the jar. Then the surface of jar is a surface of revolution of the curve $y = kx^n$, the value of n is



37. A tank has a small hole of area of cross-section A at the bottom. Water from a pipe of inner cross-section area $2A$ is entering the tank with a velocity of 5 m/s . Calculate the height (in m) upto which the water will be filled inside the tank at steady state?
38. Water is filled in a uniform container of area of cross section A . A hole of cross section area a ($\ll A$) is made in the container at a height of 20 m above the base. Water streams out and hits a small block placed at some distance from container. With

what speed (in ms^{-1}) should the block be moved such that water stream always hits the block? (Given $\frac{a}{A} = \frac{1}{20}$). (Take $g = 10 \text{ ms}^{-2}$)

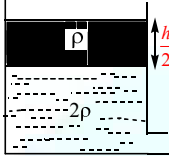
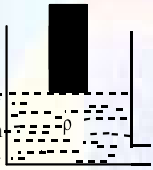


SECTION-IV
(Matrix Matching Type)

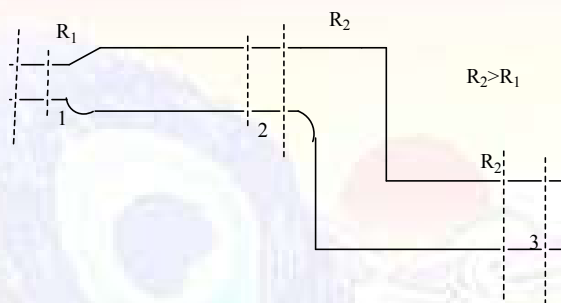
Section-IV (Matrix-Match Type, Total Marks: 16) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and five statements (p, q, r, s and t) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 2 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. Thus, each question in this section carries a maximum of 8 marks. There are no negative marks in this section.

39. For the cases in Column I match the velocity of efflux (cross section of opening is very small) in Column II

Column I		Column II	
(A)	Two immiscible liquids of densities ρ & 2ρ	(P)	$\sqrt{2gh}$

(B)	 A perfectly fitting piston made of material of density ρ which can slide without friction	(Q)	$\sqrt{2.5gh}$
(C)	 A solid cylinder of half the cross section of tank is just touching the water surface. Now it is pushed by a distance $\frac{h}{2}$ downward (An external agent holds cylinder, vessel is large).	(R)	$\sqrt{3gh}$
(D)	Arrangement is same as in option C but cylinder is pushed down by a distance h	(S)	$\sqrt{3.5gh}$
		(T)	$\sqrt{4gh}$

40. An arrangement of the pipes is shown in the Fig. The flow of water (incompressible and non-viscous) through the pipes is steady in nature. Three sections of the pipe are marked in which section (1) and section (2) are at same horizontal level, while being at a greater height than section (3). Correctly match order of the different physical parameters.



Column-I		Column- II	
(A)	Order of volume flow rate	(P)	$1 > 2 > 3$
(B)	Order of kinetic energy of a mass element	(Q)	$1 = 2 = 3$
(C)	Order of pressure in the sections	(R)	$1 > 2 = 3$
(D)	Order of flow speed in sections	(S)	$3 > 2 > 1$

SECTION-1

(SINGLE CORRECT CHOICE TYPE)

Section-I (Single Correct Answer Type, Total Marks: 24) contains 8 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**. For each question you will be awarded 3 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks if no bubble is darkened. In all other cases, minus one (-1) mark will be awarded.

41. If $\vec{a}$ and $\vec{b}$ are two unit vectors and θ is the angle between them, then the unit vector along the angular bisector of $\vec{a}$ and $\vec{b}$ will be given by

- A) $\frac{\vec{a}-\vec{b}}{2\cos\frac{\theta}{2}}$ B) $\frac{\vec{a}+\vec{b}}{2\cos\frac{\theta}{2}}$ C) $\frac{\vec{a}-\vec{b}}{\cos\frac{\theta}{2}}$ D) $\frac{\vec{a}+\vec{b}}{\cos\frac{\theta}{2}}$

42. If vectors $\vec{a}$ and $\vec{b}$ are two adjacent sides of a parallelogram, then the vectors representing the altitude of the parallelogram which is perpendicular to $\vec{a}$ is

- A) $\vec{b} + \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$ B) $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}$ C) $\vec{b} - \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$ D) $\vec{a} - \frac{\vec{b} \cdot \vec{a}}{|\vec{b}|^2} \vec{b}$

43. In a quadrilateral ABCD, $\vec{AC}$ is the bisector of $\vec{AB}$ and $\vec{AD}$, angle between $\vec{AB}$ and $\vec{AD}$ is $\frac{2\pi}{3}$, $15|\vec{AC}| = 3|\vec{AB}| = 5|\vec{AD}|$. Then the angle between $\vec{BA}$ and $\vec{CD}$ is

- A) $\cos^{-1} \frac{\sqrt{14}}{7\sqrt{2}}$ B) $\cos^{-1} \frac{\sqrt{21}}{7\sqrt{3}}$ C) $\cos^{-1} \frac{2}{\sqrt{7}}$ D) $\cos^{-1} \frac{2\sqrt{7}}{14}$

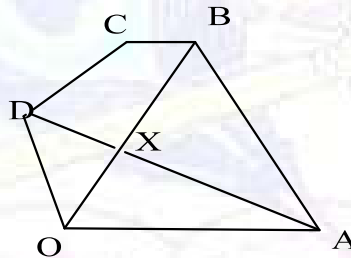
44. ABCDEF is a regular hexagon where O is its centre. The value of $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF}$ is
- A) $\overrightarrow{AO}$ B) $2\overrightarrow{AO}$ C) $4\overrightarrow{AO}$ D) $6\overrightarrow{AO}$
45. If $(x, y, z) \neq (0, 0, 0)$ and $(\vec{i} + \vec{j} + 3\vec{k})x + (3\vec{i} - 3\vec{j} + \vec{k})y + (-4\vec{i} + 5\vec{j})z = a(x\vec{i} + y\vec{j} + z\vec{k})$, then the value of a is
- A) 1 B) -1 C) 2 D) 3
46. The sides of a parallelogram are $2\vec{i} + 4\vec{j} - 5\vec{k}$ and $\vec{i} + 2\vec{j} + 3\vec{k}$. Then the unit vector parallel to one of the diagonals is
- A) $\frac{1}{\sqrt{41}}(\vec{i} + 6\vec{j} - 2\vec{k})$ B) $\frac{1}{7}(3\vec{i} - 6\vec{j} - 2\vec{k})$ C) $\frac{1}{\sqrt{69}}(\vec{i} + 2\vec{j} + 8\vec{k})$ D) $\frac{1}{\sqrt{69}}(-\vec{i} - 2\vec{j} + 8\vec{k})$
47. A vector $\vec{a}$ has the components $2p$ and 1 w.r.t. a rectangular Cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If with respect to the new system, $\vec{a}$ has components $(p+1)$ and 1 , then p is equal to
- A) -1 B) -1/3 C) 1/2 D) 2
48. If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{b} = 4\vec{i} + 3\vec{j} + 4\vec{k}$ and $\vec{c} = \vec{i} + x_1\vec{j} + x_2\vec{k}$ are linearly dependent vectors and $|\vec{c}| = \sqrt{3}$, then which of the following is/are may be true?
- A) $x_2 = -1$ B) $x_2 = 1$ C) $x_1 = -2$ D) $x_1 = \pm 2$

SECTION-III
(INTEGER TYPE)

Section-III (Integer Answer Type, Total Marks: 24) contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS. For each question you will be awarded 4 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks otherwise. There are no negative marks in this section.

53. ABCD is a parallelogram. L is a point on BC which divides BC in the ratio 1:2. AL intersects BD at P. M is a point on DC which divides DC in the ratio 1:2 and AM intersects BD in Q. Then Point P divides AL in the ratio $K:1$, then value of K is?

54. Let OABCD be a pentagon in which the sides OA and CB are parallel and the sides OD and AB are parallel. AD and OB intersect at X according to the diagram Also $OA:CB=2:1$ and $OD:AB=1:3$, Then The ratio $\frac{AX}{XD}$ is?



55. Consider the regular hexagon ABCDEF with centre at O(origin) , Then If

$\overrightarrow{AD} + \overrightarrow{EB} + \overrightarrow{FC}$ is equal to $k(\overrightarrow{AB})$, then k is?

56. Let two non – collinear vectors $\vec{a}$ and $\vec{b}$ inclined at an angle $\frac{2\pi}{3}$ be such that $|\vec{a}|=3$ and $|\vec{b}|=4$. A point P moves so that at any time t the position vector $\overrightarrow{OP}$ (where O is the origin) is given as $\overrightarrow{OP} = (e^t + e^{-t})\vec{a} + (e^t - e^{-t})\vec{b}$. If the least distance of P from origin is $\sqrt{2}\sqrt{\sqrt{a}-b}$ where $a, b \in N$, then find the value of “b” is.

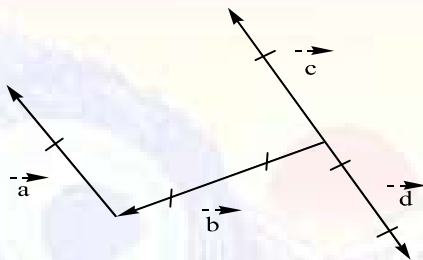
57. Let the position vectors of the points A, B, C be $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = -\hat{i} - \hat{j} + 8\hat{k}$ and $\vec{c} = -4\hat{i} + 4\hat{j} + 6\hat{k}$ respectively, then If $|\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = t|\vec{a} - \vec{b}|^2$, value of t is ?

58. Given two vectors $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$, $\vec{b} = -2\hat{i} + 2\hat{j} - \hat{k}$, and $\lambda = \frac{\text{the projection of } \vec{a} \text{ on } \vec{b}}{\text{the projection of } \vec{b} \text{ on } \vec{a}}$ then 3λ is equal to

SECTION-IV
(Matrix Matching Type)

Section-IV (Matrix-Match Type, Total Marks: 16) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and five statements (p, q, r, s and t) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 2 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. Thus, each question in this section carries a maximum of 8 marks. There are no negative marks in this section.

59. Refer to the following diagram and match the following from Column-I to Column-II.



Column I	Column II
A) Collinear Vectors	P) $\vec{a}$
B) Co-initial Vectors	Q) $\vec{b}$
C) Equal Vectors	R) $\vec{c}$
D) Unlike Vectors with same initial point	S) $\vec{d}$
	T) $2\vec{d}$

60. Suppose $\vec{a}, \vec{b}, \vec{c}$ be any three 3 dimensional unit vectors and let

$$k = |\vec{a} - \vec{b}|^2 + |\vec{c} - \vec{b}|^2 + |\vec{a} - \vec{c}|^2.$$

Column I	Column II
A) If $k = 2$, then	P) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 3$
B) If $k = 0$, then	Q) $ \vec{a} + \vec{b} + \vec{c} = 1$
C) If $k = 8$, then	R) $\vec{a} + \vec{b} + \vec{c} = \vec{0}$
D) If k has to be maximum, then	S) $ \vec{a} + \vec{b} + \vec{c} = \sqrt{7}$
	T) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -1$



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Max.Marks:240

KEY SHEET

CHEMISTRY

1	B	2	A	3	B	4	A	5	B
6	D	7	B	8	B	9	ABCD	10	ABC
11	BCD	12	BC	13	6	14	3	15	5
16	3	17	5	18	3	19	A-PS B-QR C-QS D-PS	20	A-QS B-PR C-PR D-QR

PHYSICS

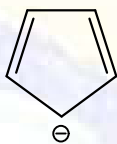
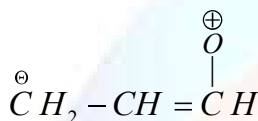
21	B	22	C	23	D	24	C	25	A
26	B	27	A	28	D	29	AB	30	ABC
31	BCD	32	ACD	33	3	34	5	35	8
36	4	37	5	38	1	39	A-Q B-Q C-R D-T	40	A-Q B-R C-S D-R

MATHS

41	B	42	C	43	C	44	D	45	B
46	D	47	B	48	B	49	ABC	50	ABD
51	BC	52	BD	53	3	54	6	55	4
56	7	57	2	58	7	59	A-PRST B-QRST C-PR D-RST	60	A-S B-P C-QT D-R

SOLUTIONS CHEMISTRY

1. 1. Among the given the hyper conjugative structures are more for 3° -butyl carbonium ion due to $3-\alpha-C$ & $9-\alpha-H$ atoms.
2. Due to localized lone pair/ in pyridine.
3. Among the given resonance structures is the least stable one EN atom, O is carrying positive charge.



4. Among the structures given  has the highest resonance energy, due to delocalization 6 electrons. Further it has aromaticity as it satisfies $4n+2$ rule.
5. Octet rule, charge separation, electronegative atom with positive charge.
6. Conceptual.
7. Conceptual.
8. Conceptual.
9. E.N $\alpha - I$.
10. Concept of mesomeric effect.
11. Conceptual.
12. Conceptual.
13. Conceptual.
14. Conceptual.
15. Conceptual.
16. Conceptual.
17. Conceptual.
18. Conceptual.
19. Conceptual.
20. Conceptual

PHYSICS

21. Conceptual.

22. Conceptual.

23. Conceptual.

24. Velocity of efflux. $v = \sqrt{2gh}$

$$\text{Velocity of top layer} = \frac{AV}{Av} = \frac{A}{A_0} \sqrt{2gh}$$

$$\therefore \text{Acceleration top layer} = \frac{d}{dx} \left[\frac{A}{Av} \sqrt{2gh} \right]$$

$$\begin{aligned} & \frac{A}{Av} \sqrt{2g} \times \frac{1}{2\sqrt{h}} \times \frac{dh}{dt} \\ &= \frac{A}{A_0} \sqrt{2g} \times \frac{1}{2\sqrt{h}} \times \frac{A}{A_0} \sqrt{2gh} \\ & \left(\frac{A}{A_0} \right)^2 g \end{aligned}$$

Correct option is (c)

25. Total pressure at the bottom = 3 atm

Pressure due to water in the tank = 3 atm – 1 atm = 2atm = 20 m of water column

Height of water in the tank is $h = 20 \text{ m}$ So, velocity of efflux = $\sqrt{2gh} = \sqrt{2 \times 10 \times 20} = \sqrt{400} \text{ m/s}$

$$26. \quad v_{rel} = \sqrt{2g_{eff} \frac{h}{2}} = \sqrt{g_{eff} h} = \sqrt{gh \cos \theta}$$

The actual velocity of the liquid, $v = (v_r - v_o)$

$$\text{Then, the time of flight, } T = \sqrt{\frac{2(h/2)}{g \cos \theta}} = \sqrt{\frac{h}{g \cos \theta}}$$

 $h = vt - \frac{1}{2}(g \cos \theta)T^2$ putting T and v , we have

$$v_0 = h \left(1 + \frac{\tan \theta}{2} \right) \sqrt{\frac{g \cos \theta}{h}} + \sqrt{gh \cos \theta}$$

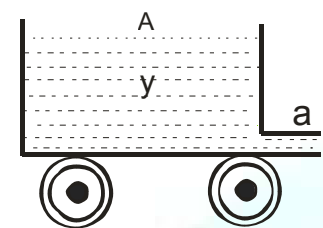
27. Conceptual.

28. Conceptual.

29. Conceptual.

30. $v = \sqrt{2gy}$

$$F = \rho a 2gy$$



$$(\rho A y) \alpha = \rho a 2gy \Rightarrow \alpha = \frac{a}{A} 2g$$

31. Since waterfalls at maximum distance from wall so the hole should be made at h .

For toppling, $\rho a v^2 \times h \geq \pi r^2 2h \rho g \times r$

$$\Rightarrow \rho a 2gh \times h \geq \pi r^2 2h \rho g r$$

$$\Rightarrow h \geq \frac{\pi r^3}{a}$$

32. P and Q are the Claussius – Clayperon equation and Vant Hoff isochore.

33. Hint: $v_1 = \sqrt{2g\left(\frac{h}{2}\right)} = \sqrt{gh} \rightarrow (1)$

From Bernouli's theorem

$$2\rho gh + 4\rho gh = \frac{1}{2}(4\rho)v_2^2$$

$$\therefore v_2 = \sqrt{3gh} \rightarrow (2)$$

$$\therefore \frac{v_2}{v_1} = \sqrt{3}$$

34. $y = \left(\frac{w^2}{2g}\right)x^2 = \frac{(2\pi \times 2)^2}{2 \times 10} x^2$

$$y = 8x^2$$

$$\therefore y = 8(0.05)^2 = 200 \times 10^{-4} \\ = 2 \times 10^{-2} m$$

$$\therefore y = 2 cm$$

35. For gases, Velocity is given by

$$v = \sqrt{\frac{2h\rho g}{d}} = \sqrt{\frac{2 \times 0.1 \times 1000 \times 10}{1.3}} = 39.22 m/s \approx 40 m/s$$

36. At steady state the rate at which the liquid is entering the tank should be equal to the rate at which the liquid is flowing out of the tank therefore

$$A\sqrt{2gh} = 2Av$$

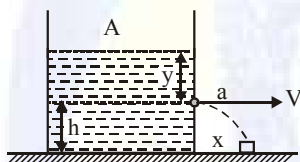
$$h = \frac{2v^2}{g} = 5m$$

37. Conceptual

38. Velocity of efflux $v = \sqrt{2gy}$

$$\text{Range } x = \sqrt{2gy} \times \sqrt{\frac{2h}{g}}$$

The velocity of the block must be $\left(\frac{dx}{dt}\right)$.



$$\therefore V_b = \frac{dx}{dt} = \sqrt{\frac{2h}{g}} \times \sqrt{2g} \times \frac{1}{2\sqrt{y}} \frac{dy}{dt}$$

$$V_b = \frac{\sqrt{h}}{\sqrt{y}} \frac{dy}{dt} \dots (i)$$

Using equation of continuity

$$\frac{A dy}{dt} = a\sqrt{2gy} \dots (ii)$$

equation (i) and (ii)

$$V_b = \sqrt{\frac{h}{y}} \times \frac{a}{A} \sqrt{2gy}$$

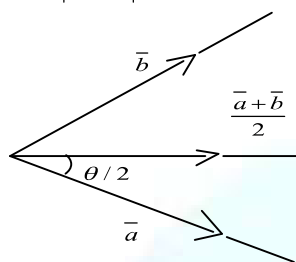
$$V_b = \sqrt{2gh} \times \frac{a}{A} = 20 \times \frac{1}{20} = 1 \text{ ms}^{-1}.$$

39. Conceptual.

40. Conceptual.

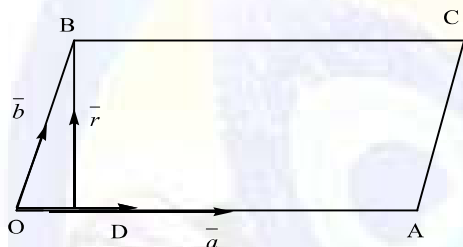
MATHS

41. But $\left| \frac{\vec{a} + \vec{b}}{2} \right| = \cos \frac{\theta}{2}$



$\therefore$ Required unit vector is $\frac{\vec{a} + \vec{b}}{2 \cos \frac{\theta}{2}}$

42.



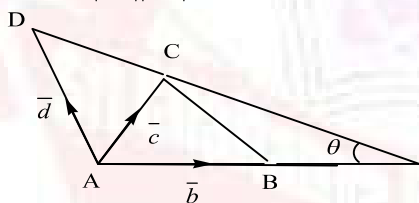
Let $\overrightarrow{OD} = x\vec{a}$, $\overrightarrow{DB} = \vec{r} = \vec{b} - x\vec{a}$

But $\vec{r} \cdot \vec{a} = 0 \Rightarrow x = \frac{(\vec{b} \cdot \vec{a})}{|\vec{a}|^2}$

43. Let $AC = \lambda \Rightarrow AB = 5\lambda$ and $AD = 3\lambda$

$\overrightarrow{AB} = \vec{b}$, $\overrightarrow{AC} = \vec{c}$, $\overrightarrow{AD} = \vec{d}$

$\cos \theta = \frac{\overrightarrow{BA} \cdot \overrightarrow{CD}}{|\overrightarrow{BA}| |\overrightarrow{CD}|}$



$= \frac{-\vec{b} \cdot (\vec{d} - \vec{c})}{|\vec{b}| |\vec{c} - \vec{d}|}$

$= \frac{\vec{d} \cdot \vec{b} - \vec{b} \cdot \vec{c}}{5\lambda \cdot \sqrt{7}\lambda}$

$= \frac{10\lambda^2}{5\sqrt{7}\lambda^2}$

$\theta = \cos^{-1} \left(\frac{2}{\sqrt{7}} \right)$

44. $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF}$

$$= \overline{ED} + \overline{AC} + \overline{AD} + \overline{AE} + \overline{CD}$$

$$(\because \overline{AB} = \overline{ED} \quad \text{and} \quad \overline{AF} = \overline{CD})$$

$$= (\overline{AC} + \overline{ED}) + (\overline{AE} + \overline{ED}) + \overline{AD}$$

$$= \overline{AD} + \overline{AD} + \overline{AD} = 3\overline{AD} = 6\overline{AO}$$

45. The given vector equation can be written as

$$[(1-a)x + 3y - 4z]\vec{i} + [x - (3+a)y + 5z]\vec{j} + [3x + y - az]\vec{k} = 0$$

$$\Rightarrow (1-a)x + 3y - 4z = 0$$

$$x - (3+a)y + 5z = 0 \quad \text{and} \quad 3x + y - az = 0$$

The above system of equation has a non-trivial solution.

$$\Rightarrow \begin{vmatrix} 1-a & 3 & -4 \\ 1 & -(3+a) & 5 \\ 3 & 1 & -a \end{vmatrix} = 0$$

$$\Rightarrow a = 0, -1$$

46. Let $\vec{a} = 2\vec{i} + 4\vec{j} - 5\vec{k}$ and $\vec{b} = \vec{i} + 2\vec{j} + 3\vec{k}$.

Then the diagonals of the parallelogram are

$$\vec{p} = \vec{a} + \vec{b} \quad \text{and} \quad \vec{q} = \vec{b} - \vec{a},$$

$$\text{i.e., } \vec{p} = 3\vec{i} + 6\vec{j} - 2\vec{k}, \quad \vec{q} = -\vec{i} - 2\vec{j} + 8\vec{k}$$

So unit vectors along the diagonals are

$$\frac{1}{7}(3\vec{i} + 6\vec{j} - 2\vec{k}) \quad \text{and} \quad \frac{1}{\sqrt{69}}(-\vec{i} - 2\vec{j} + 8\vec{k}).$$

47. We have, $\vec{a} = 2p\vec{i} + \vec{j}$

on rotation, let $\vec{b}$, be the vector with components

$(p+1)$ and 1 so that $\vec{b} = (p+1)\vec{i} + \vec{j}$.

$$\text{Now, } |\vec{a}| = |\vec{b}| \Rightarrow a^2 = b^2$$

$$\Rightarrow 4p^2 + 1 = (p+1)^2 + 1$$

$$\Rightarrow 4p^2 = (p+1)^2 \quad \Rightarrow 3p = -1 \text{ or } p = 1$$

$$\Rightarrow 2p \pm (p+1) \quad \therefore p = -1/3 \text{ or } p = 1$$

48. Clearly $\vec{a}, \vec{b}$ and $\vec{c}$ must be coplanar

$$\begin{vmatrix} 1 & 1 & 1 \\ 4 & 3 & 4 \\ 1 & x_1 & x_2 \end{vmatrix} = 0 \Rightarrow x_2 = 1$$

$$\text{Also } |\vec{c}| = \sqrt{3} \Rightarrow 3 = 1 + x_1^2 + x_2^2 \Rightarrow x_1^2 = 1 \Rightarrow x_1 = \pm 1$$

46. For Coplanar Vectors $\begin{vmatrix} 1 & 2 & 3 \\ 0 & \lambda & \mu \\ 0 & 0 & 2\lambda - 1 \end{vmatrix} = 0$

$$\Rightarrow (2\lambda - 1)\lambda = 0 \Rightarrow \lambda = 0, \frac{1}{2}$$

$$47. \quad \vec{a} = [\pm(\vec{i} - \vec{j}) \pm (\vec{j} + \vec{k})]$$

$$= \pm(\vec{i} + \vec{k}), \pm(\vec{i} - 2\vec{j} - \vec{k})$$

$$51. \quad \text{Let } \vec{\alpha} = \vec{i} + x\vec{j} + 3\vec{k}, \quad \vec{\beta} = 4\vec{i} + (4x-2)\vec{j} + 2\vec{k}$$

$$\text{Given, } 2|\vec{\alpha}| = |\vec{\beta}|$$

$$\Rightarrow 2\sqrt{10+x^2} = \sqrt{20+4(2x-1)^2}$$

$$\Rightarrow 10+x^2 = 5 = (4x^2 - 4x + 1)$$

$$\Rightarrow 3x^2 - 4x - 4 = 0$$

$$\Rightarrow x = 2, -\frac{2}{3}$$

$$52. \quad \text{Since } \vec{a} \text{ and } \vec{b} \text{ are equally inclined to } \vec{c}, \vec{c} \text{ must be of form } t \left(\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|} \right). \text{ Now,}$$

$$\frac{|\vec{b}|}{|\vec{a}|+|\vec{b}|} \vec{a} + \frac{|\vec{a}|}{|\vec{a}|+|\vec{b}|} \vec{b} = \frac{|\vec{a}||\vec{b}|}{|\vec{a}|+|\vec{b}|} \left(\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|} \right)$$

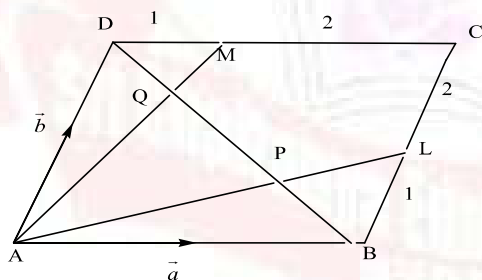
$$\text{Also, } \frac{|\vec{b}|}{2|\vec{a}|+|\vec{b}|} \vec{a} + \frac{|\vec{a}|}{2|\vec{a}|+|\vec{b}|} \vec{b} = \frac{|\vec{a}||\vec{b}|}{2|\vec{a}|+|\vec{b}|} \left(\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|} \right)$$

$$\text{Other two vector cannot be written in the form } t \left(\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|} \right)$$

$$53. \quad \overline{BL} = \frac{1}{3}\vec{b}$$

$$\therefore \overline{AL} = \vec{a} + \frac{1}{3}\vec{b}$$

Let $\overline{AP} = \lambda \overline{AL}$ and P divides DB in the ratio $\mu : 1 - \mu$.



$$\text{Then } \overrightarrow{AP} = \lambda \vec{a} + \frac{\lambda}{3} \vec{b} \quad (1)$$

$$\text{Also } \overrightarrow{AP} = \mu \vec{a} + (1-\mu) \vec{b} \quad (2)$$

From (1) and (2),

$$\lambda \vec{a} + \frac{\lambda}{3} \vec{b} = \mu \vec{a} + (1-\mu) \vec{b}$$

$$\therefore \lambda = \mu \quad \text{and} \quad \frac{\lambda}{3} = 1 - \mu$$

$$\therefore \lambda = \frac{3}{4}$$

Hence, P divides AL in the ratio 3:1

54. Let the position vectors of A, B, C and D be $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ respectively.

Then OA:CB = 2:1

$$\Rightarrow \overrightarrow{OA} = 2\overrightarrow{CB}$$

$$\Rightarrow \vec{a} = 2(\vec{b} - \vec{c}) \quad (1)$$

and OD:AB = 1:3

$$3\overrightarrow{OD} = \overrightarrow{AB}$$

$$\Rightarrow 3\vec{d} = (\vec{b} - \vec{a}) = \vec{b} - 2(\vec{b} - \vec{c}) \quad [\text{Using (1)}]$$

$$\Rightarrow 3\vec{d} = -\vec{b} + 2\vec{c} \quad (2)$$

Let OX:XC = $\lambda:1$ and AX:XD = $\mu:1$

Now, x divides OC in the ratio $\lambda:1$. Therefore,

$$\text{P.V of } X = \frac{\lambda \vec{c}}{\lambda + 1} \quad (3)$$

X also divides AD in the ratio $\mu:1$

$$\text{P.V of } X = \frac{\mu \vec{d} + \vec{a}}{\mu + 1} \quad (4)$$

$$\frac{\lambda \vec{c}}{\lambda + 1} = \frac{\mu \vec{d} + \vec{a}}{\mu + 1}$$

$$\Rightarrow \left(\frac{\lambda \vec{c}}{\lambda + 1} \right) \vec{c} = \left(\frac{\mu}{\mu + 1} \right) \vec{d} + \left(\frac{1}{\mu + 1} \right) \vec{a}$$

[using (1) and (4)]

$$\Rightarrow \left(\frac{\lambda \vec{c}}{\lambda + 1} \right) \vec{c} = \left(\frac{\mu}{\mu + 1} \right) \left(\frac{-\vec{b} + 2\vec{c}}{3} \right) + \left(\frac{1}{\mu + 1} \right) 2(\vec{b} - \vec{c})$$

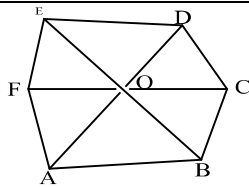
$$\Rightarrow \left(\frac{\lambda \vec{c}}{\lambda + 1} \right) \vec{c} = \left(\frac{6 - \mu}{3(\mu + 1)} \right) \vec{b} + \left(\frac{2\mu}{3(\mu + 1)} - \frac{2}{\mu + 1} \right) \vec{c}$$

$$\Rightarrow \left(\frac{6 - \mu}{3(\mu + 1)} \right) \vec{b} + \left(\frac{2\mu - 6}{3(\mu + 1)} - \frac{\lambda}{\lambda + 1} \right) \vec{c} = \vec{0}$$

$$\Rightarrow \frac{6 - \mu}{3(\mu + 1)} = 0 \quad \text{and} \quad \frac{2\mu - 6}{3(\mu + 1)} - \frac{\lambda}{\lambda + 1} = 0 \quad [As \vec{b} \text{ and } \vec{c} \text{ are non-Collinear}]$$

$$\Rightarrow \mu = 6, \quad \lambda = \frac{2}{5}$$

55. Consider the regular hexagon ABCDEF with centre at O (origin)



$$\begin{aligned}
 \overrightarrow{AD} + \overrightarrow{EB} + \overrightarrow{FC} &= 2\overrightarrow{AO} + 2\overrightarrow{OB} + 2\overrightarrow{OC} \\
 &= 2(\overrightarrow{AO} + \overrightarrow{OB}) + 2\overrightarrow{OC} \quad [\text{Since } \overrightarrow{OC} = \overrightarrow{AB}] \\
 &= 2\overrightarrow{AB} + 2\overrightarrow{AB} \Rightarrow 4\overrightarrow{AB}
 \end{aligned}$$

56 Conceptual

57. clearly $\triangle ABC$ is equilateral

$$58. \frac{\left| \begin{array}{c} \text{the projection of } \vec{a} \text{ on } \vec{b} \end{array} \right|}{\left| \begin{array}{c} \text{the projection of } \vec{b} \text{ on } \vec{a} \end{array} \right|} = \frac{|\vec{a}|}{|\vec{b}|}$$

59. CONCEPTUAL.

$$60. k = |\vec{a} - \vec{b}|^2 + |\vec{c} - \vec{b}|^2 + |\vec{a} - \vec{c}|^2 = 6 - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

Clearly k is maximum when $\vec{a}, \vec{b}, \vec{c}$ are equally inclined at 120° to each other



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

WAT-25

Date: 18-11-17

Max.Marks:183

Name of the Student: _____

H.T. NO:

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2017-PAPER -I

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

18-11-17_Jr. Super60-I_Jee-Adv_WAT-25_WEEKEND SYLLABUS

MATHS: SYLLABUS COVERED FROM_ 13-11-17 TO 17-11-2017

PHYSICS: SYLLABUS COVERD FROM _13-11-17 TO 17-11-2017

CHEMISTRY: Reactive intermediates, Strength of acids and bases

Sri Chaitanya IIT Academy

18-11-17_Jr. Super60-I_Jee-Adv_2017_P1_WAT-25_Q'P

JEE-ADVANCE-2017-P1-Model**Time: 03:00 Hr's****IMPORTANT INSTRUCTIONS****Max Marks: 183****PHYSICS:**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 8 – 12)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 13 – 18)	Questions With Single Answer Type	+3	-1	6	18
Total				18	61

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 25)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 26 – 30)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 31 – 36)	Questions With Single Answer Type	+3	-1	6	18
Total				18	61

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 43)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 44 – 48)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 49 – 54)	Questions With Single Answer Type	+3	-1	6	28
Total				18	61

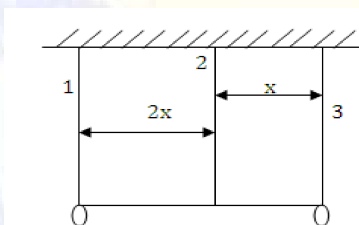
Sec: Jr. Super60-I**space for rough work****Page 2**

PHYSICS**Max. Marks:61****SECTION – I****(MULTIPLE CORRECT ANSWER TYPE)**

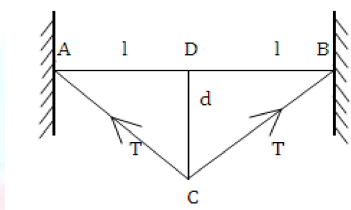
This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

1. Three vertical wires 1,2,3 are supporting a rod of mass M in horizontal position as shown. All wires are of equal length and equal area of cross-section. Given $Y_2 = Y_3$. Wires 1 and 3 are attached to extreme ends of rod. Choose the correct option(s) (T_1 , T_2 and T_3 are tensions in wires 1, 2 and 3 respectively).



- A) $T_1 = 2T_3$ B) $2T_1 = T_3$ C) $T_2 = T_3$ D) $T_1 = \frac{Mg}{5}$
2. A wire AB of length $2l$ and cross-section area a is stretched without tension between fixed points A and B. The wire is pulled at the centre into shape ACB such that $d \ll l$. The tension in the string is T and the strain produced in it is ϵ . If the Young's modulus of the material of wire is Y . Then



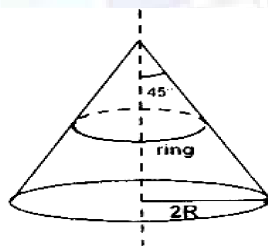
A) $\varepsilon = \frac{d}{2l}$

B) $\varepsilon = \frac{d^2}{2l^2}$

C) $T = aY \left(\frac{d}{2l} \right)$

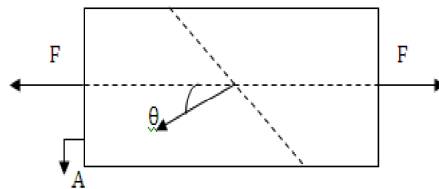
D) $T = aY \left(\frac{d^2}{2l^2} \right)$

3. A uniform metallic ring of mass m , radius R , cross sectional area 'a' and Young's modulus Y is kept on a smooth cone of radius $2R$ and semi vertical angle 45° , as shown in the figure. Assume that the extension in the ring is small.



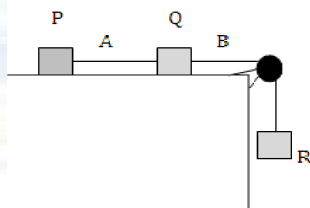
- A) The tension in the ring will be same throughout
 B) The tension in the ring will be independent of the radius of ring (of constant mass) if radius of the ring is less than $2R$
 C) The extension in the ring will be $\frac{mgR}{aY}$
 D) Elastic potential energy stored in the ring will be $\frac{m^2 g^2 R}{8\pi Ya}$

4. A bar of cross section A is subjected to equal and opposite tensile forces at its end. Consider a plane section of the bar whose normal makes an angle θ with the horizontal axis of bar



- A) The tensile stress on this plane is $\frac{F}{A} \cos^2 \theta$
 B) The shearing stress on this plane is $\frac{F}{2A} \sin 2\theta$
 C) Tensile stress is maximum when $\theta = 90^\circ$
 D) Shearing stress is maximum when $\theta = 45^\circ$

5. A copper wire ($Y = 10^{11} \text{ N/m}^2$) of length 8m and steel wire ($Y = 2 \times 10^{11} \text{ N/m}^2$) of length 4m each of 0.5 cm^2 cross-sections are fastened end to end and stretched with a tension of 500N
- A) Elongation in copper wire in 0.8mm
 B) Elongation in steel wire is $1/4^{\text{th}}$ mm the elongation in copper wire
 C) Total elongation in 1.0mm
 D) Elastic potential energy of the system in 0.25 joule
6. Each of the three blocks P, Q and R shown in figure has a mass of 3kg. Each of the wires A and B has cross-sectional area 0.005 cm^2 and Young's modulus $2 \times 10^{11} \text{ N/m}^2$. Neglect friction. Find the longitudinal strain developed in each of the wires. Take $g = 10 \text{ m/s}^2$.



- A) Strain in wire 'A' is 0.2 mm B) Strain in wire 'B' is 0.4 mm
 C) Strain in wire A is 0.1 mm D) Strain in wire B is 0.2 mm

7. A light rod of length 2m is suspended from a ceiling horizontally by means of two vertical wires A and B of equal length tied to its ends. Wire 'A' is made of steel and is of cross section 10^{-3} m^2 and the wire 'B' of brass of cross section $2 \times 10^{-3} \text{ m}^2$. 'X' is the distance from steel wire end, at which a weight may be hung.

$Y_{\text{steel}} = 2 \times 10^{11} \text{ pa}$ and $Y_{\text{brass}} = 10^{11} \text{ pa}$. Which of the following are correct?

- A) $X=1.2\text{m}$, if strains of both wires are to be equal
- B) $X=1.42\text{m}$, if stresses of both wires are to be equal
- C) $X=1\text{m}$, if strains of both wires are to be equal
- D) $X=1.33\text{m}$, if stresses of both wires are to be equal

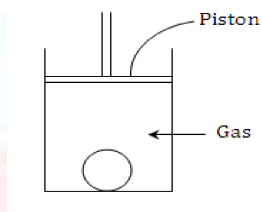
SECTION-II

(INTEGER ANSWER TYPE)

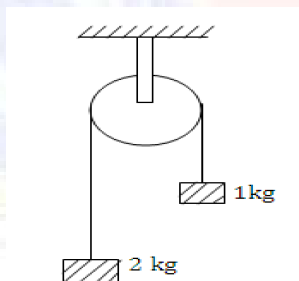
This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

8. The bulk stress on the spherical object of radius $10/\pi \text{ cm}$ if area and mass of piston are 50 cm^2 and 50 kg, respectively, for a cylinder filled with gas as shown in fig. is $x \times 10^5 \text{ Nm}^{-2}$. Then $x = \left(g = 10 \text{ ms}^{-2} \text{ and } 1 \text{ atm} = 10^5 \frac{\text{N}}{\text{m}^2} \right)$ (Assuming the object is in normal state when it is in the atmosphere on the surface of earth)



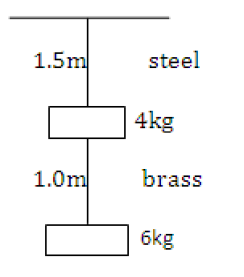
9. Two blocks of masses 1 kg and 2 kg are connected by a metal wire going over a smooth pulley as shown in fig. The breaking stress of the metal is $(40/3\pi) \times 10^6 \text{ N/m}^2$. If $g = 10 \text{ ms}^{-2}$. The minimum radius of the wire used if it is not to break, is $x \times 10^{-3} \text{ m}$. Then x is



10. Two wires of diameter 0.25 cm, one made of steel and other made of brass, are loaded as shown in the figure. The unloaded length of the steel wire is 1.5 m and that of brass

is 1.0m. Young's modulus of steel is 2×10^{11} pa and Young's modulus of brass is

10^{11} Pa. The ratio of elongations of steel and brass wires, is $\frac{x}{4}$. Then x is....



11. A uniform rod of mass m and length ℓ is rotating with constant angular velocity ω about an axis which passes through its one end and perpendicular to the length of rod. The area of cross section of the rod is A and its young's modulus is Y . Neglect gravity. The strain at the midpoint of the rod is $\frac{3m\omega^2\ell}{KAY}$. Then the value of 'K' is
12. Average depth of Indian Ocean is about 3000m. Bulk modulus of water is 2.2×10^9 pa and density of water is 1000 kgm^{-3} . The fractional compression of water of the bottom of the ocean is $\beta(0.68)\%$. Then the value of β rounded to nearest integer is.....

SECTION – III

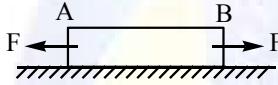
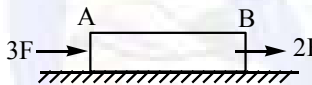
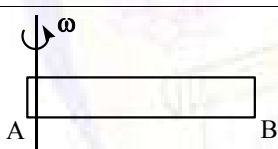
(SINGLE CORRECT ANSWER TYPE)

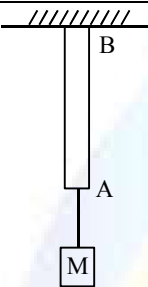
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Four identical rods (AB) each of mass m , length l , cross sectional area 'a' and

Young's modulus Y are taken in column-I in different situations. Mark the most appropriate option

	Column-I		Column-II		Column-III
A)	 Surface is smooth and F is the external force	I)	Longitudinal strain is non uniform	P)	Strain energy density is uniform
B)	 Smooth	II)	Magnitude of longitudinal strain decreases continuously from A to B	Q)	Strain energy density is not uniform
C)	 The rod is rotating about end A with constant	III)	Magnitude of longitudinal strain increases continuously from A to B	R)	Strain energy density continuously increases from A to B

	angular velocity ω on a smooth horizontal surface				
D)	 End B is fixed to the ceiling and a block of mass M is connected to end A by a massless spring	IV)	There is net compression in the rod	S)	Strain energy density continuously decreases from A to B

13. For the rod given in (B) in Column-I

- A) I, IV, P B) I, IV, Q C) I, III, R D) None of these

14. For the rod given in (C) in Column-II

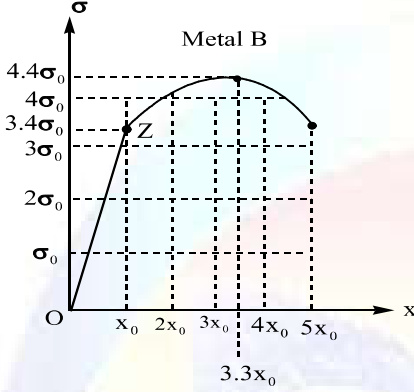
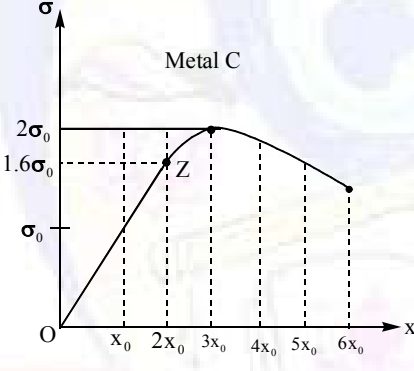
- A) I, II, P B) I, III, Q, S C) I, II, Q, S D) None of these

15. For the rod given in (D) in Column-I

- A) III, P B) I, III, P, R C) I, III, Q, R D) None of these

In column I stress (y-axis) – strain (x-axis) graph of four metallic wires A, B, C and D of same natural length and same cross-sectional area are shown. OZ is a straight line and end point of the curve represents fracture point

	Column-I		Column-II		Column-III
A)		I)	Young's modulus is maximum among the four	P)	Breaking stress is maximum among the four

B)		II)	Young's modulus is minimum among the four	Q)	Breaking stress is minimum among the four
C)		III)	More ductile among the four	R)	Within the proportionality limit strain energy that can be stored in the wire is maximum among the four

D)		IV)	More brittle among the four	S)	Within the proportionality limit strain energy that can be stored in the wire is minimum among the four
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16. Which of the following are correct for wire D

- A) I, IV, S B) II, IV, S C) IV, R D) II, Q, S

17. Which of the following are correct for wire B

- A) I, III, Q, R B) I, IV, Q, R C) I, P, R D) I, P

18. Which of the following are correct for wire C

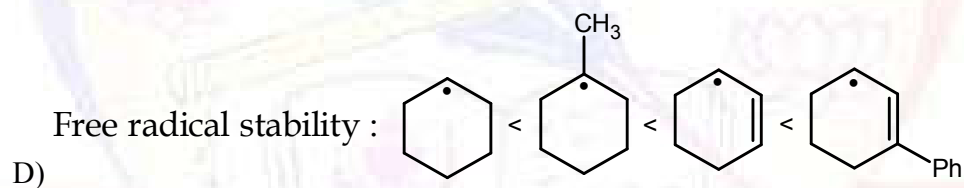
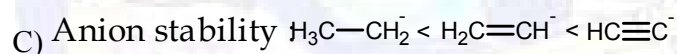
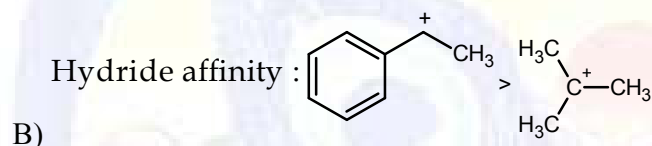
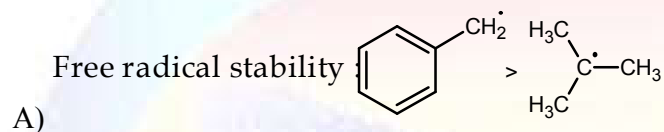
- A) III, Q, S B) IV, Q, S C) II, III, Q, S D) None of these

CHEMISTRY**Max.Marks:61****SECTION – I****(MULTIPLE CORRECT ANSWER TYPE)**

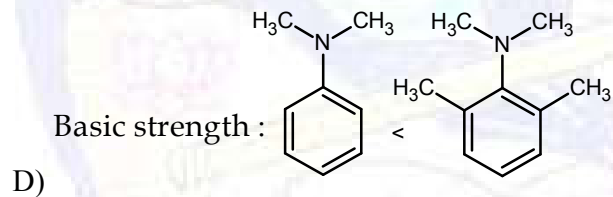
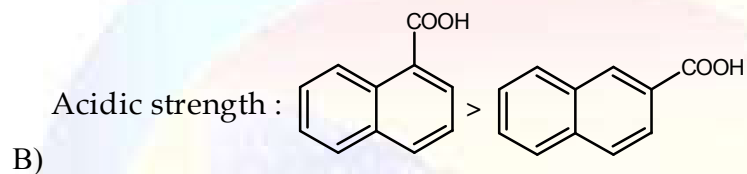
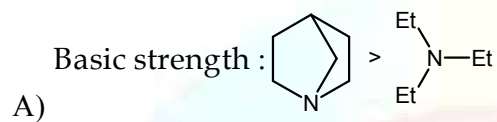
This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

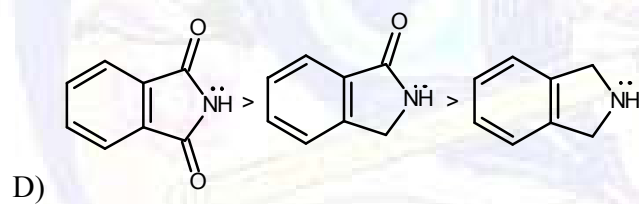
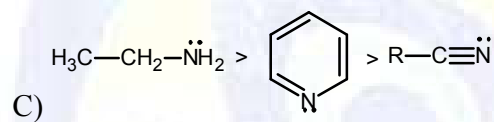
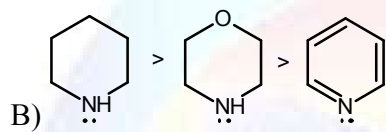
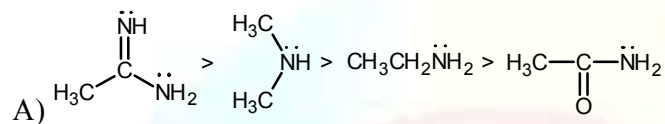
19. Which among the following statements are correct?



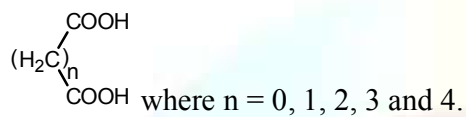
20. Select the correct option against mentioned property



21. The correct basic order (s) among the following is / are

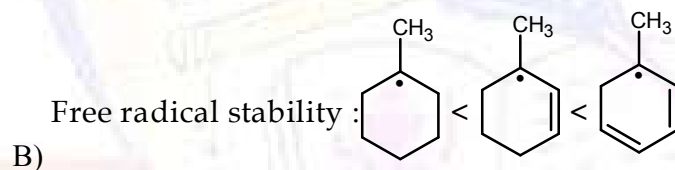
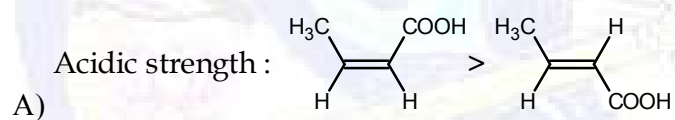


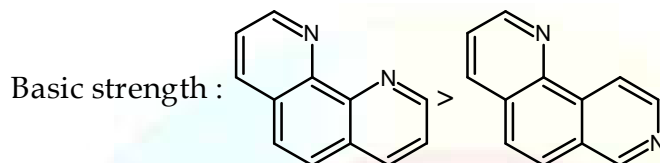
22. Which of the following is/are correct about given dicarboxylic acids.



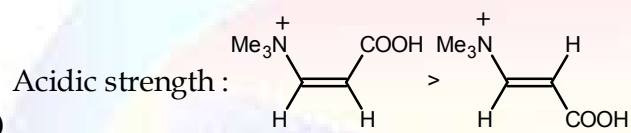
- A) pK_{a1} for all the dicarboxylic acids is smaller than the pK_a for mono carboxylic acids with the same number of carbon atoms
- B) The difference between pK_{a1} and pK_{a2} for dicarboxylic acids decreases as 'n' increases
- C) pK_{a1} for all the dicarboxylic acids is higher than the pK_{a2}
- D) K_{a1} for all dicarboxylic acid is higher than the K_{a2} .

23. Which of the following is correct?



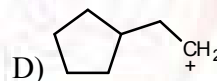
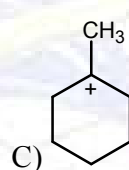
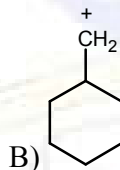
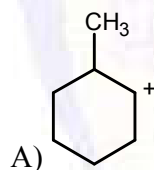
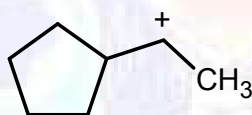


C)

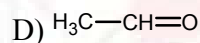
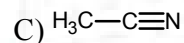
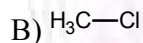
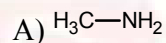


D)

24. The possible carbocations that can be formed on rearrangement for the following carbocation is/are....



25. Which of the following behave both as a nucleophile and an electrophile?



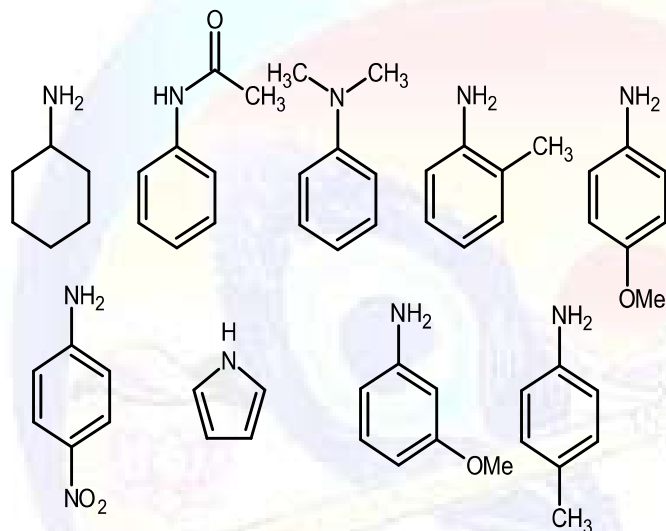
SECTION-II

(INTEGER ANSWER TYPE)

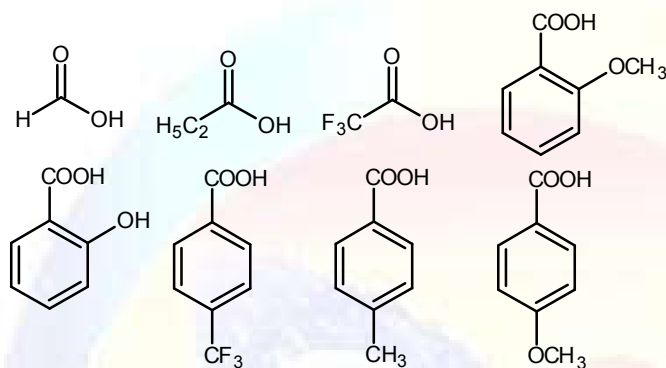
This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

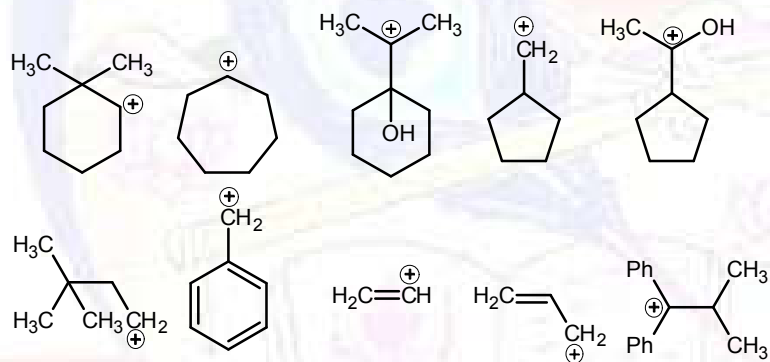
26. Among the following compounds, how many of them is more basic than aniline?



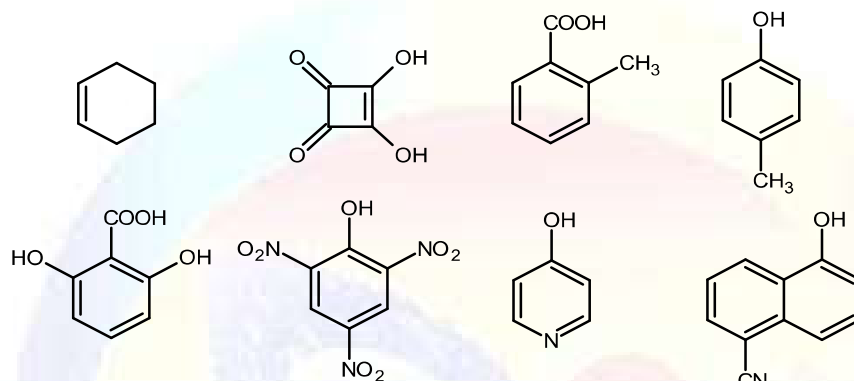
27. How many of the following acids are more acidic than benzoic acid?



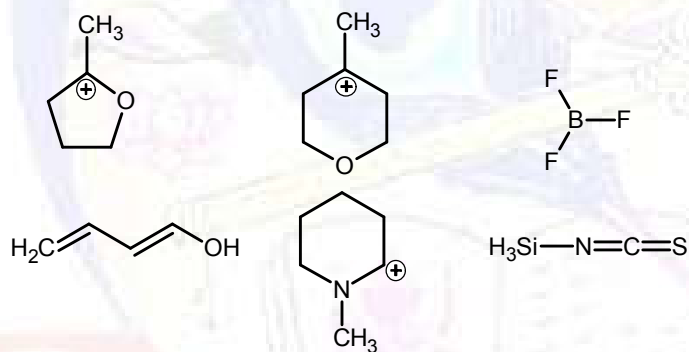
28. How many carbocations undergo rearrangements?



29. How many compounds are soluble in aqueous NaOH ?



30. Among the following, find out number of ions / molecules that can show backbonding.



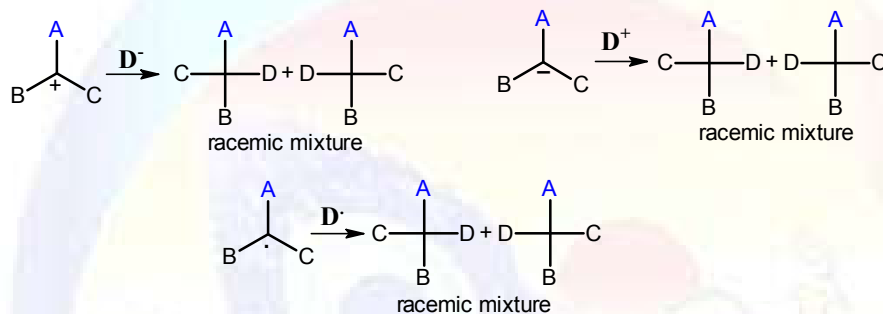
SECTION – III

(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

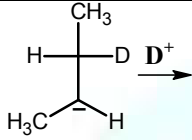
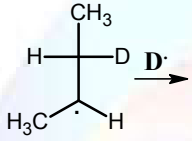
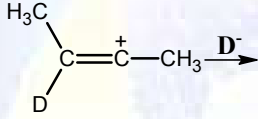
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.31, Q.32 and Q.33 by appropriately matching the information given below.



Do not consider rearrangements if possible for given intermediate in column-I. Consider the given information and solve the given problems.

	Column – I		Column – II (Regarding reactant)		Column – III
(I)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}-\text{C}-\text{D} \\ \\ \text{H}_3\text{C}-\text{C}^+-\text{H} \end{array} \xrightarrow{\text{D}^-}$	(i)	Can be stabilized by inductive effect	(P)	Racemic mixture will be formed

(II)		(ii)	Can be stabilized by hyper conjugation	(Q)	Meso is one of the product
(III)		(iii)	Destabilized by inductive effect	(R)	Diastereomeric mixture will be formed
(IV)		(iv)	sp hybridized carbon atom	(S)	One of the product is optically active

31. Which of the following option is correct?

- A) I-iii-Q B) II-ii-P C) III-iii-P D) IV-iv-R

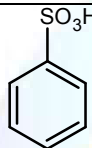
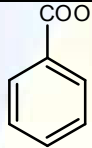
32. Which of the following option is incorrect?

- A) I-i-S B) II-iii-Q C) III-ii-R D) IV-iii-R

33. Which of the following option is correct?

- A) I-iv-R B) II-iii-P C) III-ii-S D) IV-i-Q

Answer Q.34, Q.35 and Q.36 by appropriately matching the information given in the three columns of the following table (pKa values are given relative to water).

	Column – I		Column – II(pK _a)		Column – III
(I)		(i)	16	(P)	Give effervescence with NaHCO ₃
(II)		(ii)	15.2	(Q)	React with NaOH
(III)		(iii)	-6.5	(R)	More acidic than water
(IV)	CH ₃ OH	(iv)	4.2	(S)	Gives H ₂ gas on reaction with sodium metal

34. Which of the following option is correct?
 A) I-iii-P B) II-ii-P C) III-iv-Q D) IV-i-P
35. Which of the following option is incorrect?
 A) I-iii-R B) II-i-S C) III-i-S D) IV-ii-R
36. Which of the following option is correct?
 A) I-ii-Q B) II-iii-R C) III-i-S D) IV-ii-P

SECTION – I**(MULTIPLE CORRECT ANSWER TYPE)**

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

37. Let $\vec{a}, \vec{b}$ be two vectors perpendicular to each other and $|\vec{a}| = 2, |\vec{b}| = 3, \vec{c} \times \vec{a} = \vec{b}$ such that

the least value of $|\vec{c} - \vec{a}|$ is $\frac{m}{n}$ where m and n are relatively prime, then

A) $m - n = 1$

B) $\sin^{-1}\left(\frac{m}{n} - 1\right) = \frac{-\pi}{6}$

C) $\cos^{-1}(2m - 3n) = \frac{\pi}{2}$

D) None of these

38. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-zero vectors, then which of the following statements is/are true?

A) If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then $\frac{(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c})}{(\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{c})} = -1$

B) If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} < 0$

C) $\vec{a} \times (\vec{b} \times \vec{c}), \vec{b} \times (\vec{c} \times \vec{a}), \vec{c} \times (\vec{a} \times \vec{b})$ form a right handed system

D) $\vec{c}, (\vec{a} \times \vec{b}) \times \vec{c}, \vec{a} \times \vec{b}$ form a right handed system

39. Let unit vectors $\vec{a}$ and $\vec{b}$ are perpendicular and unit vector $\vec{c}$ be inclined at an angle 'θ' to both $\vec{a}$ and $\vec{b}$. If $\vec{c} = \alpha\vec{a} + \beta\vec{b} + \gamma(\vec{a} \times \vec{b})$, then

A) $\alpha^2 - \beta^2 = \gamma^2$ B) $\alpha = \beta$ C) $1 - 2\alpha^2 = \gamma^2$ D) $2\alpha^2 = 1 + \cos 2\theta$

40. Let $\vec{a} = 2\vec{i} + \vec{k}$, $\vec{b} = \vec{i} + \vec{j} + \vec{k}$ and $\vec{c} = 4\vec{i} - 3\vec{j} + 7\vec{k}$ and $\vec{r}$ be a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then

A) $\vec{r} = -\vec{i} + 8\vec{j} - 2\vec{k}$ B) $|\vec{r}| = |\vec{b}| \sqrt{4|\vec{a}|^2 + |\vec{b}|^2}$

C) $\vec{r} = -\vec{i} - 8\vec{j} + 2\vec{k}$ D) $|\vec{r}| = |\vec{b}| \sqrt{7|\vec{b}|^2 + 2}$

41. The vectors $\vec{a}$ and $\vec{b}$ are perpendicular to each other. If $\vec{r} = l\vec{b} + m(\vec{a} \times \vec{b})$ is a vector such that $\vec{r} \cdot \vec{a} = 0$, $\vec{r} \cdot \vec{b} = 1$ and $[\vec{r} \vec{a} \vec{b}] = 1$, then

A) $l = \frac{1}{|\vec{b}|}, m = \frac{1}{|\vec{a} \times \vec{b}|^2}$ B) $l = \frac{1}{|\vec{b}|^2}, m = \frac{1}{|\vec{a} \times \vec{b}|}$

C) $l = \frac{1}{|\vec{b}|^2}, m = \frac{1}{|\vec{a} \times \vec{b}|^2}$ D) $l = m|\vec{a}|^2$

42. The volume of a right triangular prism $ABCA_1B_1C_1$ is equal to 3. If the position vectors of the vertices of the base ABC are A (1, 0, 1), B (2, 0, 0) and C (0, 1, 0), then the position vector of the vertex A_1 can be
- A) (0, 2, 0) B) (0, -2, 0) C) (0, -2, 2) D) (2, 2, 2)
43. Let $\vec{a}, \vec{b}, \vec{c}$ be the position vectors of the points A, B, C respectively. Let α, β, γ be the inclinations between, $\vec{b}, \vec{c}$; $\vec{a}, \vec{b}$ and $\vec{a}, \vec{c}$ respectively, then the volume 'V' of the tetrahedron, 'OABC' is given by
- A) $V^2 = \frac{abc}{36} \begin{vmatrix} 1 & \cos \beta & \cos \gamma \\ \cos \beta & 1 & \cos \alpha \\ \cos \gamma & \cos \alpha & 1 \end{vmatrix}$ B) $V^2 = \frac{a^2b^2c^2}{36} \begin{vmatrix} 1 & \cos \beta & \cos \gamma \\ \cos \beta & 1 & \cos \alpha \\ \cos \gamma & \cos \alpha & 1 \end{vmatrix}$
- C) $V = \frac{a^2b^2c^2}{36} \begin{vmatrix} 1 & \cos \beta & \cos \gamma \\ \cos \beta & 1 & \cos \alpha \\ \cos \gamma & \cos \alpha & 1 \end{vmatrix}$ D) None of these

SECTION-II**(INTEGER ANSWER TYPE)**

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

44. The points D, E, F respectively divide the sides BC, CA, AB of a triangle 'ABC' in the ratio 1:2. The pairs of lines AD, BE; BE, CF; CF, AD meet at P, Q, R respectively such that area of $\Delta PQR = \frac{K}{l}$ (area of ΔABC) where K and l are relatively prime, then
- $K + l =$

45. Let 'a' be the perpendicular distance of a corner of a unit cube from a diagonal not passing through it and 'θ' be the angle between any edge and a face not containing that edge of a regular tetrahedron such that $\cos \theta = b$, then $a^2 + b^2 =$
46. Let $\vec{a}, \vec{b}, \vec{c}$ and $\vec{a'}, \vec{b'}, \vec{c'}$ be the two sets of non-coplanar reciprocal system of vectors such that i) $[\vec{a} \vec{b} \vec{c}] [\vec{a'} \vec{b'} \vec{c'}] = m$ & ii) $\vec{a} \cdot \vec{a'} + \vec{b} \cdot \vec{b'} + \vec{c} \cdot \vec{c'} = n$, then $m + n =$
47. A new operation * is defined between two non-zero vectors $\vec{\alpha}$ and $\vec{\beta}$ as $\vec{\alpha} * \vec{\beta} = |\vec{\alpha}| |\vec{\beta}| \tan \frac{\theta}{2}$ where $(\vec{\alpha}, \vec{\beta}) = \theta$. Let $m = \vec{\alpha} * \vec{\alpha}$ and the condition for which $\vec{\alpha}$ and $\vec{\beta}$ are perpendicular is $n = \frac{\vec{\alpha} * \vec{\beta}}{|\vec{\alpha}| |\vec{\beta}|}$, then $m + n =$
48. Let $\vec{r}_1 = a\vec{i} + b\vec{j} + c\vec{k}$ ($a < b < c$) where a, b, c are the solutions of $2[x] = x + 2\{x\}$ and $\vec{r}_2 = d\vec{i}$ where d is the unit's digit of $\left[\frac{10^{20000}}{100^{100} + 3} \right]$ and $|3\vec{r}_1 \times \vec{r}_2| = m\sqrt{n}$ where m and n are relatively prime, then $\left[\frac{m}{n} \right] =$
 ([.] is G.I.F and { } is F.P.F)

SECTION – III

(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.49, Q.50 and Q.51 by appropriately matching the information given in the three columns of the following table.

Column:1 $\vec{a} =$	Column:2 $\vec{b} =$	Column:3 $ \vec{a} \times \vec{b} = x\sqrt{y}$ x and y are relatively prime
I) $2\vec{i} + \vec{j} - \vec{k}$	i) $-\vec{i} + 2\vec{j} - \vec{k}$	P) $x + y = 3$
II) $\vec{i} - \vec{j} + \vec{k}$	ii) $\vec{i} + 2\vec{j} + 3\vec{k}$	Q) $x + y = 5$
III) $-\vec{i} + 3\vec{j} + 2\vec{k}$	iii) $2\vec{i} - \vec{j} + \vec{k}$	R) $x + y = 15$
IV) $\vec{i} + \vec{j} + \vec{k}$	iv) $3\vec{i} - 2\vec{j} + \vec{k}$	S) $x + y = 8$

49. Which of the following options is the only correct combination?
A) (I), (iv), (S) B) (III), (ii), (R) C) (IV), (i), (P) D) (III), (iii), (Q)
50. Which of the following options is the only correct combination?
A) (III), (iii), (P) B) (I), (iv), (Q) C) (IV), (i), (Q) D) (II), (i), (R)
51. Which of the following options is the only correct combination?
A) (II), (i), (Q) B) (II), (i), (P) C) (IV), (iv), (R) D) (IV), (iv), (S)

Answer Q.52, Q.53 and Q.54 by appropriately matching the information given in the three columns of the following table.

Column:1 $\bar{a} =$	Column:2 $\bar{b} =$	Column:3 $[[\bar{a}\bar{b}\bar{c}]] = xy$ Where 'xy' is the two digit number
I) $\bar{i} + 2\bar{j} + 3\bar{k}$ II) $\bar{i} - \bar{j}$ III) $3\bar{i} - 4\bar{j} + \bar{k}$ IV) $2\bar{i} + 4\bar{k}$	i) $2\bar{i} - \bar{j} - 4\bar{k}$ ii) $\bar{i} + \bar{j} + \bar{k}$ iii) $2\bar{i} - 3\bar{j} + 4\bar{k}$ iv) $\bar{i} + 2\bar{j} + 4\bar{k}$	P) $x + y = 3$ Q) $x + y = 8$ R) $x(y) = 7$ S) $y - 2x = 8$

52. If $\bar{c} = \bar{i} + 3\bar{j} - \bar{k}$, then which of the following options is the only correct combination?

A) (I), (iii), (Q) B) (I), (iii), (P) C) (II), (iv), (P) D) (II), (iv), (Q)

53. If $\bar{c} = 2\bar{i} - 4\bar{j} + 3\bar{k}$, then which of the following options is the only correct combination?

A) (III), (i), (P) B) (III), (i), (S) C) (II), (iv), (S) D) (II), (iv), (Q)

54. If $\bar{c} = \bar{i} + 4\bar{j} + 3\bar{k}$, then which of the following options is the only correct combination?

A) (II), (i), (R) B) (III), (i), (P) C) (III), (i), (S) D) (II), (iii), (Q)



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

WAT-25

Date: 18-11-17

Time: 07:30 AM to 10:30 AM

2017-P1

Max.Marks:183

KEY SHEET

PHYSICS

1	C	2	BD	3	ABC	4	ABD	5	ABCD
6	CD	7	CD	8	1	9	1	10	5
11	8	12	2	13	B	14	C	15	C
16	C	17	D	18	C				

CHEMISTRY

19	ACD	20	ABD	21	ABC	22	ABD	23	ABCD
24	AC	25	CD	26	4	27	5	28	6
29	7	30	5	31	D	32	D	33	C
34	A	35	B	36	C				

MATHS

37	C	38	ABD	39	BD	40	BCD	41	C
42	BD	43	B	44	7	45	1	46	4
47	1	48	2	49	A	50	C	51	B
52	B	53	D	54	C				

SOLUTIONS PHYSICS

1. $T_1(2x) = T_3(x)$ [for rotational equilibrium]

$T_1 + T_2 + T_3 = Mg$ [for translational equilibrium]

2.
$$e = \sqrt{l^2 + d^2} - L = L \left[i + \frac{d^2}{2l^2} - 1 \right]$$

$$T = \frac{YAc}{L}$$

3. $2T \sin \frac{d\theta}{2} = \frac{N}{\sqrt{2}} = dm g \quad T d\theta = \frac{m}{2\pi R} (R d\theta) g \quad T = \frac{mg}{2\pi}$

Elongation = $\left(\frac{T}{aY} \right) 2\pi R = \frac{mgR}{aY}$ Energy = $\frac{1}{2} \times \text{stress} \times \text{strain} = \frac{m^2 g^2}{8\pi^2 a^2 Y}$

4. Tensile stress = $\frac{F \cos \theta}{A'}$ where $A' = \frac{A}{\cos \theta}$ Shear stress = $\frac{F \cos \theta}{A'}$

5. $\Delta L = \frac{FL}{AY} = \Delta L_c + \Delta L_s$; $U = \frac{1}{2} (\text{strain})^2 [Y_c + Y_s] A$

6. $T_1 - T_2 = 3a$; $T_2 = 3a$ and $\frac{e}{L} = \frac{\text{Stress}}{Y}$

7. Conceptual

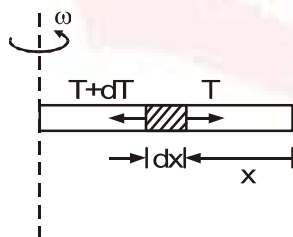
8. Bulk Stress = $\frac{Mg}{A} + P_{atm}$

$2g - T = 2a$; $P - g = a$

9. $B.\text{stress} = \frac{T}{A}$

10. $\Delta L = \frac{FL}{AY}$

11.



$$dT = dm(\ell - x)\omega^2 \quad dT = \frac{m}{\ell} dx(\ell - x)\omega^2$$

$$\int_0^T dT = \int_0^{\ell/2} \frac{m\omega^2}{\ell} (\ell - x) dx$$

$$= \frac{m\omega^2}{\ell} \left[\ell x - \frac{x^2}{2} \right]_0^{\ell/2} = \frac{m\omega^2}{\ell} \left[\frac{\ell^2}{2} - \frac{\ell^2}{8} \right]$$

∴ Tension at mid point is :s

$$T = \frac{3}{8} m\ell \omega^2 \Rightarrow stress = \frac{3m\ell \omega^2}{8A} \Rightarrow strain = \frac{3m\ell \omega^2}{8AY}$$

12. $B = \frac{\frac{\Delta P}{\Delta V}}{\frac{\Delta V}{V}} = \frac{h\rho g}{\frac{\Delta V}{V}}$

13 to 18: Conceptual

CHEMISTRY

19. conceptual
20. conceptual
21. basic character $\propto$ electron density on nitrogen
22. pKa2 of dicarboxylic acids increases due to the formation of intramolecular H-bond.
23. conceptual
24. conceptual
25. π -bonded electrons are transferred to more electronegative atom.
26. 1,3,5,9.
27. in VIIIth molecule hyper conjugative effect, in IXth molecule electron donating resonance effect decreases acidity
28. (i),(ii),(iii),(iv),(vi),&(x).
29. except 'a' all are soluble in sodium hydroxide
30. 1,3,4,5,6.
31. conceptual
32. conceptual
33. conceptual
34. conceptual
35. conceptual
36. conceptual

MATHS

$$37. \quad |\vec{c} \times \vec{a}| = |\vec{b}|, (\vec{c}, \vec{a}) = \theta$$

$$|\vec{c} - \vec{a}|^2 = \frac{9}{4} + \left(\frac{3}{2} \cot \theta - 2\right)^2 \geq \frac{9}{4}$$

$$38. \quad \vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \text{ \& } a^2 + b^2 + c^2 = -2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

$$\therefore \vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b}) = \vec{0}$$

$\Rightarrow$ they are coplanar

$$\text{Similarly } [\vec{c}, (\vec{a} \times \vec{b}), \vec{a} \times \vec{b}] > 0$$

$\Rightarrow$ they form a RHS

$$39. \quad \vec{a} \cdot \vec{b} = 0$$

$$\vec{a} \cdot \vec{c} = \vec{b} \cdot \vec{c} = \cos \theta$$

$$\text{Also } \vec{a} \cdot \vec{c} = \alpha, \vec{b} \cdot \vec{c} = \beta \text{ and } \vec{c} \cdot \vec{c} = 2\alpha^2 + \gamma^2$$

$$40. \quad (\vec{r} - \vec{c}) \times \vec{b} = \vec{0}$$

$$\vec{r} \cdot \vec{a} = 0$$

$$\Rightarrow \vec{r} = -\vec{i} - 8\vec{j} + 2\vec{k}$$

$$41. \quad \vec{r} = \frac{(\vec{r} \cdot \vec{a})}{|\vec{a}|^2} \vec{a} + \frac{(\vec{r} \cdot \vec{b})}{|\vec{b}|^2} \vec{b} + \frac{\{\vec{r} \cdot (\vec{a} \times \vec{b})\}}{|\vec{a} \times \vec{b}|^2} (\vec{a} \times \vec{b})$$

42. Conceptual

$$43. \quad V^2 = \frac{1}{36} [\vec{a} \vec{b} \vec{c}]^2 = \frac{1}{36} \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$44. \quad \text{Take A as origin of reference and } \vec{AB} = \vec{b}, \vec{AC} = \vec{c} \text{ then position vector of P is } \frac{2}{7}(\vec{b} + \vec{c}).$$

$$\text{Similarly position vectors of Q and R are respectively } \frac{\vec{b} + 4\vec{c}}{7}, \frac{2\vec{b} + \vec{c}}{7}.$$

$$45. \quad a = \frac{\sqrt{2}}{\sqrt{3}}, b = \cos \theta = \frac{1}{\sqrt{3}}$$

$$46. \quad [\vec{a} \vec{b} \vec{c}] = \frac{1}{[\vec{a}' \vec{b}' \vec{c}']}$$

$$\vec{a} \cdot \vec{a}' = \vec{b} \cdot \vec{b}' = \vec{c} \cdot \vec{c}' = 1$$

$$47. \quad \frac{\vec{\alpha} \cdot \vec{\beta}}{|\vec{\alpha}| |\vec{\beta}|} = 1 \text{ and } \vec{\alpha} \cdot \vec{\alpha} = 0$$

$$48. \quad a = 0, b = \frac{4}{3}, c = \frac{8}{3} \text{ and } d = 3$$

49 to 54. Conceptual.



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A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

Date: 25-11-17

Max.Marks:180

Name of the Student: _____

I.D. NO:

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2014_PAPER-II

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

25-11-17_Jr.Super60-I_Jee-Adv_WAT-26_WEEKEND SYLLABUS

MATHS: SYLLABUS COVERED FROM_ 20-11-17 TO 24-11-17

PHYSICS: SYLLABUS COVERD FROM 20-11-17 TO 24-11-2017

CHEMISTRY: Tautomerism, Preparation of alkanes from different compounds

JEE-ADVANCE-2014-P2-Model

Time: 3:00 Hours

IMPORTANT INSTRUCTIONS

Max Marks: 180

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec- I(Q.N: 01 – 10)	Questions with Single Correct Choice	3	-1	10	30
Sec- II(Q.N: 11 – 16)	Questions with Comprehension Type (3 Comprehensions – 2 +2+2 = 6Q)	3	-1	6	18
Sec- III(Q.N: 17 – 20)	Matrix Matching Type	3	-1	4	12
Total				20	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 21 – 30)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N : 31 – 36)	Questions with Comprehension Type (3 Comprehensions – 2 +2+2 = 6Q)	3	-1	6	18
Sec – III(Q.N : 37 – 40)	Matrix Matching Type	3	-1	4	12
Total				20	60

MATHEMATICS:

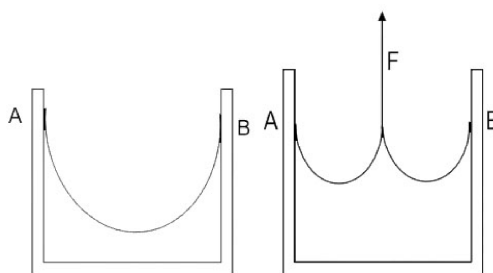
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 41 – 50)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N: 51 – 56)	Questions with Comprehension Type (3 Comprehensions – 2 +2+2 = 6Q)	3	-1	6	18
Sec – III(Q.N : 57 – 60)	Matrix Matching Type	3	-1	4	12
Total				20	60

PHYSICS**Max Marks: 60****SECTION – I
(SINGLE CORRECT ANSWER TYPE)**

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

1. When a film of soap solution is created inside a loop formed by rectangular wire frame and an inextensible light thread AB of length l , the thread assumes shape of a semicircle and remains equilibrium. By pulling the midpoint of thread with a force F , the thread can be given shape of two semicircles as shown in Figure. Calculate surface tension σ of the soap solution.

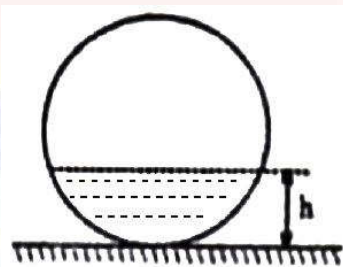


- A) $\frac{\pi F}{\ell}$ B) $\frac{\pi F}{2\ell}$ C) $\frac{\pi F}{4\ell}$ D) None
2. A wire forming a loop is dipped into soap solution and taken out, so that a film of soap solution is formed. A loop of 6.28 cm long thread is gently put on the film and pricked

with a needle inside the loop. The thread loop takes the shape of a circle. The tension in the thread is (*Surface tension of soap solution* = 0.03 N/m .)

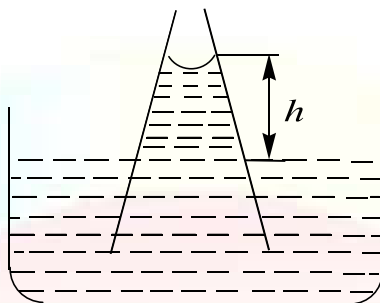
- A) $3 \times 10^{-4} \text{ N}$ B) $6 \times 10^{-4} \text{ N}$ C) $2 \times 10^{-4} \text{ N}$ D) $1.5 \times 10^{-4} \text{ N}$

3. A liquid is filled in a spherical container of radius R till a height h . At this positions the liquid surface at the edges is also horizontal. The contact angle is



- A) 0 B) $\cos^{-1}\left(\frac{R-h}{R}\right)$ C) $\cos^{-1}\left(\frac{h-R}{R}\right)$ D) $\sin^{-1}\left(\frac{R-h}{R}\right)$

4. A capillary of the shape as shown is dipped in a liquid. Contact angle between the liquid and the capillary is 0° and effect of liquid inside the meniscus is to be neglected. T is surface tension of the liquid, r is radius of the meniscus, g is acceleration due to gravity and ρ is density of the liquid then height h in equilibrium is



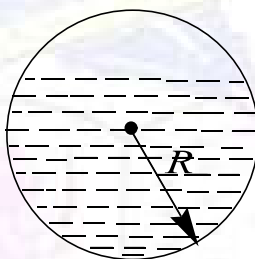
A) Greater than $\frac{2T}{r\rho g}$

B) Equal to $\frac{2T}{r\rho g}$

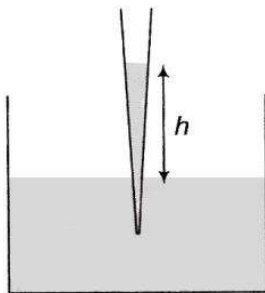
C) Less than $\frac{2T}{r\rho g}$

D) $\frac{4T}{r\rho g}$

5. A spherical glass vessel filled with liquid is kept in uniform gravity. Horizontal surface represents meniscus of liquid. Now complete system is taken to gravity free space. C is the center of sphere.

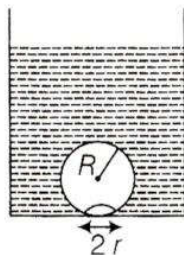


- A) Vessel + liquid is a wetting combination
B) Liquid forms drop & leaves contact of vessel
C) Finally, liquid forms a drop touching vessel
D) Finally, liquid spreads over entire surface of vessel
6. The following observations were taken for determining surface tension T of water by capillary method. Diameter of capillary $d = 1.25 \times 10^{-2} m$, rise of water $h = 1.45 \times 10^{-2} m$.
Using $g = 9.80 m/s^2$ and the simplified relation $T = \frac{r h g}{2} \times 10^3 N/m$, the possible error in surface tension is closest to
A) 1.5% B) 2.4% C) 10% D) 0.15%
7. A glass capillary tube is of the shape of truncated cone with an apex angle α so that its two ends have cross-sections of different radii. When dipped in water vertically, water rises in it to a height h , where the radius of its cross-section is b . If the surface tension of water is S , its density is ρ , and its contact angle with glass is θ , the value of h will be (g is the acceleration due to gravity)



A) $\frac{2S}{b\rho g} \cos(\theta - \alpha)$ B) $\frac{2S}{b\rho g} \cos(\theta + \alpha)$ C) $\frac{2S}{b\rho g} \cos(\theta - \alpha/2)$ D) $\frac{2S}{b\rho g} \cos(\theta + \alpha/2)$

8. On heating water, bubbles being formed at the bottom of the vessel detach and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If $r \ll R$ and the surface tension of water is T , value of r just before bubbles detach is (density of water is ρ)



A) $R^2 \sqrt{\frac{2\rho g}{3T}}$ B) $R^2 \sqrt{\frac{\rho g}{6T}}$ C) $R^2 \sqrt{\frac{\rho g}{T}}$ D) $R^2 \sqrt{\frac{3\rho g}{T}}$

9. Assume that a drop of liquid evaporates by decrease in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible? The surface tension is T , density of liquid is ρ and L is its latent heat of vaporization
- A) $\frac{\rho L}{T}$ B) $\sqrt{\frac{T}{\rho L}}$ C) $\frac{T}{\rho L}$ D) $\frac{2T}{\rho L}$
10. A cube of mass $m = 3.2\text{kg}$ floats on the surface of water. Water wets it completely. The cube is 0.2m on each edge. By what additional distance it is buoyed up or down by surface tension? Surface tension of water = 0.07Nm^{-1} .
- A) $2.8 \times 10^{-4} m$ B) $1.4 \times 10^{-4} m$ C) $3.2 \times 10^{-6} m$ D) $6.1 \times 10^{-2} m +$

SECTION - II
(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 11 & 12

A glass capillary sealed at the upper end is of length 0.11 m and internal diameter $2 \times 10^{-5} \text{ m}$. The tube is immersed vertically into a liquid of surface tension $5.06 \times 10^{-2} \text{ N/m}$ (Assume complete wetting liquid)

11. To what length has the capillary to be immersed so that the liquid level inside and outside the capillary becomes the same.
- A) $0.01m$ B) $0.11m$ C) $0.02m$ D) $0.20m$
12. The length of water column inside the capillary tube when the seal is broken is
- A) $0.01m$ B) $0.10m$ C) $0.11m$ D) $1.01m$

Paragraph for Question Nos. 13 & 14

When liquid medicine of density ρ is to be put in the eye, it is done with the help of a dropper. As the bulb on the top of the dropper is pressed, a drop forms at the opening of the dropper. We wish to estimate the size of the drop.

We first assume that the drop formed at the opening is spherical because that requires a minimum increase in its surface energy. To determine the size, we calculate the net vertical force due to the surface tension T when the radius of the drop is R . When this force becomes smaller than the weight of the drop, the drop gets detached from the dropper.

13. If the radius of the opening of the dropper is r , the vertical force due to the surface tension on the drop of radius R (assuming $r \ll R$) is

A) $2\pi rT$ B) $2\pi RT$ C) $\frac{2\pi r^2T}{R}$ D) $\frac{2\pi R^2T}{r}$

14. If $r = 5 \times 10^{-4} m$, $\rho = 10^3 kg m^{-3}$, $g = 10 ms^{-2}$, $T = 0.11 Nm^{-1}$, the radius of the drop when it detaches from the dropper is approximately

A) $1.4 \times 10^{-3} m$ B) $3.3 \times 10^{-3} m$ C) $2.0 \times 10^{-3} m$ D) $4.1 \times 10^{-3} m$

Paragraph for Question Nos. 15 & 16

A long capillary tube of radius $0.2 mm$ is placed vertically inside a beaker of water

15. If the surface tension of water is $7 \times 10^{-2} N/m$ and the angle of contact between glass of water is zero, then determine the height of water column in the tube above the surface of water in beaker

A) $3 cm$ B) $9 cm$ C) $7 cm$ D) $5 cm$

16. If the tube is now pushed into water so that only $5.0 cm$ of its length is above the surface, then determine the angle of contact between the liquid and glass surface.

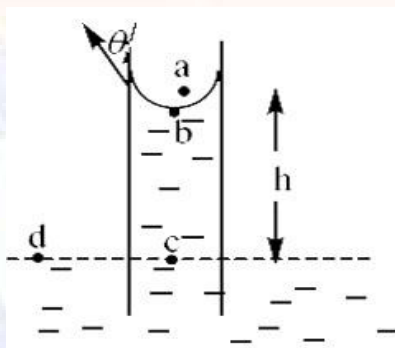
A) $\cos^{-1}\left(\frac{4}{5}\right)$ B) $\cos^{-1}\left(\frac{5}{7}\right)$ C) $\cos^{-1}\left(\frac{3}{5}\right)$ D) $\cos^{-1}\left(\frac{5}{4}\right)$

SECTION - III
(Matching List Type)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

17. One end of a capillary tube dipped into a liquid of surface tension T , as shown in Figure.



If the rise in liquid in capillary tube is ' h ' and angle of contact $\theta = 37^\circ$. If the points ' a ' and ' b ' are lying just above & just below the meniscus and point ' c ' and ' d ' are on the same horizontal level as shown in the Figure. The radius of the capillary tube is r , and atmospheric pressure is P_0 and ρ is the density of liquid Match Column – I & II based on above paragraph.

Column – I

I) P_a (Pressure at 'a')II) P_b (pressure at 'b')III) P_c (Pressure at 'c')IV) P_d (Pressure at 'd')

Column – II

P) P_0 Q) $P_0 - \frac{8T}{5r}$ R) ρgh S) $> \left(\rho gh - \frac{8T}{5r} \right)$ A) $I \rightarrow P, S; II \rightarrow Q; III \rightarrow P, S; IV \rightarrow P, S$ B) $I \rightarrow P; II \rightarrow Q, R, S; III \rightarrow P, S; IV \rightarrow S$ C) $I \rightarrow P; II \rightarrow R, S; III \rightarrow P, S; IV \rightarrow S$ D) $I \rightarrow P; II \rightarrow S; III \rightarrow P, S; IV \rightarrow S$

18. A capillary tube a glass of radius R is placed inside a tub containing water. Capillary rise is h . Assume that $2h$ length is outside the water. Column I contain conditions and Column – II contains the corresponding capillary rise. Contact angle for glass and water is 0° .

Column – I		Column – II	
I)	If the arrangement is placed in an elevator going up with acceleration $g/2$	P)	$3h/2$
II)	If the whole arrangement is falling freely	Q)	$2h$
III)	If a glass rod of radius $R/2$ is inserted symmetrically in the capillary tube	R)	$2h/3$
IV)	If water in tub is replaced by fluid having density double and surface tension 3 times the values for water keeping other things same.	S)	Water spills out of the capillary

A) $I \rightarrow R; II \rightarrow P; III \rightarrow Q; IV \rightarrow P$ B) $I \rightarrow R; II \rightarrow Q, S; III \rightarrow Q; IV \rightarrow P$ C) $I \rightarrow R; II \rightarrow Q; III \rightarrow P; IV \rightarrow P$ D) $I \rightarrow P; II \rightarrow R; III \rightarrow Q; IV \rightarrow P$

19. Changes made

Most probable effect of changes made on the surface tension of liquid

A	Increase in temperature of liquid	Q	decrease
B	Addition of soluble impurities to liquid	P	Increase
C	Addition of insoluble impurities to liquid	R	No change
D	Changing container in which liquid is there	S	Cannot predict probable effect on surface tension

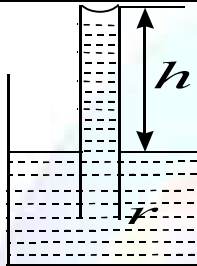
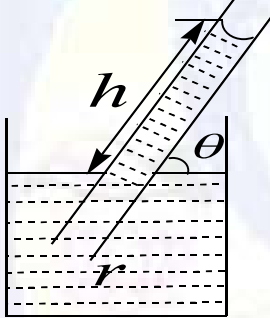
A) A-Q, B-P, C-R, D-S

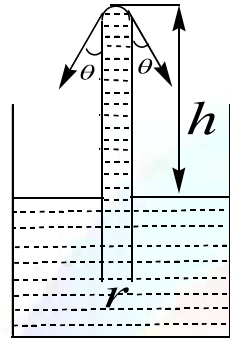
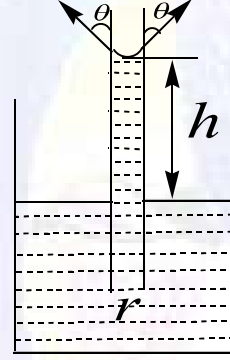
B) A-Q, B-Q, C-P, D-S

C) A-Q, B-P, C-Q, D-R

D) A-P, B-S, C-R, D-R

20. Assume angle of contact to be zero if not specified, r – radius of capillary
(T is surface tension) density is ρ

A) 	P) $h = \frac{2T}{r\rho g} \operatorname{cosec} \theta$
B) 	Q) $h = \frac{2T}{r\rho g}$

 C)	R) $h = \frac{2T \cos \theta}{r \rho g}$
 D)	S) Not possible

A) A – Q : B – P : C – S : D – R

B) A – P : B – Q : C – R : D – S

C) A – Q : B – P : C – R : D – S

D) A – S : B – R : C – P : D – Q

CHEMISTRY**Max Marks: 60****SECTION – I
(SINGLE CORRECT ANSWER TYPE)**

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

21. $(\text{CH}_3)_3 \text{CMgCl}$ on reaction with D_2O produces

- A) $(\text{CH}_3)_3 \text{CD}$ B) $(\text{CH}_3)_3 \text{COD}$ C) $(\text{CD}_3)_3 \text{CD}$ D) $(\text{CD}_3)_3 \text{COD}$

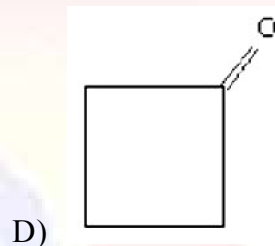
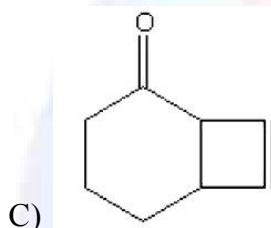
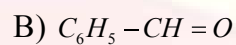
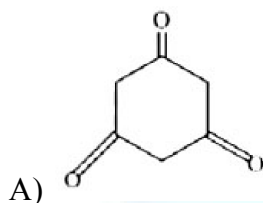
22. How many structural isomers (carbonyl compounds) of $\text{C}_5\text{H}_{10}\text{O}$ show ENOLIZATION?

- A) 4 B) 5 C) 6 D) 7

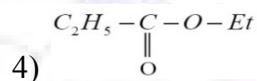
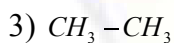
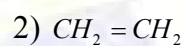
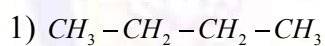
23. In which of the following compounds the methylene hydrogens are most acidic?

- A) $\text{CH}_3\text{COCH}_2\text{CH}_3$ B) $\text{CH}_3\text{CH}_2\text{COOEt}$
C) $\text{CH}_3\text{CH}_2\text{CH}(\text{COOEt})_2$ D) $\text{CH}_3\text{COCH}_2\text{CN}$

24. Tautomerism is not exhibited by:



25. An equimolar mixture of propanoic acid and KOH is electrolysed in presence of water. Which of these products are obtained.



A) 1,3

B) 1,2,3

C) 1,2,3,4

D) 3,4

26. A mixture of two organic compounds was treated with sodium metal in ether solution. Isobutane was obtained as one of chief product. The two chlorine compounds are
- A) methyl chloride and propyl chloride
 - B) methyl chloride and ethyl chloride
 - C) Isopropyl chloride and methyl chloride
 - D) Isopropyl chloride and ethyl chloride
27. Methane cannot be synthesised as chief product under which reaction
- 1. wurtz reaction
 - 2. Kolbe's reaction
 - 3. De- Carboxylation of salts of fatty acids
- A) 1,2 B) 2,3 C) 1,2,3 D) 1,3
28. When sodium acetate is heated with sodalime the reaction is called
- A) Dehydration B) Decarboxylation
 - C) Dehydrogenation D) Dehydrohalogenation
29. Ethane cannot be obtained by the following
- A) Heating methyl iodide with sodium metal in ether
 - B) Hydrogenation of ethene
 - C) Sodium acetate on Kolbe's electrolysis
 - D) Hydrolysis of Al_4C_3

30. Select the incorrect statement among the following
- A) alkyl halides give alkane with $n - \text{Bu}_3\text{SnH}$
 - B) Decarboxylation of formic acid gives methane
 - C) Methane is formed from decarboxylation of sodium acetate with NaOH/CaO
 - D) Grignard reagent gives alkane with ethyl alcohol.

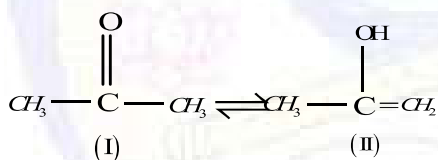
SECTION - II
(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

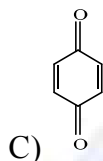
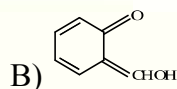
Paragraph for Question Nos. 31 & 32

Tautomerism is the phenomenon in which two structural isomers differing in the relative position of their atoms are spontaneously interconvertible and can exist in dynamic equilibrium.



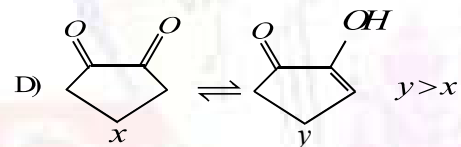
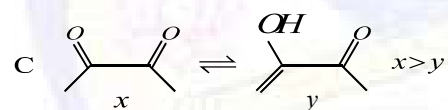
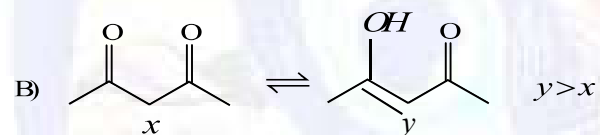
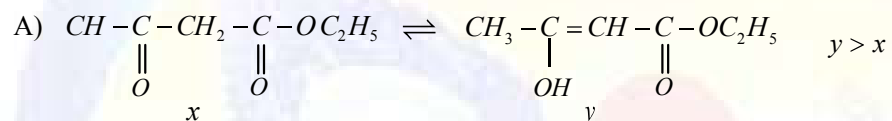
Tautomers I and II are structural isomers that are related only by the shift of a hydrogen atom and one or more π bond.

31. Identify the one which does not exhibit tautomerism



D) All of these

32. Which of the following is not correct? (x,y represent % of the two in equilibrium)



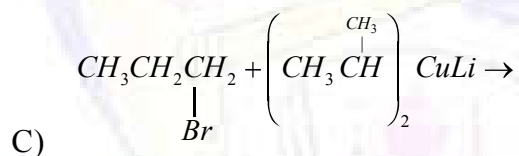
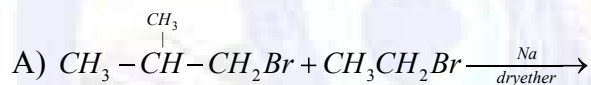
Paragraph for Question Nos. 33 & 34

Wurtz reaction: Alkyl halide reacts with metallic sodium in the presence of dry ether to form alkanes containing the double the number of C atoms present in the alkyl halide. Corey-House Synthesis: Dialkyl copper lithium reacts with Alkyl halide gives alkane

33. A mixture of ethyl iodide and methyl iodide is subjected to wurtz reaction the product which is not possible is

A) ethane B) butane C) propane D) 2-methyl propane

34. The most convenient method to prepare 2-methyl pentane among the following is

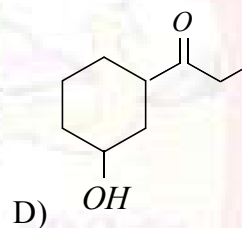
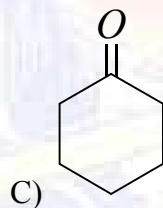
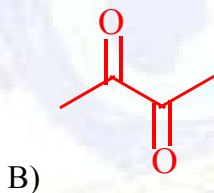
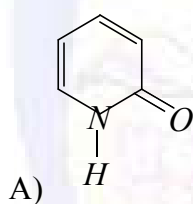


D) All of these

Paragraph for Question Nos. 35 & 36

Tautomerism: It is a phenomenon in which a single compound exists in two readily interconvertible structures that differs in the relative position of at least one atom (hydrogen). In simple aldehydes and ketones the amount of enolic form is negligibly small. However, the enolic form is stabilised by intramolecular H-bonding and some other factors.

35. In which of the following the enol form is dominant over keto form



36. The extent of enolization is greatly affected by

A) Solvent

B) Concentration

C) Temperature

D) All of these

SECTION - III
(Matching List Type)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

37. Match the following

Column I (Carbonyl compound)

Column II (Enol content %)

(A) $\text{PhCOCH}_2\text{COCH}_3$

(1) 6×10^{-5}

(B) $\text{EtOOCCH}_2\text{COOEt}$

(2) 2×10^{-1}

(C) $\text{NCCH}_2\text{COOEt}$

(3) 89

(D) CH_3CHO

(4) 7.7×10^{-3}

The correct combination is

A) A-3 B-4 C-2 D-1

B) A-3 B-4 C-1 D-2

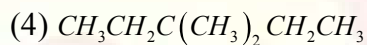
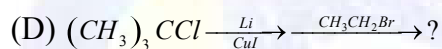
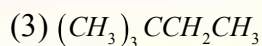
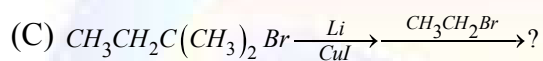
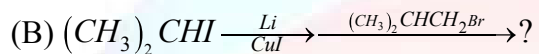
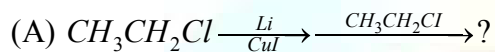
C) A-1 B-2 C-3 D-4

D) A-4 B-3 C-1 D-2

38. Match the following

Column I [Reaction]

Column II [Product]



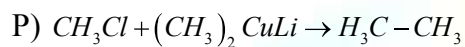
The correct match is

A) A-2 B-4 C-3 D-1 B) A-3 B-4 C-1 D-2

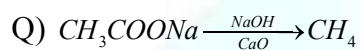
C) A-2 B-1 C-4 D-3 D) A-4 B-2 C-1 D-3

39. Column-I

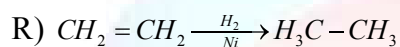
Column-II



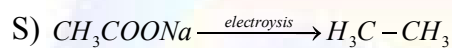
1) Free radical intermediate



2) Carbanion intermediate



3) Carbocation intermediate



4) No intermediate

A) P-4 Q-2 R-4 S-1

B) P-2 Q-1 R-4 S-3

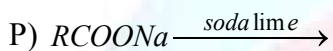
C) P-1 Q-3 R-2 S-4

D) P-4 Q-2 R-3 S-1

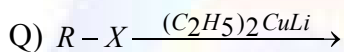
40. Preparation methods for alkanes is given in list –I. Some information about bond formation and cleavages of each preparation method is given in list II.

List – I

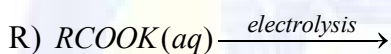
List - II



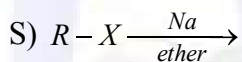
1) C – C bond formation



2) C – C bond cleavage



3) both C – C bond formation and cleavage



4) neither C – C bond formation nor cleavage

Correct Matching is

	P	Q	R	S
A)	2	1	3	1
C)	4	3	2	1

	P	Q	R	S
B)	3	4	2	1
D)	2	4	3	1

MATHEMATICS**Max Marks: 60****SECTION – I**
(SINGLE CORRECT ANSWER TYPE)

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

41. If L, M are the feet of the perpendiculars from (2, 4, 5) to xy - plane and yz – plane respectively, then distance LM is -----units
A) $9\sqrt{2}$ B) $2\sqrt{2}$ C) $\sqrt{29}$ D) $\sqrt{31}$
42. Let A (4, 7, 8), B (2, 3, 4) and C(2, 5, 7) be the vertices of ΔABC . The length of the median AD is-----units
A) $\sqrt{2}$ B) $\frac{\sqrt{2}}{2}$ C) $\frac{\sqrt{77}}{2}$ D) $\frac{\sqrt{89}}{2}$
43. If A(3, 2, 0), B(5, 3, 2), C(-9, 6, -3) are three points forming a triangle then the coordinates of a point in which the bisector of $\angle BAC$ meets BC are given by-----
A) $\left(\frac{88}{16}, \frac{57}{16}, \frac{27}{16}\right)$ B) $\left(\frac{38}{16}, \frac{57}{16}, \frac{17}{16}\right)$
C) $\left(\frac{-88}{16}, \frac{-57}{16}, \frac{-27}{16}\right)$ D) $\left(\frac{38}{16}, \frac{57}{16}, \frac{27}{16}\right)$

44. The lines $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{-1}$, $\frac{x-1}{2} = \frac{y}{1} = \frac{z+1}{4}$ are
A) parallel lines B) intersecting lines
C) perpendicular skew lines D) none of the above
45. If the d.c's l, m, n of two lines are connected by the relations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$, then the angle between the lines is
A) 30° B) 45° C) 90° D) 60°
46. The lines $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{\lambda}$ and $\frac{x}{1} = \frac{y}{2} = \frac{z+1}{-1}$ are
A) coplanar for all λ
B) coplanar for $\lambda = 19/3$
C) intersect at $\left(+\frac{1}{5}, -\frac{2}{5}, -\frac{4}{5}\right)$
D) intersect at $\left(\frac{1}{2}, -\frac{1}{2}, -1\right)$
47. The point on the line $\frac{x-2}{1} = \frac{y+3}{-2} = \frac{z+5}{-2}$ at a distance of 6 from the point $(2, -3, -5)$ is
A) $(3, -5, -3)$ B) $(4, -7, -9)$ C) $(0, 2, -1)$ D) $(-3, 5, 3)$

48. If lines $x = y = z$ and $x = \frac{y}{2} = \frac{z}{3}$, and third line passing through $(1, 1, 1)$ from a triangle of area $\sqrt{6}$ units, then point of intersection of third line with second line can be
- A) $(1, 2, 3)$ B) $(2, 4, 6)$ C) $\left(\frac{4}{3}, \frac{8}{3}, \frac{12}{3}\right)$ D) $(0, 2, 3)$
49. If a variable line in two adjacent positions has directional cosines l, m, n and $l + \delta l, m + \delta m, n + \delta n$ and $\delta\theta$ is small angle between two positions then $(\delta l)^2 + (\delta m)^2 + (\delta n)^2 =$
- A) $(\delta\theta)^2$
B) $3(\delta\theta)^2$
C) $2(\delta\theta)^2$
D) constant that does not depend on $\delta\theta$
50. The point of intersection of the lines $\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1}$ and $\frac{x+3}{-36} = \frac{y-3}{2} = \frac{z-6}{4}$ is
- A) $\left(21, \frac{5}{3}, \frac{10}{3}\right)$ B) $(2, 10, 4)$ C) $(-3, 3, 6)$ D) $(5, 7, -2)$

SECTION - II
(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct. **Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

Paragraph for Question Nos. 51 & 52

A line L_1 with direction ratios -3, 2, 4 passes through the point A(7, 6, 2) and a line L_2 with direction ratios 2, 1, 3 passes through the point B(5, 3, 4). A line L_3 with direction ratios 2, -2, -1 intersects L_1 and L_2 at C and D.

51. The length CD is equal to

- A) 4 B) 6 C) 9 D) 11

52. Centroid of $\triangle BCD$ is -----

- A) $\left(\frac{13}{3}, \frac{4}{3}, \frac{7}{3}\right)$ B) $\left(\frac{2}{3}, \frac{1}{3}, 7\right)$ C) $\left(\frac{7}{3}, 12, \frac{5}{3}\right)$ D) $\left(\frac{13}{3}, \frac{17}{3}, 7\right)$

Paragraph for Question Nos. 53 & 54

In a parallelogram OABC with position vectors of A is $3\hat{i} + 4\hat{j}$ and C is $4\hat{i} + 3\hat{j}$ with reference to O as origin. A point E is taken on the side $\overline{BC}$ which divides it in the ratio of 2: 1. Also, the line segment AE intersects the line bisecting the $\angle AOC$ internally at P.

53. The position vector of P is

- A) $\hat{i} + \hat{j}$ B) $\frac{2}{3}(\hat{i} + \hat{j})$ C) $\frac{13}{3}(\hat{i} + \hat{j})$ D) $\frac{21}{5}(\hat{i} + \hat{j})$

54. The equation of line parallel to $\overline{CP}$ and passing through (2,3,4) is

- A) $\frac{x-2}{1} = \frac{y-3}{5} = \frac{z-4}{0}$ B) $\frac{x-2}{1} = \frac{y-3}{6} = \frac{z-4}{0}$
C) $\frac{x-2}{2} = \frac{y-3}{5} = \frac{z-3}{0}$ D) $\frac{x-2}{3} = \frac{y-3}{5} = \frac{z-3}{0}$

Paragraph for Question Nos. 55 & 56

Consider a triangular pyramid ABCD the position vectors of whose angular points are $A(3,0,1)$, $B(-1,4,1)$, $C(5,2,3)$ and $D(0,-5,4)$. Let G be the point of intersection of the medians of the $\triangle BCD$

55. The length of the vector $\overline{AG}$ is----- units

- A) $\sqrt{17}$ B) $\sqrt{51}/3$ C) $\sqrt{51}/9$ D) $\sqrt{59}/4$

56. Area of the $\triangle ABC$ (in sq units) is

- A) 24 B) $8\sqrt{6}$ C) $4\sqrt{6}$ D) None of these

SSECTION-3
(MATCHING LIST TYPE)

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

57. Match the column I with column II

Column I	Column II
(P) The co-ordinates of a point on the line $\frac{x-5}{4} = \frac{y-0}{1} = \frac{z+6}{3}$ at a distance 3 from the point (5, 3, -6) is	(1) (-1, -2, 0)
(Q) The point of intersection of the lines $\frac{x+2}{1} = \frac{y+5}{3} = \frac{z+1}{1}$ and $\frac{x}{1} = \frac{y-2}{4} = \frac{z-3}{3}$ is	(2) (5, 0, -6)
(R) A line passes through two points A(2, -3, -1) and B(8, -1, 2). The co-ordinates of a point on this line nearer to the origin at a distance of 14 units from A is	(3) (2, 5, 7)
(S) The co-ordinates of the foot of the perpendicular from the point (3, -1, 11) on the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is	(4) (-10, -7, -7)

CODE

	P	Q	R	S		P	Q	R	S
A)	2	1	4	3	B)	4	1	3	2
C)	3	2	4	1	D)	1	4	2	3

58. Consider the lines given by $L_1 : x + 3y - 5 = 0$, $L_2 : 3x - ky - 1 = 0$ and

$L_3 : 5x + 2y - 12 = 0$ in xy plane of 3dimensional space

	Column – I		Column – II
P	L_1, L_2 and L_3 are concurrent, iff	1	$k = -9, -\frac{6}{5}, 5$
Q	One of L_1, L_2 and L_3 is parallel to atleast one of the other two, iff	2	$k = -\frac{6}{5}, -9$
R	L_1, L_2 and L_3 form a triangle for $k = \text{-----}(k < 1)$	3	$k = \frac{5}{6}$
S	L_1, L_2 and L_3 do not form a triangle for $k = \text{-----}(k < 0)$	4	$k = 5$

CODE

	P	Q	R	S		P	Q	R	S
A)	2	1	4	3	B)	4	2	3	2
C)	3	2	4	1	D)	1	4	2	3

59. A line L cuts 4 main diagonals L_1, L_2, L_3 and L_4 of a unit cube at angles $\theta_1, \theta_2, \theta_3$ and θ_4 respectively. Then Match the following

List – A	List – B
P) Acute angle between L_1 and L_3 is α . Then $\cos \alpha =$	1) $\frac{1}{3}$
Q) $\sum_{k=1}^4 \sin^2 \theta_k =$ ____	2) $\frac{16}{3}$
R) $\sum_{k=1}^4 \cos^2 \theta_k =$ ____	3) $\frac{4}{3}$
S) Acute angle between L_2 and L_4 is β . Then $6 \sin^2 \beta =$ ____	4) $\frac{8}{3}$

CODE

	P	Q	R	S
A)	2	1	4	3
B)	4	1	3	2
C)	3	2	4	1
D)	1	4	3	2

60. l, m, n are directional cosines of a line in 3 – dimensional space, Then Match the following

List – A	List – B
P) Maximum value of $54l^2m^2n^2 = \underline{\hspace{1cm}}$	1) 4
Q) $(l+m-n)^2 + (m+n-l)^2 + (n+l-m)^2 + (l+m+n)^2 = \underline{\hspace{1cm}}$	2) 3
R) Maximum value of $3(lm+mn+nl)$ is $\underline{\hspace{1cm}}$	3) 2
S) Maximum value of $ 4l+4m+2n $ is $\underline{\hspace{1cm}}$	4) 6

CODE

	P	Q	R	S
A)	2	1	4	3
B)	4	1	3	2
C)	3	1	2	4
D)	1	4	2	3



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

2014_P2

Date: 25-11-17

Max.Marks:180

KEY SHEET

PHYSICS

1	B	2	B	3	B	4	C	5	C
6	A	7	D	8	A	9	D	10	B
11	A	12	C	13	C	14	A	15	C
16	B	17	A	18	B	19	C	20	A

CHEMISTRY

21	A	22	C	23	D	24	B	25	C
26	C	27	A	28	B	29	D	30	B
31	C	32	A	33	D	34	C	35	A
36	D	37	A	38	C	39	A	40	A

MATHS

41	C	42	C	43	B	44	C	45	D
46	B	47	B	48	B	49	A	50	A
51	C	52	D	53	D	54	B	55	B
56	C	57	A	58	B	59	D	60	C

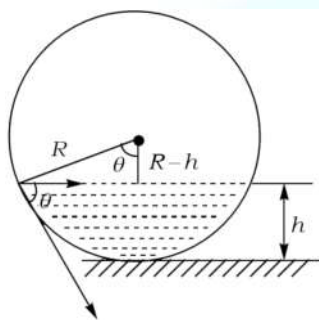
SOLUTIONS
PHYSICS

1. $2F = 8rT$

$$r = \frac{2l}{\pi}$$

2. Conceptual

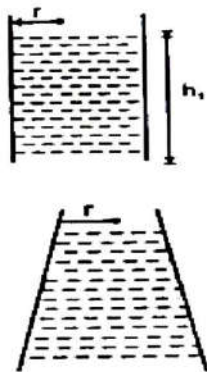
3.



$$\cos \theta = \frac{R-h}{R}$$

From the Figure

4. (3) As weight of liquid in capillary is balanced by surface, then $T \times 2\pi r = \pi^2 h_1 \rho g$ (From uniform r radius tube) $h_1 = \frac{2T}{r\rho g}$



But weight of liquid in tapered tube is more than uniform tube of radius r , then order

to balance $h < h_1$ $h < \frac{2T}{r\rho g}$

5. The bubble will detach if, Buoyant force $\geq$ surface tension force

$$\int \sin \theta T \times dl = T(2\pi r) \sin \theta$$

$$\frac{4}{3} \pi R^3 \rho_w g \geq \int T \times dl \sin \theta$$

$$(\rho_w) \left(\frac{4}{3} \pi R^3 \right) g \leq (T)(2\pi r) \sin \theta$$

$$\Rightarrow \sin \theta = \frac{r}{R}$$

$$\text{Solving, } r = \sqrt{\frac{2\rho_w R^4 g}{3T}} = R^2 \sqrt{\frac{2\rho_w g}{3T}}$$

6. $\frac{\Delta T}{T} = \frac{\Delta d}{d} + \frac{\Delta h}{h}$

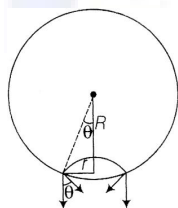
7. Using geometry

$$\frac{b}{R} = \cos \left(\theta + \frac{\alpha}{2} \right)$$

Using pressure equation

$$p_0 - \frac{25}{R} + h\rho g = p_0$$

8. The bubble will detach if,



$$\int \sin \theta T \times dl = T(2\pi r) \sin \theta$$

Buoyant force $\geq$ Surface tension force

$$\frac{4}{3} \pi R^3 \rho_w g \geq \int T \times dl \sin \theta$$

$$(\rho_w) \left(\frac{4}{3} \pi R^3 \right) g \geq (T)(2\pi r) \sin \theta$$

$$\Rightarrow \sin \theta = \frac{r}{R}$$

$$\text{Solving } r = \sqrt{\frac{2\rho_w R^4 g}{3T}}$$

$$= R^2 \sqrt{\frac{2\rho_w g}{3T}}$$

No option matches with the correct answer.

9. Conceptual

10. If surface tension is neglected, by the law of floatation $mg = (a^2 x \rho)g$ where a is the side the density of water and m is the mass of cube.

$$3.2g = 0.2^2 \times (10^3)g$$

$$\text{The height to which the cube is immersed } x = \frac{3.2}{(0.2)^2 \times 10^3} = 0.08m$$

Since water wets the cube, the angle of contact is zero and the force of surface tension acts vertically downwards. So its is buoyed down the surface tension $mg + 4aT^2 x' \rho g$,

where T is surface tension of water $3.2g + 4(0.2)(0.07) = (0.2)^2 x' (10^3)g$

$$x' = 0.8 + \frac{4(0.2)(0.7)}{(0.2)^2 (10^3)(0.8)} = (0.08 + 1.4 \times 10^{-4})m$$

The additional distance it is buoyed down by surface tension $= 1.4 \times 10^{-4}m$

11. $P_2 - \frac{2T}{r} = P_0 \text{ ie } P_2 = P_0 + \frac{2T}{r}$

If temperature is constant

$$P_1 v_1 = P_2 v_2$$

$$P_0 AL = \left[P_0 + \frac{2T}{r} \right] A(L - x) \Rightarrow x = \frac{0.11}{11} = 0.01m$$

12. If the seal is broken pressure inside the capillary will become atmospheric ie. P_0 in

$$\text{raise in the height } h = \frac{2T}{r \rho g} = \frac{2 \times 5.06 \times 10^{-2}}{10^{-5} \times 10^3} = 1.03m$$

However the length of the tube outside water is $0.11 - 0.01 = 0.1m$

Liquid will rise to top of the tube

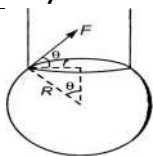
13. Vertical force due to surface tension,

$$F_v = F \sin \theta = (T 2\pi r)(r/R)$$

$$= \frac{2\pi r^2 T}{R}$$

$\therefore$ Correct option is (c)

14. $\frac{2\pi r^2 T}{R} = mg = \frac{4}{3} \pi R^3 \cdot \rho \cdot g$



$$\therefore R^4 = \frac{3r^2 T}{2\rho g} = \frac{3 \times (5 \times 10^{-4})^2 (0.11)}{2 \times 10^3 \times 10}$$

$$= 4.125 \times 10^{-12} m^4$$

$$\therefore R = 1.425 \times 10^{-3} m = 1.4 \times 10^{-3} m$$

$\therefore$ Correct option is (a)

15. Conceptual

16. Conceptual

$$17. R = \frac{r}{\cos \theta} = \frac{5r}{4}$$

$$P_d = P_c = P_a = P_0$$

$$P_a - \frac{2T}{R} = P_b$$

$$18. h = \frac{2T}{r\rho g_{eff}}$$

If $g = 0$ tube has in sufficient length

$$2\pi(r_1 + r_2)T = \pi(r_2^2 - r_1^2)\rho hg$$

19. Conceptual

20. Conceptual

CHEMISTRY

21. $\text{RMgX} + \text{Acidic H} \longrightarrow \text{R-H}$

22. aldehydes----- THREE Ketones-----THREE

23. $\text{CH}_3\text{COCH}_2\text{CN}$

24. (b) no α -H's

25. 1,2,3,4 are possible

26. $(\text{CH}_3)_2\text{CHCl} + \text{CH}_3\text{Cl} \xrightarrow{\text{Na}} (\text{CH}_3)_2\text{CH}-\text{CH}_3$

27. CH_4 can't be prepared by wurtz and kolbe's reaction

28. De carboxylation

29. Al_4C_3 given CH_4

30. (b) given H_2
31. $C_6H_5CH=CH-OH \rightleftharpoons C_6H_5-CH_2-CH=O$
32. In acetoacetic ester keto form predominates end so $x > y$
33. ethane, butane and propane are formed
34. Wurtz reaction is not suitable for the preparation of unsymmetrical alkanes
35. Enol has aromatic character
36. solvent, concentration and temperature
37. $PhCOCH_2COCH_3$ -----89 $EtOOCCH_2COOEt$ ----- 7.7×10^{-3}
 $NCCH_2COOEt$ ----- 2×10^{-1} CH_3CHO ----- 6×10^{-5}
38. Corey-House synthesis
39. Conceptual
40. P-Decarboxylation Q-Corey House R-Kolbe's S-Wurtz

MATHS

41. $L = (2, 4, 0)$ $M = (0, 4, 5)$ $LM = \sqrt{29}$
42. $D = (2, 4, 11/2)$ $AD = \sqrt{\frac{77}{2}}$
43. Use $BD : DC = c : b$
44. Conceptual
45. Eliminate n and solve for dc's. We get $\frac{l}{1} = \frac{m}{0} = \frac{n}{-1}$ or $\frac{l}{0} = \frac{m}{1} = \frac{n}{-1} \therefore \theta = 60^\circ$
46. Use the condition for coplanarity $\begin{vmatrix} 1 & -1 & 4 \\ 2 & -1 & \lambda \\ 1 & 2 & -1 \end{vmatrix} = 0 \Rightarrow \lambda = \frac{19}{3}$
47. Required point $P = (r+2, -2r-3, -2r-5)$ and $r = 2$
 $\therefore P = (4, -7, -9)$
48. Verify from options the given data
49. Use $l^2 + m^2 + n^2 = (l + \delta l)^2 + (m + \delta m)^2 + (n + \delta n)^2 = 1$
50. $(3\lambda + 5, -\lambda + 7, \lambda - 2) = (-36\mu - 3, 2\mu + 3, 4\mu + 6)$ solving for λ & μ we have
 $\lambda = \frac{16}{3}, \mu = \frac{-2}{3}$
 $\therefore$ Required point $= (21, 5/3, 10/3)$
- 51&52. $L_1 = \frac{x-7}{-3} = \frac{y-6}{2} = \frac{z-2}{4}$ and $L_2 = \frac{x-5}{2} = \frac{y-3}{1} = \frac{z-4}{3}$

$$\therefore C(-3\lambda + 7, 2\lambda + 6, 4\lambda + 2) \text{ and } D(2\mu + 5, \mu + 3, 3\mu + 4)$$

$$\text{So } \frac{2-3\lambda-2\mu}{2} = \frac{3+2\lambda-\mu}{-2} = \frac{-2+4\lambda-3\mu}{-1}$$

$$\therefore \lambda = 2, \mu = 1$$

$$\text{So } C(1, 10, 10) \text{ and } D(7, 4, 7) \Rightarrow l(CD) = 9$$

$$53. \quad \overline{OB} = 7\hat{i} + 7\hat{j}, \overline{OE} = 5\hat{i} + \frac{13}{3}\hat{j}, \overline{OP} = \frac{21}{5}(\hat{i} + \hat{j})$$

54. Direction ratio of CP is (1, 6, 0) then equation of line passing through (2, 3, 4) and

$$\text{parallel to CP is } \frac{x-2}{1} = \frac{y-3}{6} = \frac{z-4}{0}$$

55&56. Conceptual

57. (A) The given line is $x = 4y + 5, z = 3y - 6$,

$$\text{or } \frac{x-5}{4} = y, \frac{z+6}{3} = y \text{ or } \frac{x-5}{4} = \frac{y}{1} = \frac{z+6}{3} = \lambda \text{ (say)}$$

any point on this line is of the form of $(4\lambda + 5, \lambda, 3\lambda - 6)$.

the distance between $(4\lambda + 5, \lambda, 3\lambda - 6)$ and $(5, 3, -6)$ is 3 units (given). Therefore

$$(4\lambda + 5 - 5)^2 + (\lambda - 3)^2 + (3\lambda - 6 + 6)^2 = 9 \Rightarrow \lambda = 0, \frac{3}{13}.$$

so point is $(5, 0, -6)$

(B) The equation of the required plane is

$$\begin{vmatrix} x-2 & y+3 & z+5 \\ 1 & 4 & 7 \\ 3 & 5 & 7 \end{vmatrix} = 0 \Rightarrow x - 2y + z - 3 = 0$$

point $(-1, -2, 0)$ lies on this plane.

(C) The line passing through points $A(2, -3, -1)$ and $B(8, -1, 2)$ is

$$\frac{x-2}{8-2} = \frac{y+3}{-1+3} = \frac{z+1}{2+1} \text{ or } \frac{x-2}{6} = \frac{y+3}{2} = \frac{z+1}{3} = \lambda \text{ (say)}$$

any point on this line is of the form $P(6\lambda + 2, 2\lambda - 3, 3\lambda - 1)$ whose distance from point

$A(2, -3, -1)$ is 14 units. So $PA = 14, (PA)^2 = (14)^2 \Rightarrow (6\lambda)^2 + (2\lambda)^2 + (3\lambda)^2 = 196$

$$\Rightarrow \lambda^2 = 4 \Rightarrow \lambda = \pm 2.$$

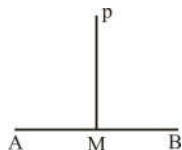
The required points are $(14, 1, 5)$ and $(-10, -7, -7)$.

The point nearer to origin is $(14, 1, 5)$.

(D) Any point on line AB, $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4} = \lambda$ is

$M(2\lambda, 3\lambda + 2, 4\lambda + 3)$.

Therefore the direction ratios of PM are $2\lambda - 3, 3\lambda + 3$ and $4\lambda - 8$



$$\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$$

$$\therefore 2(2\lambda - 3) + 3(3\lambda + 3) + 4(4\lambda - 8) = 0 \Rightarrow \lambda = 1$$

so foot of the perpendicular is $M(2, 5, 7)$

58,59,60. Conceptual

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

A.P., TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr.Super60-I

Time: 07:30 AM to 10:30 AM

Date: 02-12-17

Max. Marks: 210

Name of the Student: _____

H.T. NO:

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2012_PAPER-I

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

02-12-17_Jr.Super60-I_Jee-Adv_WAT-27_WEEKEND SYLLABUS

MATHS: SYLLABUS COVERED FROM 20-11-17 TO 01-12-2017

PHYSICS: SYLLABUS COVERED FROM 20-11-17 TO 01-12-2017

CHEMISTRY: Physical properties of alkanes , reactions of alkanes, Mechanism of Free radical Substitution, Preparation, Mechanism of Elimination reactions (E1, E2, E1CB) (Including Hoffman elimination and pyrolysis of ester

IIT-JEE-2012-P1-Model

Time: 07:30 A.M to 10:30A.M

IMPORTANT INSTRUCTIONS

Max Marks: 210

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 10)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N : 11 – 15)	Questions with Multiple Correct Choice	4	0	5	20
Sec – III(Q.N : 16 – 20)	Questions with Integer Answer Type	4	0	5	20
Total				20	70

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 21 – 30)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N : 31 – 35)	Questions with Multiple Correct Choice	4	0	5	20
Sec – III(Q.N : 36 – 40)	Questions with Integer Answer Type	4	0	5	20
Total				20	70

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 41 – 50)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N : 51 – 55)	Questions with Multiple Correct Choice	4	0	5	20
Sec – III(Q.N : 56 – 60)	Questions with Integer Answer Type	4	0	5	20
Total				20	70

Sec: Jr.Super60-I

space for rough work Page

Page 2

PHYSICS

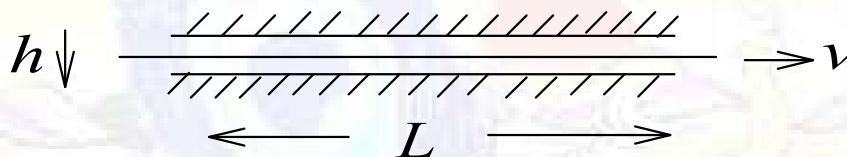
Max.Marks:70

SECTION I

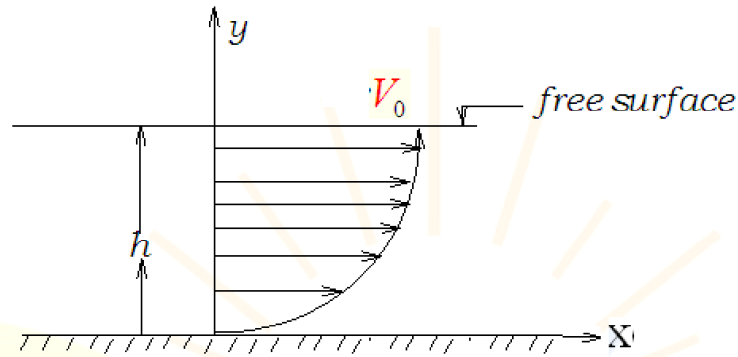
Single Correct Answer Type

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. During a painting process, a thin, flat tape of width b [dimension perpendicular to the plane of the figure] is pulled through a paint filled channel of length L . The density and viscosity of the paint liquid is ρ and η respectively. The tape is pulled at constant speed v and width of the channel is h . Find the minimum force needed to pull the tape.



- A) $\frac{2\eta bLv}{h}$ B) $\frac{4\eta bLv}{h}$ C) $\frac{8\eta bLv}{h}$ D) None of these
2. A liquid is flowing through a horizontal channel. The speed of the flow (V) depends on height (y) from the floor as $V = V_0 \left[2\left(\frac{y}{h}\right) - \left(\frac{y}{h}\right)^2 \right]$ where V_0 is the speed of the top layer. Co efficient of viscosity is η . Find the shear stress that liquid exerts on the floor.

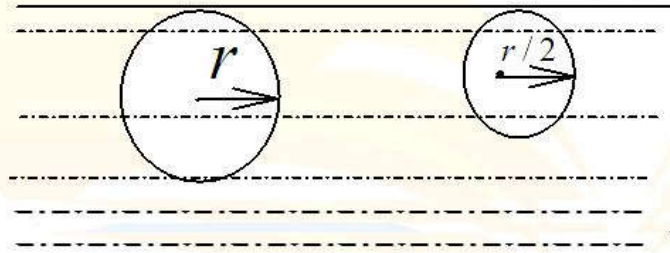


- A) Zero B) $\frac{3\eta V_0}{2h}$ C) $\frac{2\eta V_0}{h}$ D) None of these

3. A car having cross-sectional area of its front equal to A is travelling on a highway at a speed V . The viscous drag force acting on the car is known to be given by $F_v = CA\rho V^2$, where ρ is density of air and C is a constant which depends on the shape of the car. The petrol used by the car produces E joules of energy per kg of its burnt. Calculate the mileage (in km/kg) of the car if the combined efficiency of its engine and transmission is f .

- A) $\frac{fE}{CA\rho V}$ B) $\frac{fE}{CA\rho V^2}$ C) $\frac{CV^2}{A\rho f}$ D) $\frac{C\rho}{AV^2E}$

4. Two balls of radii r and $\frac{r}{2}$ are released inside a deep water tank. Their initial accelerations are found to be $\frac{g}{2}$ and $\frac{g}{4}$ respectively. Find the velocity of smaller ball relative to the larger ball, a long time after the two balls are released. Coefficient of viscosity is η and density of water is ρ .

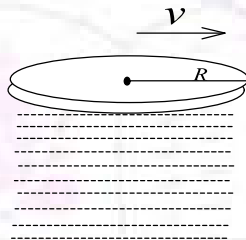


- A) $\frac{11}{54} \frac{r^2 \rho g}{\eta}$ downwards B) $\frac{11}{54} \frac{r^2 \rho g}{\eta}$ upwards
 C) $\frac{5}{27} \frac{r^2 \rho g}{\eta}$ downwards D) $\frac{5}{27} \frac{r^2 \rho g}{\eta}$ upwards
5. A ball of mass ' m ' and density ρ is being released inside a fluid of density $\frac{\rho}{2}$ and coefficient of viscosity η . Find the net force on the ball when speed achieved by ball is 60% of the terminal speed.
- A) $0.6 mg$ B) mg C) $0.4 mg$ D) $0.2 mg$

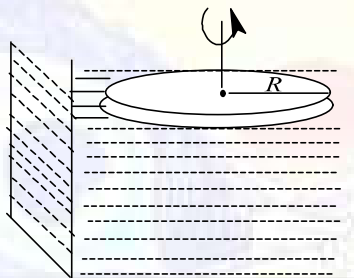
6. A marble of mass x and diameter $2r$ is gently released in a tall cylinder containing honey. If the marble displaces mass $y (< x)$ of the liquid, the terminal velocity is proportional to
- A) $x+y$ B) $x-y$ C) $\frac{x+y}{r}$ D) $\frac{x-y}{r}$
7. The rising smoke of a cigarette in the still air of a room illustrates which type of flow?
- A) Stream line near cigarette
B) Vortices indicating onset of a turbulence at 0.3m or above the cigarette
C) Both a and b
D) only turbulent flow

PARAGRAPH (8, 9 & 10):-

Consider a disk of mass M , radius R lying on a liquid layer of thickness T and coefficient of viscosity η as shown in figure.



8. The coefficient of viscosity varies as $\eta = \eta_0 x$ (x is measured along velocity line from centre of the disk). At the given instant the disk is floating towards right with a velocity v as shown. Find the force required to move the disk slowly at the given instant.
- A) $\frac{4\eta_0 R^3 v}{3T}$ B) $\frac{2\eta_0 R^3 v}{3T}$ C) $\frac{2\pi\eta_0 R^3 v}{3T}$ D) $\frac{16\eta_0 R^3 v}{T}$
9. Find the torque required to rotate the disk at a constant angular velocity ω , given the viscosity is uniformly η .

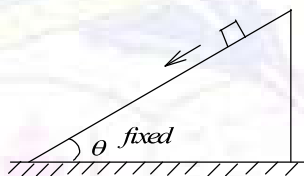


- A) $\frac{4\pi\omega\eta R^4}{T}$ B) $\frac{\pi\omega\eta R^4}{2T}$ C) $\frac{2\pi\omega\eta R^4}{T}$ D) $16\pi\omega\eta R^4$
10. A disk rotating with angular velocity ω_0 is placed on the viscous liquid. Find the angle rotated by the disc before it comes to rest. (Viscosity= η , mass of disc= M , radius of disc= R)
- A) $\frac{4\omega_0 TM}{\eta\pi R^2}$ B) $\frac{2\omega_0 TM}{\eta\pi R^2}$ C) $\frac{\omega_0 TM}{\eta\pi R^2}$ D) $\frac{\omega_0 TM}{2\eta\pi R^2}$

SECTION II**Multiple Correct Answer(s) Type**

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

11. An oil drop falls through air with a terminal velocity of $5 \times 10^{-4} \text{ m/s}$. Viscosity of air is $1.8 \times 10^{-5} \text{ Ns/m}^2$ and density of oil is 900 kg/m^3 . Neglect density of air as compared to that of oil.
- A) The radius of drop is $4.18 \times 10^{-6} \text{ m}$
 B) The radius of drop is $2.14 \times 10^{-6} \text{ m}$
 C) Terminal velocity of a drop of half of this radius is $1.25 \times 10^{-4} \text{ m}$
 D) Terminal velocity of a drop of half of this radius is $2.5 \times 10^{-4} \text{ m}$
12. A cubical block of side a and density ρ slides over a fixed inclined plane with constant velocity v . There is thin film of viscous fluid of thickness t between the plane and the block. Then, the coefficient of viscosity of the thin film cannot be



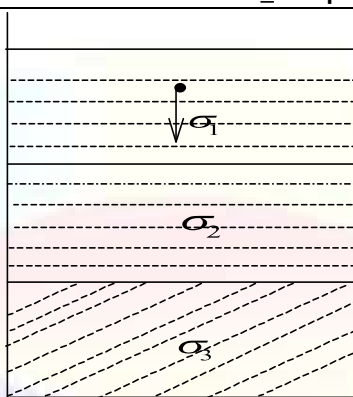
A) $\frac{\rho a g t \sin \theta}{v}$

B) $\frac{\rho a g t}{v \sin \theta}$

C) $\frac{\rho a g t \sin \theta}{\sqrt{2} v}$

D) $\frac{\rho a g t}{\sqrt{2} v \sin \theta}$

13. A tube of length l and radius R carries a steady flow of fluid whose density is ρ and viscosity η . The velocity v of flow is given by $v = v_0(1 - r^2/R^2)$, Where r is the distance of flowing fluid from the axis.
- A) The volume of fluid, flowing across the section of the tube, in unit time $2\pi v_0(R^2/4)$
- B) The Kinetic energy of the fluid within the volume of the tube is $K.E = \pi\rho l v_0^2(R^2/6)$
- C) The frictional force exerted on the tube by the fluid is $F = 4\pi\eta l v_0$
- D) The pressure difference at the ends of tube is $p = \frac{4\eta l v_0}{R^2}$
14. viscous force is somewhat like friction as it opposes the relative motion and is non-conservative but not exactly so, because;
- A) It is velocity dependent while friction not
- B) It is velocity independent while friction is not
- C) It is dependent on area of contact while friction is not
- D) It is independent of area like surface tension while friction depends.
15. A ball moves successively through three liquids, at rest as shown, of densities σ_1, σ_2 and σ_3 and viscosity coefficient η_1, η_2, η_3 and respectively with the same terminal velocity. Then



A) $\eta_3 > \eta_2 < \eta_1$

B) $\frac{\sigma_1}{\eta_1} = \frac{\sigma_2}{\eta_2} = \frac{\sigma_3}{\eta_3}$

C) $\frac{\eta_1}{\eta_3} > \frac{\eta_3}{\eta_2}$

D) $\frac{\eta_2\sigma_1 - \eta_1\sigma_2}{\eta_3\sigma_1 - \eta_1\sigma_3} = \frac{\eta_2 - \eta_1}{\eta_3 - \eta_1}$

SECTION- III

Integer Answer Type

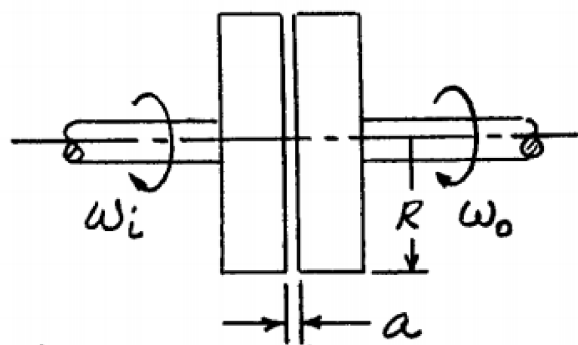
This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

16. A spherical ball of radius r and density d is dropped from rest in a viscous, fluid having density ρ and coefficient of viscosity is η . Then the power of gravitational

force acting on the ball at a time t after it is dropped is $\frac{x\pi}{27} \frac{d(d-\rho)}{\eta} g^2 r^5 \left[1 - e^{-\frac{9\eta t}{2dr^2}} \right]$.

Find x .

17. A plate of area $2m^2$ is made to move horizontally with a speed of $2ms^{-1}$ by applying a horizontal tangential force over the free surface of a liquid. The depth of the liquid is 1 m and the liquid in contact with the bed is stationary. Coefficient of viscosity of liquid = 0.01 poise. The tangential force needed to move the plate is $\alpha \times 10^{-3}$ N find the value of α
18. A viscous clutch as shown in figure transmits torque. Radius of each clutch plate is R and separation between the plates is a and is completely filled with liquid of coefficient of viscosity μ , If ω_i and ω_o are angular velocities of plates connected to input and output respectively. The torque transmitted is $\frac{\pi \mu (\omega_i - \omega_o) R^4}{xa}$ find x?



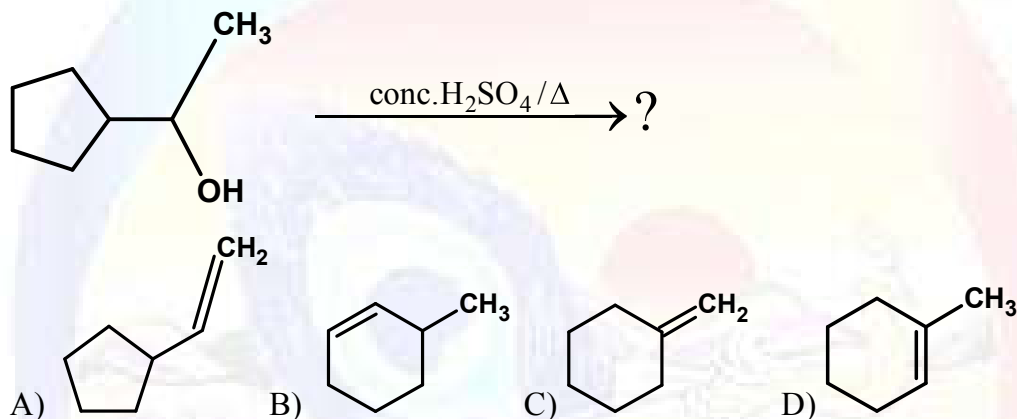
19. Eight equal drops of water each of radius $r=2\text{mm}$ are falling through air with the terminal velocity of 16cm/s . The eight drops combine to form a big drop. The terminal velocity of the big drop is $16 \times N \text{ cm/s}$, find N .
20. A small spherical ball falling under gravity in a viscous medium heat the medium due to viscous drage force. The rate of heating is proportional to r^n . (r =radius of the sphere). Find n .

SECTION I

Single Correct Answer Type

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

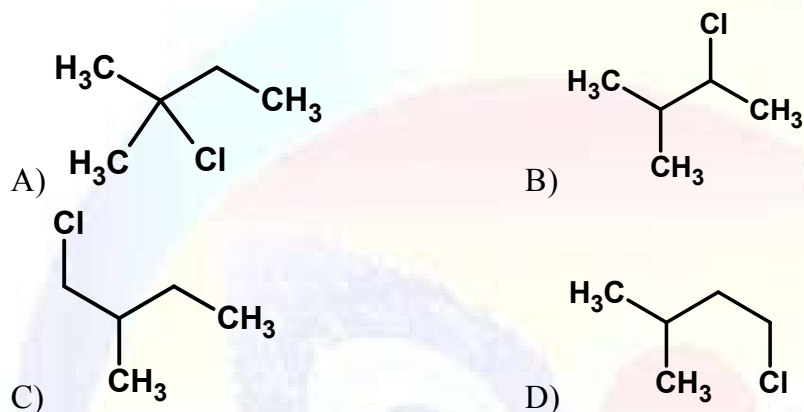
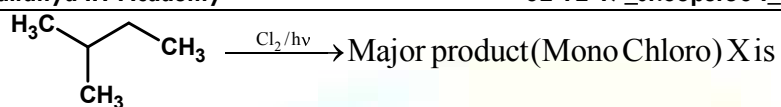
21. The major product of the following reaction is



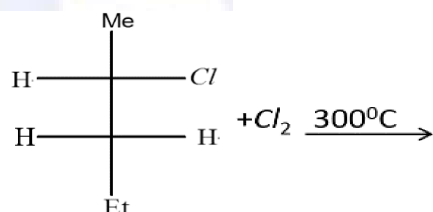
22. Amongst the isomeric pentanes the decreasing order of boiling points is

- A) Neopentane > n-pentane > isopentane
B) isopentane > neopentane > n-pentane
C) n-pentane > isopentane > neopentane
D) n-pentane > neopentane > isopentane

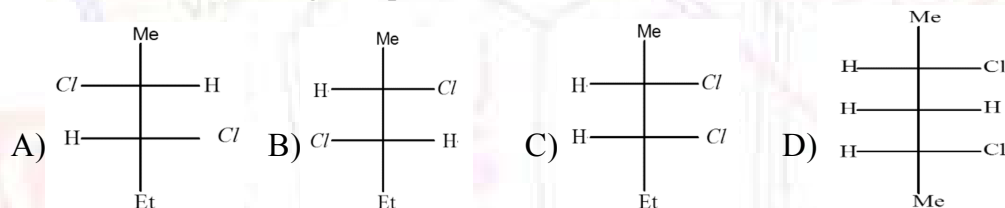
23.



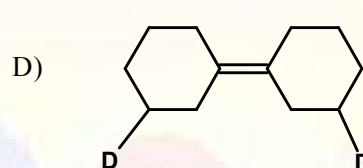
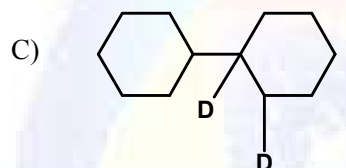
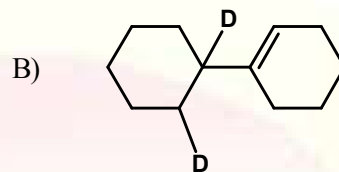
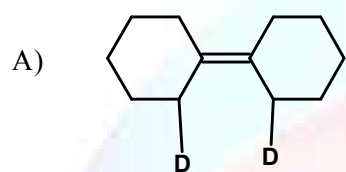
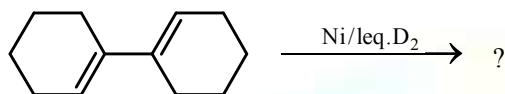
24.



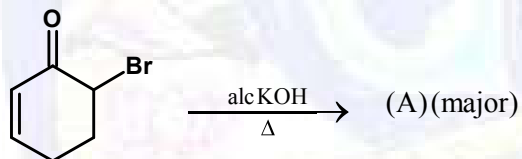
Which of the following compounds will not be obtained in the above reaction?



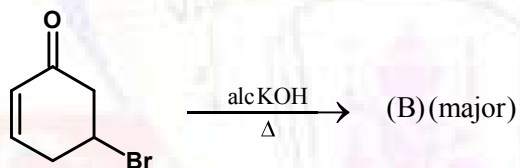
25.



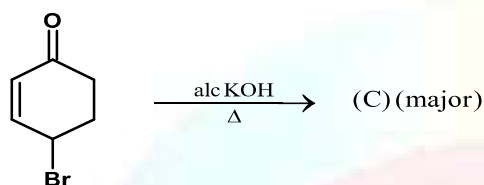
26. Reaction (1)



Reaction (2)



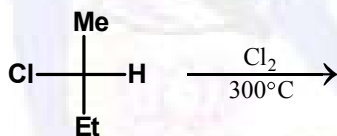
Reaction (3)



Product obtained in above reactions (1) (2) and (3) is-

- A) A = B but C is different
- B) A = B but B is different
- C) B = C but A is different
- D) A = B = C all products are identical

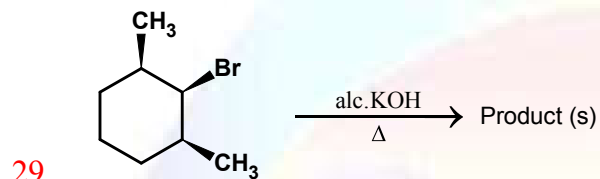
27. Pick the correct statement for monochlorination of R-2-chlorobutane-



- A) There are four possible products; three are optically active, one is optically inactive
- B) There are five possible products; three are optically inactive & two are optically active.
- C) There are five possible products; two are optically inactive & three are optically active.
- D) There are four possible products; two are optically active & two are optically inactive.

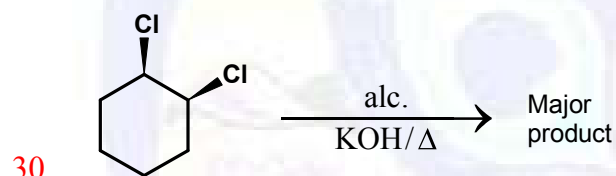
28. $\text{Ph}-\text{CH}_2-\text{CH}_2\text{Br}$ on treatment with $\text{C}_2\text{H}_5\text{OD}/\text{C}_2\text{H}_5\text{O}^-$ gives $\text{Ph}-\text{CD}=\text{CH}_2$ as the product. Which of the following is the most appropriate mechanism for this reaction.

- A) E_1 B) E_2 C) $\text{E}_{1\text{cb}}$ D) E_i

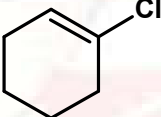


Number of E_2 alkene products (including stereoisomers) is/are-

- A) 1 B) 2 C) 3 D) 4



Will be

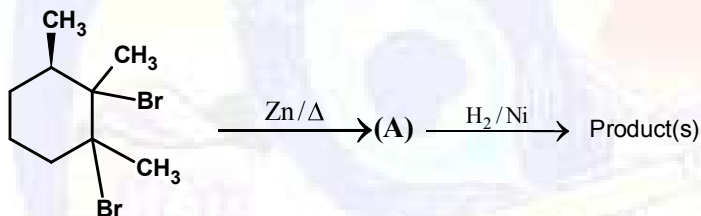
- A)  B)  C)  D) 

SECTION II**Multiple Correct Answer(s) Type**

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

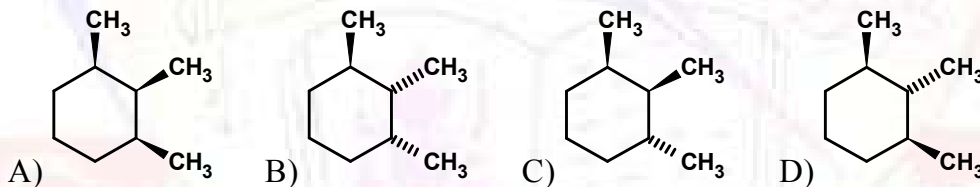
31. Photochemical fluorination is explosive while iodination is too slow to occur. The reason for this is

- A) Bond dissociation energy of I_2 is high
- B) Formation of $CH_3 - F$ is most exothermic
- C) Formation of $H - F$ is most exothermic while formation of $H - I$ is endothermic
- D) F_2 has lower bond dissociation energy than Cl_2 or Br_2

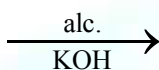
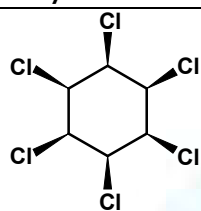


32.

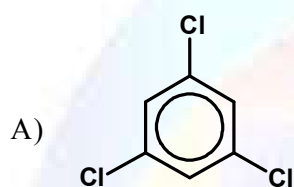
Identify the products



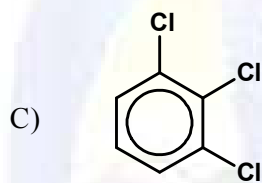
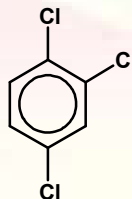
33.



Product mixture will have:



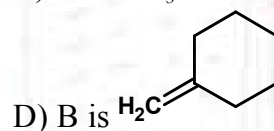
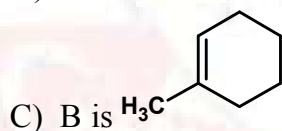
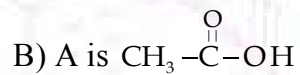
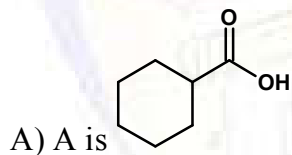
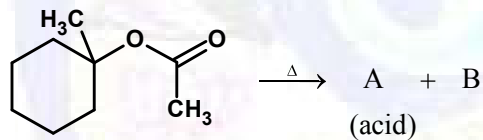
B)



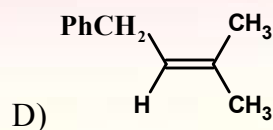
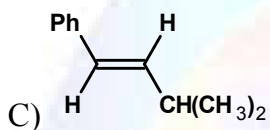
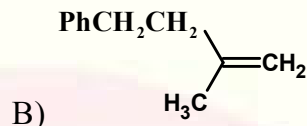
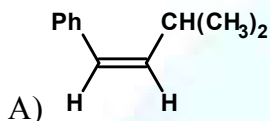
D)



34.



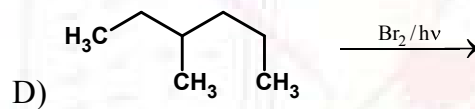
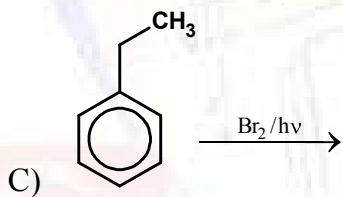
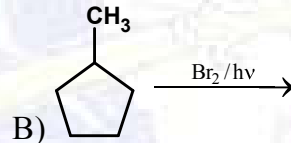
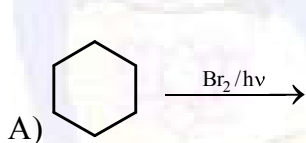
35. The main (formed in maximum percentage) product of the following reaction is

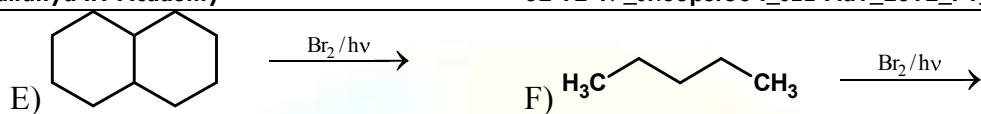


SECTION III Integer Answer Type

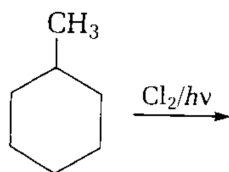
This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

36. Among the following reactions select those in which 2° halide is the major product?

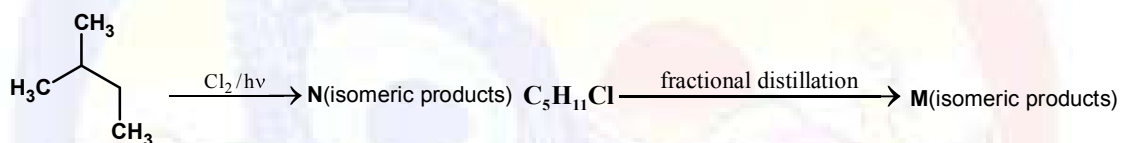




37. Find the total number of isomeric products X in the following reaction and mark (X-5) in your response sheet

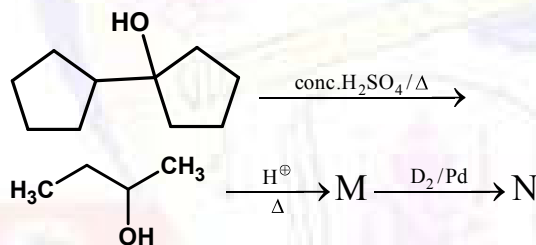


38.



Find the value of M

39. How many transition states are involved in the following dehydration of alcohol.



40.

Find the value of M+N?

MATHEMATICS**Max.Marks:70****SECTION I****Single Correct Answer Type**

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

41. The plane $4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $5x + 3y + 10z = 25$. The equation of the plane in its new position is $x - 4y + 6z = K$ where K is
- A) 106 B) -89 C) 73 D) 37
42. Let P be the image of the point (3,1,7) with respect to the plane $x - y + z = 3$, then the equation of the plane passing through P and containing the straight line $\frac{x}{1} = \frac{y}{2} = \frac{z}{1}$ is
- A) $x + y - 3z = 0$ B) $3x + z = 0$ C) $x - 4y + 7z = 0$ D) $2x - y = 0$
43. The length of the projection of the line segment joining the points (1,-1,0) and (-1,0,1) on to the plane $2x + y + 6z = 1$ is equal to
- A) $\frac{\sqrt{255}}{41}$ B) $\sqrt{\frac{237}{41}}$ C) $\frac{\sqrt{137}}{41}$ D) $\frac{\sqrt{155}}{41}$
44. The three planes $4y + 6z = 5$, $2x + 3y + 5z = 5$, $6x + 5y + 9z = 10$ are such that they
- A) meet in a point B) have a line in common
C) form a triangular prism D) none

45. The volume of the tetrahedron included between the plane $5x + 3y + 4z - 30 = 0$ and the three coordinate planes is
A) 150 B) 225 C) 75 D) 600
46. Perpendiculars are drawn from points on the line $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}$ to the plane $x + y + z = 3$. The feet of the perpendiculars lie on the line
A) $\frac{x}{5} = \frac{y-1}{8} = \frac{z-2}{-13}$ B) $\frac{x}{2} = \frac{y-1}{3} = \frac{z-2}{-5}$ C) $\frac{x}{4} = \frac{y-1}{3} = \frac{z-2}{-7}$ D) $\frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}$
47. The distance of the point $(1, -2, 3)$ from the plane $x - y + z - 5 = 0$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z-1}{-6}$ is
A) 1 B) 2 C) 4 D) 10
48. Which one of the following is best condition for the plane $ax + by + cz + d = 0$ to intersect the x and y axes at equal angles is
A) $|a| = |b|$ B) $a = -b$ C) $a = b$ D) $a^2 + b^2 = 1$
49. The projection of the line $\frac{x+1}{-1} = \frac{y}{2} = \frac{z-1}{3}$ on the plane $x - 2y + z = 6$ is the line of intersection of the given plane with the plane whose possible equation is
A) $2x + y + 2 = 0$ B) $3x + y - z = 2$ C) $2x - 3y + 8z = 3$ D) $x - 3y + 8z = 2$

50. The perpendicular distance from origin to the line of intersection of the planes $\vec{r} \cdot \vec{a} = 1, \vec{r} \cdot \vec{b} = 1$ is

A) $\left| \frac{\vec{a} + \vec{b}}{\vec{a} \times \vec{b}} \right|$ B) $\left| \frac{\vec{a} - \vec{b}}{(\vec{a} \times \vec{b})^2} \right|$ C) $\sqrt{\left| \frac{\vec{a} - \vec{b}}{\vec{a} \times \vec{b}} \right|}$ D) $\left| \frac{\vec{a} - \vec{b}}{\vec{a} \times \vec{b}} \right|$

SECTION - II

Multiple Correct Answer(s) Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

51. In R^3 space consider the planes $P_1: y=0$ and $P_2: x+z=1$. Let P_3 be a plane different from P_1 and P_2 which passes through the intersection of P_1 and P_2 . If the distance of the point $(0,1,0)$ from P_3 is 1 and the distance of a point (α, β, γ) from P_3 is 2, then which of the following relation (s) is / are true ?
- A) $2\alpha + \beta + 2\gamma + 2 = 0$ B) $2\alpha - \beta + 2\gamma + 4 = 0$
 C) $2\alpha + \beta - 2\gamma - 10 = 0$ D) $2\alpha - \beta + 2\gamma - 8 = 0$
52. Projection of the line $\frac{x+1}{2} = \frac{y+1}{-1} = \frac{z+3}{4}$ on the plane $x+2y+z=6$ has the possible equation is
- A) $x+2y+z-6=0=9x-2y-5z-8$ B) $x+2y+z+6=0=9x-2y+5z-8$
 C) $\frac{x-1}{4} = \frac{y-3}{-7} = \frac{z+1}{10}$ D) $\frac{x+3}{4} = \frac{y-2}{4} = \frac{z-7}{-10}$

53. A ray is sent along the line $\frac{x-0}{2} = \frac{y-2}{2} = \frac{z-1}{0}$ and is reflected by the plane $x=0$ at a point A. The reflected ray is again reflected by the plane $x+2y=0$ at point B. The initial ray and final reflected ray meets at point J then which of the following is/are true
- A) The coordinates of the point B is (4, -2, 1)
B) The coordinates of the point J is (-3, -1, 1)
C) The centroid of $\triangle ABJ$ is (0, 0, 0)
D) The coordinates of J is (2, -1, 1)
54. The equation of the plane(s) through the origin which are parallel to the line $\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z+1}{-2}$ and at a distance $5/3$ from is/are
- A) $2x+2y+z=0$ B) $x+2y+2z=0$ C) $x-2y+z=0$ D) $x-2y+2z=0$
55. A rod of length 2 units whose one end is (1, 0, -1) and other end touches the plane $x-2y+2z+4=0$ then which of the following is/are true
- A) The rod sweeps the figure with the given plane whose volume is π cubic units
B) The area of the region which the rod traces on the plane is 2π
C) The length of projection of the rod on the plane is $\sqrt{3}$ units
D) The centre of the region which the rod traces on the plane is $\left(\frac{2}{3}, \frac{2}{3}, \frac{-5}{3}\right)$

SECTION III
Integer Answer Type

This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

56. Let $P(a,b,c)$ be any point on the plane $3x+2y+z=7$ then the least value of $2(a^2+b^2+c^2)$ is
57. The distance of z-axis from the image of the point $(2,-3,3)$ in the plane $x-2y-z+1=0$ is
58. The position vectors of four angular points of a tetrahedron OABC are $(0,0,0), (0,0,2), (0,4,0)$ and $(6,0,0)$ respectively. A point P inside the tetrahedron is at the same distance r from the four plane faces of the tetrahedron. The value of $9r$ is
59. If the line $3x-2y+z+3=0=4x-3y+4z+1$ is parallel to the plane $2x-y+mz-2=0$ then the value of $|m|$ is
60. A square ABCD of diagonal $2a$ is folded along the diagonal AC so that planes DAC, BAC are at right angles. If the shortest distance between DC and AB is $\frac{Ka}{\sqrt{3}}$ then K is equal to



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Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

2012_P1

Date: 02-12-17

Max.Marks:210

KEY SHEET

PHYSICS

1	B	2	C	3	B	4	B	5	D
6	D	7	C	8	A	9	B	10	C
11	BC	12	BCD	13	ABCD	14	AC	15	CD
16	8	17	4	18	2	19	4	20	5

CHEMISTRY

21	D	22	C	23	B	24	A	25	A
26	D	27	C	28	C	29	B	30	C
31	BC	32	ABC	33	AB	34	BD	35	C
36	3	37	7	38	4	39	5	40	8

MATHS

41	A	42	C	43	B	44	B	45	C
46	D	47	A	48	A	49	A	50	D
51	BD	52	AC	53	AB	54	AD	55	ACD
56	7	57	1	58	6	59	2	60	2

SOLUTIONS

PHYSICS

1. $F_{\text{viscous}} = \eta b L v \left[\frac{h}{x(h-x)} \right]$ (force is minimum at $x = \frac{h}{2}$)

$$F_{\text{viscous}} = \frac{4\eta b L v}{h}$$

2. viscous force per unit area $= \eta \frac{dv}{dy}$

3. Conceptual

4. $u_{01} = \frac{2}{9} \frac{r^2 \rho g}{\eta}$ $u_{02} = \frac{r^2 \rho g}{54\eta}$

5. Conceptual

6. Conceptual

7. Conceptual

8. Conceptual

9. Conceptual

10. Conceptual

11. Conceptual

12. viscous force $= m g \sin \theta$

$$\eta a^2 \frac{v}{t} = a^3 \rho g \sin \theta$$

13. The value of fluid flowing through this section per second

$$dV = (2\pi r dr) v_0 (1 - r^2 / R^2)$$

$$\text{Total volume } V = \int_0^R (2\pi r dr) v_0 (1 - r^2 / R^2)$$

$$= 2\pi v_0 (R^2 / 4)$$

ii) The Kinetic energy of the fluid within the volume element of thickness dr

K.E of fluid within the tube is

$$= \frac{1}{2} (2\pi l) \rho v_0^2 \int_0^R (1 - r^2 / R^2)$$

$$\text{We get K.E } \pi \rho l v_0^2 (R^2 / 6)$$

Iii) The viscous drag exerts a force on the tube

$$F = -\eta A \left(\frac{dv}{dr} \right)_{r=R}$$

$$\text{Here } \left(\frac{dv}{dr} \right)_{r=R} = V_0 \left(-2r / R^2 \right)_{r=R} = -2F = V_0 / R$$

$$\therefore F = 4\pi\eta l v_0$$

$$\Delta P = P_2 - P_1 = P$$

$$\text{iv) where } P_1 = 0 \text{ \& } P_2 = P$$

$$P = \frac{\text{force}(F)}{\text{area}(\pi R^2)} = \frac{4\eta l v_0}{R^2}$$

14. Conceptual

$$\frac{p - \sigma_1}{\eta_1} = \frac{p - \sigma_2}{\eta_2} = \frac{p - \sigma_3}{\eta_3}$$

$$15. \sigma_1 < \sigma_2 < \sigma_3 \Rightarrow \eta_1 > \eta_2 > \eta_3$$

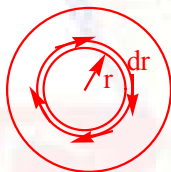
$$\text{Also } \frac{\eta_1}{\eta_3} = \frac{p - \sigma_1}{p - \sigma_3} > \frac{\eta_3}{\eta_2} = \frac{p - \sigma_3}{p - \sigma_2} < 1 \Rightarrow (C)$$

Eliminating P gives (D)

16. Conceptual

17. velocity gradient=2

$$F = 2\eta A \frac{dv}{dy} = 0.004 N$$



18.

$$d\tau = r \cdot (v \cdot F)$$

$$= r \left[\frac{\mu \cdot 2\pi r dr \cdot r (\omega_i - \omega_o)}{a} \right]$$

$$19. v_{\text{bigger}} = n^{\frac{2}{3}} v_{\text{small}} \text{ Conceptual}$$

$$20. \text{Rate of heating} = \dot{Q} \text{ so } F_v \text{ proportional to } r^3 \text{ hence rate of heating is proportional to } r^5$$

CHEMISTRY

21. Carbocation rearrangement then saytzeff elimination.

22. With increase in branching boiling point decreases

$$23. \%(2^\circ) = \frac{7.6}{21.6} \times 100 \approx 35\% (\text{major})$$

- CC(=O)OC1CCCCC1

 $\downarrow$

CC(=O)O + C1=CCCCC1
- The reaction shows the thermal decomposition of methyl cyclohexylcarbamate into methacrylic acid (A) and cyclohexene (B).

- ## MATHS

- Page 4

$$\therefore \text{the plane is } -x + 4y - 6z + 106 = 0 \Rightarrow x - 4y + 6z = 106$$

$$\Rightarrow K = 106$$

42. Image of $(3, 1, 7)$ is $(-1, 5, 3)$

$$\text{Equation of the plane is } \begin{vmatrix} x & y & z \\ 1 & 2 & 1 \\ -1 & 5 & 3 \end{vmatrix} = 0 \Rightarrow x - 4y + 7z = 0$$

43. Let A be the acute angle between the normal of the plane and the line segment joining the points

$$\therefore \cos A = \frac{3}{\sqrt{6}\sqrt{41}} \Rightarrow \text{length of projection} = \sqrt{6} \sin A$$

$$= \sqrt{6} \sqrt{1 - \frac{9}{246}}$$

$$= \sqrt{6 \cdot \frac{237}{246}} = \sqrt{\frac{237}{41}}$$

44. $\begin{vmatrix} 0 & 4 & 6 \\ 2 & 3 & 5 \\ 6 & 5 & 9 \end{vmatrix} = -4(-12) + 6(-8) = 0$

$$\text{And } \Delta_1 = \begin{vmatrix} 5 & 4 & 6 \\ 5 & 3 & 5 \\ 10 & 5 & 9 \end{vmatrix} = 0$$

45. $\frac{x}{6} + \frac{y}{10} + \frac{z}{15/2} = 1$

$$\therefore \text{Volume} = \frac{1}{6} \times 6 \times 10 \times \frac{15}{2} = 75$$

46. Any point P on line is $(2\lambda - 2, -\lambda - 1, 3\lambda)$. Point P lies on the plane

$$\Rightarrow \lambda = \frac{3}{2} \Rightarrow P = \left(1, -\frac{5}{2}, \frac{9}{2}\right)$$

The foot of the perpendicular from $(-2, -1, 0)$ on the plane is $(0, 1, 2)$

$$\therefore \text{the line is } \frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}$$

47. Equation of line passing through $(1, -2, 3)$ and parallel to the given line is

$$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-6} \text{ any point on the line is } (2\lambda+1, 3\lambda-2, -6\lambda+3). \text{ It lies on the plane}$$

$$\Rightarrow \lambda = \frac{1}{7} \therefore \text{the point is } \left(\frac{9}{7}, \frac{-11}{7}, \frac{15}{7}\right)$$

$$\therefore \text{ distance} = \sqrt{\left(1 - \frac{9}{7}\right)^2 + \left(-2 + \frac{11}{7}\right)^2 + \left(3 - \frac{15}{7}\right)^2} = 1$$

48. Let α be the angle between x-axis and plane

$$\Rightarrow \cos(90^\circ - \alpha) = \frac{|a|}{\sqrt{a^2 + b^2 + c^2}} \text{ similarly angle between y-axis and plane is}$$

$$\alpha \Rightarrow \sin \alpha = \frac{|b|}{\sqrt{a^2 + b^2 + c^2}} \Rightarrow |a| = |b|$$

49. Equation of the plane containing the given line and perpendicular to the plane

$$x - 2y + z - 6 = 0 \text{ is } \begin{vmatrix} x+1 & y & z-1 \\ -1 & 2 & 3 \\ 1 & -2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow (x+1)(8) - y(-4) + (z-1)(0) = 0$$

$$\Rightarrow 2x + y + 2 = 0$$

50. Conceptual

51. Here P_3 is $(x+z-1) + \lambda y = 0$ Distance from $(0, 1, 0)$ is 1

$$\therefore |\lambda - 1| = \sqrt{\lambda^2 + 2} \Rightarrow \lambda = \frac{-1}{2}$$

$$\therefore \text{Equation of } P_3 \text{ is } 2x - y + 2z - 2 = 0$$

$$\Rightarrow |2\alpha - \beta + 2\gamma - 2| = 6 \Rightarrow 2\alpha - \beta + 2\gamma - 2 = \pm 6$$

52. It is the line of intersection of the given plane and a plane perpendicular to given plane and containing the given line

53. $A = (0, 2, 1)$

54. We can easily get it from options

55. The rod sweeps out the figure which is a cone. The distance of the point $A(1, 0, -1)$ from the plane is 1 unit. The slant height of the cone is 2 units. The radius of base is $\sqrt{3}$

$$\therefore \text{ volume of the cone is } \frac{\pi}{3} (\sqrt{3})^2 \cdot 1 = \pi \text{ C.C}$$

$\therefore$ Area of the circle on the plane which the rod traces is 3π

56. Clearly minimum value of $a^2 + b^2 + c^2 = \left(\frac{|-7|}{\sqrt{14}}\right)^2 = \frac{49}{14} = \frac{7}{2}$

$\therefore$ minimum value of $2(a^2 + b^2 + c^2) = 7$

57. $\frac{a-2}{1} = \frac{b+3}{-2} = \frac{c-3}{-1} = -2 \Rightarrow$ the image is $(0, 1, 5)$

58. Let $P(r, r, r)$ be the point which is equidistant from the four faces. Its distance is r from the plane $2x + 3y + 6z = 12$ also

$\therefore |11r - 12| = 7r \Rightarrow r = \frac{2}{3} \text{ or } 3 \text{ but } r < 2 \Rightarrow r = 2/3$

59. Dir's of the line are $5, 8, 1$. It is perpendicular to the normal to the plane

$\therefore 10 - 8 + m = 0 \Rightarrow m = -2$

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

Date: 09-12-17

Max.Marks:186

Name of the Student: _____

H.T. NO:

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2016_PAPER-II

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

09-12-17_Jr.Super60-I_Jee Adv_WAT-28_Weekend Syllabus

MATHS: SYLLABUS COVERED FROM 04-12-17 to 08-12-2017

PHYSICS: SYLLABUS COVERED FROM 04-12-17 to 08-12-2017

CHEMISTRY: Reactions of alkenes: Addition of hydrogen halide, halogen, halogen/water, halogen/alcohol, NOCl, RCl, RCOCl, RH, water, oxymercuration - demercuration, hydroboration - oxidation; Mechanism of electrophilic and free radical addition, Hydroxylation reactions; Oxidative cleavage - KMnO₄, ozonolysis; allyl substitution, allyl oxidation; Isomerization; Reaction with carbenes and carbenoids, 1-2 and 1-4 addition on unsaturated systems

JEE-ADVANCE-2016-P2-Model**Time: 3:00 hours****IMPORTANT INSTRUCTIONS****Max Marks: 186****PHYSICS:**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 7 – 14)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 15 – 18)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	3	0	4	12
Total				18	62

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 25 – 32)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 33 – 36)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	3	0	4	12
Total				18	62

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 43 – 50)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	3	0	4	12
Total				18	62

Sec: Jr. Super60-I**space for rough work****Page 2**

PHYSICS

Max Marks: 62

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

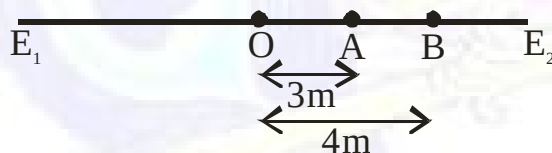
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

1. The displacement of two identical particles executing SHM are represented by equations $x_1 = 4 \sin[10t + [\pi/6]]$ and $x_2 = 5 \cos \omega t$. For what value of ω energy of both the particles is same? (At the mean position potential energy is taken as zero).

A) 16 units B) 6 units C) 4 units D) 8 units

2. A particle is performing SHM. Its amplitude is 5 m and time period of oscillation is 4 s. The minimum time taken by the particle to move from point A to B is (O is mean position E_1 and E_2 are extreme positions)



A) 0.12 s B) 0.16 s C) 0.18 s D) 0.19 s.

3. The mass of the astronauts in a space station cannot be checked with the scales used on the Earth. Instead we use a spring balance, which oscillates when a mass is suspended

from it. Imagine a seat of a mass of 20 kg, which oscillates with a frequency of 2Hz. After the astronaut sits in the seat the frequency decreases to 1 Hz. How much is the mass of the astronaut?

- A) 40kg B) 60 kg C) 80 kg D) 100 kg

4. A particle performs simple harmonic motion on y-axis with amplitude A and time period T. The time taken by the particle to travel a distance A/5 starting from rest is

- A) $\frac{T}{20}$ B) $\frac{T}{2\pi} \cos^{-1}\left(\frac{4}{5}\right)$ C) $\frac{T}{2\pi} \cos^{-1}\left(\frac{1}{5}\right)$ D) $\frac{T}{2\pi} \sin^{-1}\left(\frac{1}{5}\right)$

5. A particle of mass m is located at a place where it is acted upon by a one dimensional conservative force and its potential energy is given by:

$U(x) = A(1 - \cos px)$, where A and p are constants. The period of small oscillations of the particle is:

- A) $2\pi \sqrt{\frac{m}{Ap}}$ B) $2\pi \sqrt{\frac{m}{Ap^2}}$
C) $2\pi \sqrt{\frac{m}{A}}$ D) $\frac{1}{2\pi} \sqrt{\frac{Ap}{m}}$

6. The displacement x of a particle in motion is given in terms of time by

$$x(x-4) = 1 - 5 \cos \omega t$$

- A) The particle executes SHM
- B) The particle executes oscillatory motion which is not SHM
- C) The motion of the particle is neither oscillatory nor simple harmonic
- D) The particle is not acted upon by a force when it is at $x = 4$

SECTION – II
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases

7. The x -coordinate of a particle moving on x -axis is given by $x = 3\sin 100t + 8\cos^2 50t$ where x is in cm and t is time in seconds. Which of the following is/are correct about this motion.
- A) The motion of the particle is not S.H.M
 - B) The amplitude of the S.H.M of the particle is 5 cm
 - C) The amplitude of the resultant S.H.M is $\sqrt{73}$ cm
 - D) The maximum displacement of the particle from the origin is 9 cm

8. A particle is executing SHM with a time period T and amplitude A . At $t = 0$ displacement of the particle was $+A/2$ and it was moving away from the mean position. $\vec{x}$, $\vec{v}$ and $\vec{a}$ are the displacement from the mean position, velocity and acceleration of the particle at any instant. Choose the correct statement(s).
- A) at $t = T/4$, $\vec{a} \cdot \vec{v} > 0$ B) at $t = T/4$, $\vec{x} \cdot \vec{v} < 0$
- C) at $t = T/2$, $\vec{a} \cdot \vec{v} < 0$ D) at $t = T/2$, $\vec{x} \cdot \vec{v} > 0$
9. A coin is placed on a horizontal platform, which undergoes vertical simple harmonic motion of angular frequency ω . The amplitude of oscillation is gradually increased. The coin will leave contact with the platform for the first time
- A) at the highest position of the platform
- B) at the mean position of the platform
- C) for an amplitude of g/ω^2
- D) for an amplitude of $\sqrt{g}/\omega$

10. A particle is executing SHM between points $-X_m$ and X_m , as shown in figure – 1. The velocity $V(t)$ of the particle is partially graphed and shown in figure – II. Two points A and B corresponding to time t_1 and time t_2 respectively are marked on the $V(t)$ curve,

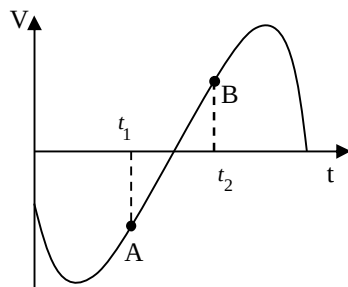
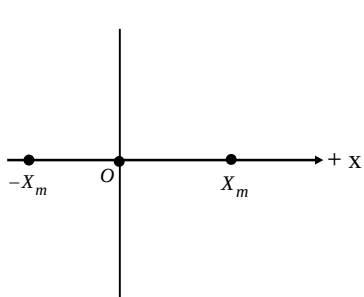


Figure – I

Figure – II

- A) At time t_1 , it is going towards $+X_m$
- B) At time t_1 , its speed is decreasing
- C) At time t_2 , its position lies in between $-X_m$ and O
- D) The phase difference $\Delta\phi$ between points A and B must be expressed as $90^\circ < \Delta\phi < 180^\circ$